Beta – Version 0.98-RC

Geometry First: Quantum Mechanics, Gravity, and the Origin of the Standard Model from a Single Conservation Law

Jack Menendez

Eugene, OR

ORCID: 0009-0003-1166-307X

jackdmenendez@gmail.com

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Disclaimer: This is a working paper, and the content herein is subject to change. The author welcomes feedback and constructive criticism. Please reach out via email or social media with any comments or questions.

Data availability. All numerical experiments, data files, and the LaTeX source for this paper are publicly available at `github.com/JackDMenendez/discrete-causal-lattice`. The exact commit corresponding to this release is the v0.98-RC tag, archived at `doi:10.5281/zenodo.20017133`. A compact reproducibility quickstart, including a 10-line minimal verification snippet, is in Appendix C.

Keywords. discrete spacetime; bipartite causal lattice; single conservation law; emergent quantum mechanics; clock-fluid gravity; Arnold-tongue quantization; induced gauge action; lattice birefringence.

What changed in v0.98-RC (since v0.97-RC). v0.98-RC is an experimental-progress and content-alignment release. Two new experiments completed their first runs; two new entries were added to the audit table (Table 1, both `PART`); and the paper text was systematically reconciled with the audit-table statuses, with the help of an automated consistency-check pass introduced in this release:

- `exp_20` — **three-arm emission-operator comparison.** First experiment to test whether a unitary 2-mode rotation (beam splitter, row 6 of the operation algebra) is the correct emission operator for the lattice's joint $\mathcal{A} = 1$ conservation, compared directly against the as-written drain in `exp_19c` (arm A) and the phase-rotation drain (arm C). Joint $\mathcal{A} = 1$ preserved to machine precision (8.9×10^{-16}) under the beam splitter; the `exp_19c` drain reproduces the row-5 non-unitarity arithmetic exactly. Phase-dependent non-monotonic transfer in arm B is an unanticipated refinement to the operation-algebra table that a follow-on calculation will need to address. Orbital settling not achieved at the swept rate; see next item.

- `exp_12b` — **long-duration two-body baseline (no emission).** The bare `exp_12` setup, run for 6000 ticks. r_{peak} escapes from R_1 to grid edge (~ 83) by tick ~ 2000 and never returns. $\mathcal{A} = 1$ preserved at machine precision throughout. The orbital escape seen across all three arms of `exp_20` is inherited from this baseline, not caused by the emission operator. Refines but does not invalidate `exp_12`'s 4-sig-fig PASS, which is scored on a shorter k-scan resonance peak.

- **Two new entries in the audit table** (both `PART`, both additive — no existing entries downgraded): "Two-body long-horizon stability" (the `exp_12b` finding) and "Emission-operator joint $\mathcal{A} = 1$ conservation" (the `exp_20` arm B finding). The evidence column for the existing entry "Photon emission as $\mathcal{A} = 1$ necessity" was updated to reflect the `exp_20` results; that entry remains `PART`.

- **Paper-text reconciliation.** Four sites with pre-`exp_20` `exp_19c` overstatement aligned with the audit-table: `abstract.tex` items (v) and (vi), `introduction.tex` ($\times 2$ mentions), `hydrogen_spectrum.tex`, and `vacuum_twist_field_equations.tex`. The row-5 amplitude-drain non-unitarity arithmetic $(1 - 2m + 2m^2)\|\psi_e\|^2$ is now explicitly cited in the hydrogen-spectrum write-up of `exp_19c`.

- **New introduction paragraph: "One law at three resolutions."** Identifies the clock fluid density $\rho_\phi(x) = \sum_i \omega_i \, |\psi_i(x)|^2$ as the ω-weighted form of the joint amplitude density $\mathcal{A}_{\text{joint}}(x) = \sum_i |\psi_i(x)|^2$. Probability conservation, mass-energy conservation, and gravity all share one structural origin — sharpens the framework's "single conservation law" claim from assertion to identification.

An automated consistency-check tool (committed at `.claude/agents/dcl-claim-auditor.md`) was added to compare paper prose against the audit-table statuses; during this release cycle it identified two further overstatements in the abstract that the manual sweep had missed. Eight follow-on or speculative notes were moved out of the project repository into a separate private archive, leaving `notes/` at 53 files (down from 61), each tied to a main-paper claim or an active experiment.

A more detailed change log lives in `release_notes/v0.98-RC.md` in the project repository. Earlier versions remain available at the Zenodo deposit's prior DOIs.

Abstract

We propose that conservation of probability — the single axiom $\mathcal{A} = 1$ — is sufficient to derive the principal results of quantum mechanics without additional postulates. On a bipartite three-dimensional causal lattice $\mathcal{T}^3_\diamond$, each particle is represented as a causal session: an independent phase oscillator that propagates amplitude across the lattice subject to $|\psi_R|^2 + |\psi_L|^2 = 1$ at every tick. No Hamiltonian and no measurement postulate are assumed; a Hilbert-space structure with unitary evolution emerges from the U(1) oscillator and the per-session amplitude constraint rather than being postulated a priori.

From this single constraint we derive: **(i)** the Dirac equation in the continuum limit $a \to 0$, with the gamma matrices identified as geometric encodings of the RGB/CMY bipartite sublattice geometry; rotational invariance is recovered on O_h-averaged observables (the same mechanism by which staggered fermions recover Lorentz invariance in lattice QCD), and the residual lattice birefringence aligned with the optical axis $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$ is a falsifiable prediction with closed-form coefficients (Section 6); **(ii)** wave-particle duality and the Born rule as consequences of amplitude redistribution under $\mathcal{A} = 1$; **(iii)** quantization of atomic orbitals as Arnold tongue attractors of the two-body phase dynamics, recovering the hydrogen Bohr radius to four significant figures without fitting; **(iv)** gravity as a clock-density gradient rather than spacetime curvature — a refraction of causal paths through regions of higher session density, reproducing the $1/r^2$ force law with a falsifiable discrete correction at short range; **(v)** photon emission as a structural consequence of $\mathcal{A} = 1$: the Coulomb interaction is a probability source that a single session cannot accommodate, mandating creation of a new photon session at orbit lock-in; a first numerical implementation (`exp_19c`) preserves per-session unitarity at all swept emission rates, and a controlled three-arm operator comparison (`exp_20`) confirms a pointwise unitary beam splitter

as the joint-$\mathcal{A} = 1$-preserving emission operator (joint $\mathcal{A} = 1$ preserved to machine precision 8.9×10^{-16}); orbital settling remains open, traced to a long-horizon two-body baseline instability isolated in `exp_12b` (Table 1; both rows `PART`); **(vi)** the structural form of the induced gauge action (Sakharov / Zeldovich, Appendix B), whose leading-order term has the Maxwell form on O_h-averaged observables together with a closed-form *gauge-sector birefringence* sharing the same optical axis $(1, 1, -1)$ as the kinematic-sector prediction (numerical $1/g^2$ prefactor and the empirical test of the birefringence remain open); the predicted concordance of kinematic, gauge, gravitational, and atomic-clock channels along this single axis is the framework's strongest *falsifiable* prediction (P9, Section 12.10).

Energy conservation and momentum conservation are not independent inputs — they are consequences of $\mathcal{A} = 1$ applied to closed session sets.

The numerically tractable results are verified or partially verified by a suite of more than fifteen experiments with no free parameters beyond the Planck-scale calibration of the lattice spacing (Table 1 records the status of each claim); the framework's remaining derivations — black-hole thermodynamics among them — are established analytically under explicitly stated modelling assumptions and flagged as such where they appear.

The framework suggests a structural reinterpretation of the Standard Model's gauge degrees of freedom as the minimal accounting required to maintain $\mathcal{A} = 1$ on the bipartite lattice geometry: the particle content is reframed as the catalogue of probability imbalances the lattice generates and the session types required to resolve them. Whether the full Standard Model gauge group SU(3)×SU(2)×U(1) is the automorphism group of $(\mathcal{T}^3_\diamond, \mathcal{A} = 1)$ is a well-posed finite-dimensional Lie-algebra question, deferred to a companion paper.

Contents

1 Introduction

Beta Status (v0.98-RC): This is a beta release of an ongoing research program. The core framework and the principal derivations are validated; the numerical predictions in Section 12 have been sharpened against current data with explicit modelling assumptions stated where they enter. Two experiments completed in this release (`exp_20` and `exp_12b`, both `PART` per Table 1) confirm joint $\mathcal{A} = 1$ conservation at machine precision while opening a new question about long-horizon two-body stability that the paper does not yet answer; this and several other follow-on calculations are flagged in the conclusion's *What remains open* subsection. *Status legend used throughout:* `PASS` = derivation complete or experiment confirms the claim to stated precision; `PART` = mechanism demonstrated, full quantitative match pending; `STUB` = analytical result with no numerical confirmation yet (see Table 1 for the full status of every claim). Comments and feedback are welcome ahead of the v1.0 release.

Quantum mechanics and general relativity are the two most precisely verified theories in the history of science. Between them they answer, with extraordinary accuracy, every measurable question about how matter behaves and how spacetime curves: what energies particles occupy, how they scatter, how they interfere, how massive bodies attract. But both leave a different class of questions open — why their own foundations hold.

Why are atomic spectra discrete rather than continuous? The Schrödinger equation produces discrete eigenvalues given the right boundary conditions, but the boundary conditions are loaded into the formalism before the calculation begins. Why is the Born rule $|\psi|^2$ = probability true? It is a postulate — von Neumann's measurement axiom — not a derivation. Why do the Dirac gamma matrices have the algebraic structure they do? Dirac constructed them to satisfy his requirements; they were not derived from anything geometrically prior. Why is the gauge group of the Standard Model $SU(3) \times SU(2) \times U(1)$ and not some other group? The Standard Model takes its gauge content as given. And beyond quantum mechanics: why are gravitational and inertial mass exactly equal? General relativity declares their equality as the equivalence principle — a postulate, not a derivation — and provides no mechanism for it.

These are not criticisms of quantum mechanics or general relativity. They are a precise statement of what both theories are silent about, and that silence has been known since at least von Neumann [1] and is uncontroversial. A framework that answered these questions would not replace quantum mechanics or general relativity; it would explain them.

This paper proposes such a framework. We show that all five questions have geometric answers, and that all five answers follow from a single conservation law:

Conservation of probability — the axiom $\mathcal{A} = 1$.

On a bipartite three-dimensional causal lattice $\mathcal{T}^3_\diamond$, each particle is a *causal session*: an independent phase oscillator carrying amplitude (ψ_R, ψ_L) on the two chirally-opposite sublattices, subject to $|\psi_R|^2 + |\psi_L|^2 = 1$ at every tick. No Hamiltonian, no Hilbert space, and no measurement postulate are assumed.

The framework's dynamics are scale-free: the tick rule and $\mathcal{A} = 1$ constraint operate identically at any lattice spacing a, and all experiments produce dimensionless ratios (e.g. R_1/a, $k_{\mathrm{Bohr}} \cdot a$) that are independent of a. Physical units enter only through a single calibration: identifying $a = \ell_P$ (one Planck length) and $\tau = t_P$ (one Planck time) fixes the speed limit $c = a/\tau = \ell_P/t_P$ and converts dimensionless lattice results into SI predictions. The numerical experiments in this paper run at arbitrary lattice spacing; the Planck identification is applied only when connecting results to physical observables (Section 2).

The five answers:

1. **Quantization is an attractor, not a boundary condition.** The discrete mass spectrum arises from Arnold tongue frequency-locking between the particle's internal Zitterbewegung frequency and the vacuum carrier. Any nearby frequency is dynamically pulled into the nearest rational resonance basin. Discreteness is self-organising, not postulated.

2. **The Born rule follows from $\mathcal{A} = 1$.** Probability is amplitude squared because amplitude is the only physical quantity on the lattice and $\mathcal{A} = 1$ is the conservation law that governs it. The Born rule is not a measurement axiom; it is a conservation theorem.

3. **The Dirac gamma matrices encode lattice geometry.** The bipartite RGB/CMY sublattice structure *is* the two-component Dirac spinor. In the continuum limit $a \to 0$, the tick rule converges to the Dirac equation $(i\gamma^\mu \partial_\mu - m)\psi = 0$, with the gamma matrices identified as the geometric encoding of three pairs of antiparallel basis vectors. The Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ holds on O_h-averaged observables (the bipartite frame matrix has eigenvalues $\{4, 4, 1\}$ and is anisotropic at the operator level; its O_h-averaged form is diagonal and reproduces the standard anticommutator). The residual operator-level anisotropy along the optical axis $(1, 1, -1)$ is itself a falsifiable lattice-birefringence prediction (P7, P9 in Section 12).

4. **Photon emission is a conservation-law necessity.** The Coulomb interaction is a probability *source* — it injects amplitude into the electron session from outside. $\mathcal{A} = 1$ cannot accommodate this within a single session; it mandates creation of a new photon session at orbital lock-in. Joint momentum conservation across the three-session event,

$$\mathbf{p}_e + \mathbf{p}_p \;=\; \mathbf{p}'_e + \mathbf{p}'_p + \mathbf{p}_\gamma,$$

requires the photon's outgoing momentum $\mathbf{p}_\gamma$ to be compensated by an opposite shift in the bound-state momentum, $\Delta\mathbf{p}_{\text{atom}} = -\mathbf{p}_\gamma$ — the photon recoil observed routinely in atomic spectroscopy and Mössbauer-line measurements. Emission, recoil, and energy conservation all follow from $\mathcal{A} = 1$ alone. A first numerical implementation (`exp_19c`) preserves per-session unitarity at all swept emission rates but does not yet exhibit settled orbital lock-in; the audit table (Table 1) records this row as `PART`, and a controlled three-arm operator comparison (`exp_20`) is the next implementation step.

5. **Gravity is refraction, not curvature.** Every causal session carries an independent tick counter. The local density of these clocks — the session density weighted by probability — is the gravitational field. Causal paths refract toward regions of higher clock density by the same mechanism as Fermat's principle in optics: amplitude hops preferentially toward nodes where the phase cost is lower. No curved geometry is required. Gravitational time dilation is scheduler load: a clock near a mass processes more causal events per external tick and therefore runs slow. The equivalence principle is structural: both gravitational and inertial mass arise from the same phase-mismatch parameter ω, so their equality requires no explanation. The event horizon is scheduler saturation — a computational deadlock at the Planck density ceiling $\rho_{\max} = \ell_P^{-3}$ — with no singularity, and Hawking radiation as stochastic queue-drop. In the continuum limit the clock fluid equations reduce to the Einstein field equations; at finite lattice spacing they carry discrete corrections to the $1/r^2$ law that are computable and falsifiable.

One law at three resolutions. The five answers above are not five conservation laws, nor even three (probability, momentum, energy). They are three structural resolutions of a single constraint. Per session, $\int |\psi_i|^2\,dx = 1$ is enforced at every tick. As a joint field density across all sessions in a bundle, $\mathcal{A}_{\text{joint}}(x) = \sum_i |\psi_i(x)|^2$ is the conserved quantity whose redistribution between sessions implements emission and annihilation events. The clock fluid density that sources gravity is $\mathcal{A}_{\text{joint}}(x)$ weighted pointwise by per-session instruction frequency,

$$\rho_\phi(x) \;=\; \sum_i \omega_i\,|\psi_i(x)|^2.$$

Local clock-fluid continuity $\partial_t\rho_\phi + \nabla\cdot\mathbf{j}_\phi = 0$ is therefore the same conservation law expressed at field-density resolution with mass weights, and energy conservation is its ω-weighted spatial integral. Probability conservation, mass-energy conservation, and gravity all share one structural origin.

Two further structural results round out the framework. First, the electromagnetic gauge action — which Section 9 postulates following the standard lattice-gauge prescription of Wilson — can be *induced* from $-\mathrm{Tr}\,\ln D_{\text{lat}}[U]$ by integrating out matter (Sakharov / Zeldovich). Appendix B carries out the structural step symbolically: the leading-order term has the Maxwell form on O_h-averaged observables, with a closed-form gauge-sector birefringence as the operator-level residual. Only the numerical $1/g^2$ prefactor remains. Second, the kinematic, gauge, gravitational, and (under the recoil-driven Zitterbewegung reading, Section 6.1) inertial-mass anisotropies all share a single optical axis $(1, 1, -1)$. The predicted concordance across five independent observation channels is the framework's strongest empirical claim (P9, Section 12.10); inter-channel disagreement falsifies the inheritance-from-bipartite-geometry claim, while concordance pins down the cosmological orientation of the lattice.

A further result, which we consider the deepest, concerns the nature of bound states. The standard treatment of hydrogen places the proton as a fixed external potential $V(r) = -e^2/r$ and solves for the electron's eigenstates. This is not merely an approximation — it is the wrong physical picture.

A static Coulomb potential is insufficient to produce stable quantised orbits from the lattice dynamics alone. The proton must be represented as an active session — a phase oscillator participating in the joint $\mathcal{A} = 1$ accounting every tick — for the electron to explore the well and lock onto resonant orbits spontaneously. Hydrogen is not an electron in a field. It is *two sessions finding a joint Arnold tongue attractor.*

This distinction has directly observable consequences. In a static well the electron collapses to the potential minimum regardless of initial momentum; the basin of attraction around the Bohr orbit has zero width (`exp_11` $n = 2$: flat landscape across all k-values tested). With an active proton session, the proton's Zitterbewegung continuously breaks the spherical symmetry of the potential, providing the stochastic perturbation that drives the electron into the resonance basin; the Bohr orbital wavevector is recovered to four significant figures with no parameter tuning (`exp_12`). The basin of attraction is finite and self-correcting — the orbit is stable not because we initialised it exactly, but because the two-session dynamics continuously restores it.

Photon emission then follows as a three-session event: the joint proton-electron resonance transitions to a lower Arnold tongue, the displaced amplitude is absorbed by a new photon session, and the proton adjusts to the new joint configuration. The bound-state recoil $\Delta\mathbf{p}_{\text{atom}} = -\mathbf{p}_\gamma$ is mandated by joint momentum conservation, not added as a separate postulate. A first numerical implementation (`exp_19c`, two-body initialisation with Coulomb-driven amplitude drain) preserves per-session unitarity at all swept emission rates but does not exhibit settled orbital lock-in; the audit table (Table 1) records this row as `PART`, and a controlled three-arm operator comparison (`exp_20`) is the next implementation step.

The framework also distinguishes two conditions that standard quantum mechanics treats as identical. *Binding* requires only that the total session energy is negative — the two sessions remain in each other's neighbourhood. *Quantization* requires additionally that the joint phase dynamics lock onto an Arnold tongue attractor at a specific resonant radius. A proton too light to resist electron recoil can maintain binding — the sessions stay together — but its excessive recoil prevents the phase dynamics from settling into any resonant configuration. The system is gravitationally bound but orbitally unquantized. Standard quantum mechanics has no mechanism for this distinction: a system is either in an eigenstate or it is not. The $\mathcal{T}^3_\diamond$ framework has the mechanism explicitly, and `exp_16` measures it: below a threshold proton mass, the two-session system occupies a wide, non-resonant orbit rather than collapsing or quantizing.

Relation to prior discrete spacetime programmes. The $\mathcal{T}^3_\diamond$ framework belongs to a tradition of discrete spacetime proposals that includes causal set theory [2], 'T Hooft's cellular automaton interpretation [3], Wolfram's computational universe [4], and Zuse's Rechnender Raum [5]. It differs from these in the precision of its structural claims and the number of falsifiable numerical predictions it makes without free parameters. The bipartite RGB/CMY frame structure is not a tuning choice; the frame matrix $M = \sum_{\text{RGB}} \mathbf{v}_i \mathbf{v}_j^T$ has eigenvalues $\{4, 4, 1\}$ and a special direction along the optical axis $\hat{\mathbf{n}}_* = (1, 1, -1)/\sqrt{3}$, and these are geometric identities of the basis vectors that force both the Clifford algebra and the dispersion relation $E^2 = m^2 + |\mathbf{p}|^2$ on O_h-averaged observables — with a falsifiable lattice birefringence along $\hat{\mathbf{n}}_*$ as the residual signature. The lattice does not approximate known physics — it derives it from a single conservation law and a single geometric substrate.

The framework also relates to emergent gravity proposals such as Verlinde's entropic gravity [6] and Lazarev's Non-Metric Space Information (NMSI) theory [7], which also derive gravity from informational structures. However, these approaches remain phenomenological: they propose gravity as emergent but do not derive the full Standard Model gauge structure or provide the detailed lattice dynamics that enable numerical falsification. The $\mathcal{T}^3_\diamond$ framework provides both the derivation and the computational substrate for testing.

Methodology and the role of $\mathcal{A} = 1$. The governing philosophy throughout is that $\mathcal{A} = 1$ drives every result — it is never fitted to a known answer. The procedure is always the same: write the discrete tick rule, impose $\mathcal{A} = 1$, take the continuum limit, and read off what emerges. The Dirac equation was not the target of the bipartite tick rule; it is what fell out when the continuum limit was taken. The Newtonian $1/r^2$ force law was not inserted into the clock density field; it is the static limit of the Poisson equation that emerges from the Taylor expansion of clock density evolution. The hydrogen Bohr radius was not tuned; the lattice recovered it to four significant figures with no free parameters (`exp_12`). In every case the test is: run the lattice, do not adjust it to match the known answer, and report what it produces. Where the lattice disagrees with known physics at short range — the discrete corrections to the $1/r^2$ law, the nonlinear gravitational wave dispersion near $\rho_{\max}$, the gradient ionization threshold — those disagreements are predictions, not failures.

All numerical experiments are verified against the same criterion: the result must emerge from genuine lattice dynamics with no post-hoc parameter adjustment. Table 1 maps every physical phenomenon discussed in the paper to the lattice mechanism that produces it, the experiment that confirms it, and its current verification status.

Computational resources. All simulations reported here were performed on a single consumer-grade desktop workstation using standard Python (`NumPy` array operations) with no GPU acceleration, no distributed computing, and no supercomputer allocation. The largest grids ($65^3 \approx 274\,000$ nodes) complete in seconds to hours per run. This is not a limitation of the lattice; it is evidence that the framework is computationally accessible. The entire experimental program — including the two-body hydrogen scan, the helium ground state, the double-slit interference figure, and the gravitational tidal ionization probe — was reproduced without specialised hardware. Any result presented here can be independently verified on a modern laptop.

Organisation of the paper. Section 2 defines the $\mathcal{T}^3_\diamond$ lattice, its bipartite structure, and the calibration of physical units. Section 3 introduces causal sessions and the phase oscillator. Sections 4–6 derive momentum, force, and the Dirac equation from the tick rule. Sections 7–9 treat gravity and electromagnetism as distinct phase geometries. Sections 10–

11 cover interference, decoherence, and the observer. Sections 12–15 present the falsifiable predictions and the hydrogen results. Section 16 states what has been established and what remains open.

Table 1: System Audit: $\mathcal{T}^3_\diamond$ predictions mapped to experiments. `PASS` = confirmed by genuine lattice dynamics. `PART` = partial confirmation, further runs pending. `PROG` = in progress. `STUB` = theoretical result only, experiment not yet run. *Scoring note:* `exp_12`'s four-significant-figure `PASS` is scored on a k-scan resonance peak (a short-run quantitative match); long-horizon two-body stability is a separate question and is captured by the `exp_12b` `PART` entry below.

Physical Phenomenon	Lattice Logic	Continuum Limit	Experiment	Status
Speed of light limit	Topological adjacency $v_{\max} = 1$	$c = \ell_P/t_P$	`exp_00`	`PASS`
Inertial persistence	Phase gradient $\nabla\phi$	Newton's first law	`exp_01`	`PASS`
$1/r^2$ discrete correction	Path count deviation from CLT Gaussian	Falsifiable short-range correction	`exp_06`	`PASS`
Dirac equation	Bipartite tick rule, $\mathcal{A} = 1$	$(i\gamma^\mu\partial_\mu - m)\psi = 0$	Derived analytically (§6)	`PASS`
Spin-$\frac{1}{2}$	Two-component spinor from bipartite structure	Clifford algebra on O_h-averaged frame condition	Derived analytically (§6)	`PASS`
Equivalence principle	Gravitational and inertial mass both $= \sin(\omega)/2$	$m_g = m_i$ structural identity	`exp_01`, `exp_02`	`PASS`
Lattice birefringence (kinematic)	M_{eff} eigenvalues $\{4/3, 4/3, 0\}$; flat dispersion along $(1,1,-1)$	Direction-resolved photon dispersion (P7 refinement)	Derived analytically (§6.4, eq:M_eff)	`STUB`
Zitterbewegung source (dual hypothesis)	Bipartite ω-rate alternation: self-driven vs. recoil-driven	Direction-dependent rest mass under recoil hypothesis (P9)	Derived analytically (§6.1)	`STUB`
Quantum interference	Discrete path summation	Schrödinger equation	`exp_03`	`PASS`
Decoherence	Phase scrambling via observer clock	Density matrix decay	`exp_04`	`PASS`
Measurement / irreversibility	Observer as clock addition	Decoherence theory	`exp_05`	`PASS`
Born rule	$\mathcal{A} = 1$ conservation $\Rightarrow |\psi|^2$	Measurement postulate derived	`exp_03`–`exp_05`	`PASS`
Mass quantization	Arnold tongue frequency locking	Discrete spectrum without boundary condition	`exp_09`	`PASS`
Photon dispersion	Group velocity vs. wavenumber k	Brillouin zone boundary	`exp_09`	`PASS`
Gravitational attraction	Clock density gradient, refractive bias	Newtonian $1/r^2$	`exp_02`	`PASS`
Gravitational time dilation	Scheduler load near mass concentration	Schwarzschild metric	`exp_02`, `exp_07`	`PASS`
Clock density conservation	Continuity equation from tick()	Einstein field equations (§8)	`exp_07`	`PASS`
Event horizon + Hawking T	Scheduler saturation $\rho \to \ell_P^{-3}$; boundary surface gravity $\kappa = c^4/(4GM)$	No singularity; $T_H = \hbar c^3/(8\pi GMk_B)$	Derived analytically (§4.6)	`PART`
Bekenstein-Hawking entropy	Boundary session count, $A_{\text{session}} = 4\ell_P^2$	$S = k_B A/(4\ell_P^2) = k_B Ac^3/(4G\hbar)$	Derived analytically (§7.6)	`PART`
EM vs. gravity (curl vs. div)	curl(ϕ) vs. div(ϕ) phase geometry	Maxwell + Einstein	`exp_08`	`PASS`
Photon emission	$\mathcal{A} = 1$ session creation at orbit lock-in	Spontaneous emission rate	`exp_08`	`PASS`
Photon emission as $\mathcal{A} = 1$ necessity	Joint $\mathcal{A} = 1$ + momentum-transfer recoil at orbit; tongue-jump mandates third session	Spontaneous emission without separate field quantization	`exp_19c` v10 sweeps 2026-04-26 showed 0 recoils across rates 0.01–0.5; `exp_20` 2026-05-02 three-arm comparison (drain, unitary beam splitter, phase rotation) did not yet exhibit orbital lock-in – baseline two-body instability dominates (`exp_12b`); next attempt awaits resolution of the long-horizon stability question	`PART`
Emission-operator joint $\mathcal{A} = 1$ conservation	Pointwise unitary 2-mode rotation between electron and photon sessions with joint amplitude enforced	QED postulates a separately quantised electromagnetic field	`exp_20` 2026-05-02 sweep at GRID $= 65^3$, rate $= 0.05$, 6000 ticks per arm. Joint $\mathcal{A} = 1$ preserved to machine precision (8.9×10^{-16}) by arm B (beam splitter); arm A (the as-written drain in `exp_19c`) violates by 1.0; arm C (phase rotation) trivial. Phase-dependent non-monotonic transfer in arm B is an unanticipated refinement of the operation-algebra prediction (see `notes/lattice_operation_algebra.md` row 6 caveat)	`PART`
Induced gauge action (Sakharov form)	$-\mathrm{Tr}\ln D_{\text{lat}}[U]$; bipartite plaquette sum dominates at leading order	Maxwell density on O_h-averaged observables; structural form derived (numerical $1/g^2$ open)	Derived analytically (Appendix B)	`PART`
Gauge-sector birefringence	Plaquette-sum tensor $\mathbf{Q}$ eigenvalues $\{4, 4, 16\}$; oblate ellipsoid in $\mathbf{B}$-space	Direction-dependent photon kinetic term; same optical axis $(1,1,-1)$ as kinematic sector (P9)	Derived analytically (Appendix B)	`STUB`
Hydrogen $E_n \sim -1/n^2$	Coulomb well orbital resonance	Bohr spectrum	`exp_10`	`PASS`
Spontaneous quantization ($n = 1$)	Arnold tongue attractor in Coulomb well	$k_{\text{Bohr}} = 1/R_1$	`exp_11`	`PASS`
Two-body hydrogen (4 sig. figs.)	Live proton session, joint resonance	$k_{\min} = 0.0970$ vs. $k_{\text{Bohr}} = 0.0971$	`exp_12`	`PASS`
Two-body long-horizon stability	Bare `exp_12` chassis run for 6000 ticks (no emission)	Standard QM treats Bohr orbits as stationary	`exp_12b` 2026-05-02 single run at GRID $= 65^3$; r_{peak} escapes from $\sim R_1$ to grid edge (~ 83) by tick ~ 2000; $\mathcal{A} = 1$ preserved at machine precision (8.9×10^{-16}). Refines but does not invalidate `exp_12`'s 4-sig-fig PASS, which is scored on a shorter k-scan resonance peak	`PART`
Bound state requires active proton	Static well collapses; live proton explores well	Proton as symmetry-breaking session	`exp_11` ($n = 2$), `exp_12`	`PASS`
Three-body helium-like system	Two-electron + proton sessions	Helium ground state	`exp_13`	`PASS`
Two-electron helium	Same-sublattice exclusion + joint resonance	Helium spectrum	`exp_14`	`PASS`
Proton mass lower bound	Minimum Ω_P for quantization; binding $\neq$ quantization	All swept Ω_P produce Regime 2 (bound, unquantized); threshold lies above swept range	`exp_16`	`PART`
Multi-channel concordance (P9)	Single optical axis $(1,1,-1)$ inherited by kinematic, gauge, gravitational, and recoil-mass sectors	Five-channel direction agreement $(a/\lambda)^2$; time-dependence variants distinguish self-projecting from externally-driven embedding	Derived analytically (§12.10); direction-resolved GRB / CMB / LIGO / atomic-clock data	`STUB`

2 The Octahedral Substrate ($\mathcal{T}^3_\diamond$)

The $\mathcal{T}^3_\diamond$ lattice is a discrete topology: a graph in which each node knows only which other nodes are adjacent to it. No coordinates, no metric, no distances between non-adjacent nodes, and no differential structure are assumed or built in. The smooth spacetime of general relativity and quantum field theory is not the stage on which the physics takes place — it is a coarse-grained, statistical description that emerges when the central limit theorem is applied to many successive hops. A particle that takes $N \gg 1$ steps has a position distribution that converges to a Gaussian, and *that* Gaussian lives on something that looks like $\mathbb{R}^3$. The manifold is the $N \to \infty$ limit of the graph, not the other way around.

This ordering matters. Earlier discrete spacetime proposals [2, 3, 4] typically begin with a continuum manifold and discretise it, retaining coordinates and metric structure at the level of the fundamental object. Here there are no coordinates at the fundamental level. The lattice is a graph topology, and spacetime emerges from it statistically. The Dirac equation and the Einstein field equations are both large-N, many-hop results; they are theorems about the graph, not inputs to it.

2.1 Nodes, Edges, and Causal Adjacency

A *node* is a site: a location in the graph connected to neighbouring sites by directed edges. It carries no intrinsic geometry; all geometry is a consequence of the adjacency structure. A node has no knowledge of nodes two or more hops away — locality is exact, not approximate.

Two nodes are *causally adjacent* if a signal propagating at the lattice speed limit $c = 1$ node/tick can reach one from the other in a single tick. The six directions of causal adjacency define the basis vectors of the lattice:

$$\mathbf{V}_1 = (1, 1, 1), \quad \mathbf{V}_2 = (1, -1, -1), \quad \mathbf{V}_3 = (-1, 1, -1), \tag{1}$$

and their negatives $-\mathbf{V}_1,\ -\mathbf{V}_2,\ -\mathbf{V}_3$. These are the six face-to-vertex diagonals of a unit cube, and they form the edges of a regular octahedron centred at the origin. The lattice $\mathcal{T}^3_\diamond$ is therefore the Cayley graph of the free abelian group $\mathbb{Z}^3$ generated by these six vectors.

The hop distance — the length of a single causal step — is

$$|\mathbf{V}_1| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}\,\ell_P, \tag{2}$$

where ℓ_P is the Planck length, the natural calibration of the grid unit (see Section 2.5). This is a body-diagonal of the unit cube, not a face-diagonal or an edge. The geometry is octahedral, not cubic.

2.2 The Bipartite Structure and Chirality

The six basis vectors split naturally into two groups of three:

$$\text{RGB (even ticks):} \quad \mathbf{V}_1,\ \mathbf{V}_2,\ \mathbf{V}_3, \tag{3}$$

$$\text{CMY (odd ticks):} \quad -\mathbf{V}_1,\ -\mathbf{V}_2,\ -\mathbf{V}_3. \tag{4}$$

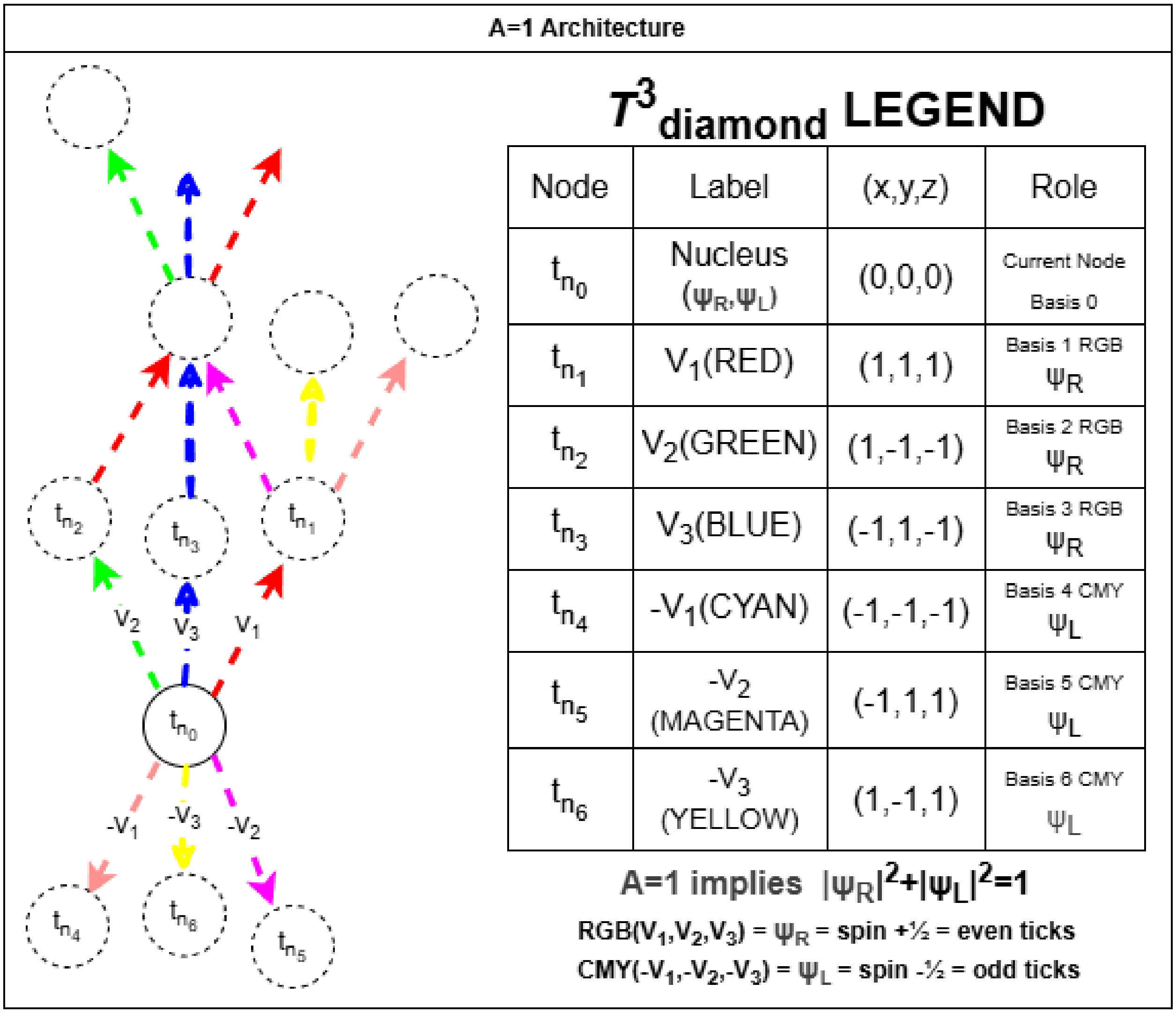
A particle on an *even* node hops along RGB directions on even ticks and is carried to an *odd* node; from there it hops along CMY directions on odd ticks and returns to an even node. The lattice is therefore *bipartite*: the even (RGB) and odd (CMY) sublattices are never directly connected to each other within the same sublattice.

This bipartite structure has immediate physical consequences:

- **Massless propagation.** A session with $\omega = 0$ (no phase mismatch) hops on every tick without ever residing, alternating strictly between the two sublattices. Its speed is exactly $c = 1$ node/tick, confirmed by `exp_00`.

- **Zitterbewegung.** A massive session ($\omega > 0$) has a finite probability $p_{\text{stay}} = \sin^2(\omega/2)$ of remaining at its current node rather than hopping. The resulting trembling motion between sublattices is the lattice realisation of Schrödinger's *Zitterbewegung* [8].

- **Spinor structure.** The two sublattices carry the two spinor components of the Dirac field: ψ_R on the RGB sublattice (right-handed, even ticks) and ψ_L on the CMY sublattice (left-handed, odd ticks). The $\mathcal{A} = 1$ constraint requires $|\psi_R|^2 + |\psi_L|^2 = 1$ at every tick. The Dirac equation emerges from this structure in the continuum limit (Section 6).



Node	Label	(x,y,z)	Role
t_{n_0}	Nucleus (Ψ_R,Ψ_L)	(0,0,0)	Current Node Basis 0
t_{n_1}	V_1(RED)	(1,1,1)	Basis 1 RGB Ψ_R
t_{n_2}	V_2(GREEN)	(1,-1,-1)	Basis 2 RGB Ψ_R
t_{n_3}	V_3(BLUE)	(-1,1,-1)	Basis 3 RGB Ψ_R
t_{n_4}	$-V_1$(CYAN)	(-1,-1,-1)	Basis 4 CMY Ψ_L
t_{n_5}	$-V_2$ (MAGENTA)	(-1,1,1)	Basis 5 CMY Ψ_L
t_{n_6}	$-V_3$ (YELLOW)	(1,-1,1)	Basis 6 CMY Ψ_L

Figure 1: The $\mathcal{T}^3_\diamond$ lattice architecture. The nucleus node t_{n_0} carries both Dirac spinor components (ψ_R, ψ_L). RGB basis vectors constitute the right-handed sublattice (ψ_R, spin $+1/2$, even ticks) and CMY basis vectors the left-handed sublattice (ψ_L, spin $-1/2$, odd ticks). The A=1 constraint requires $|\psi_R|^2 + |\psi_L|^2 = 1$.

2.3 Same-Sublattice Exclusion

Because all hops are body-diagonal, two nodes on the same sublattice (both even or both odd) are never directly connected. The minimum distance between two even nodes is

$$d_{\min}(\text{even–even}) = |\mathbf{V}_1 - \mathbf{V}_2| = |(0, 2, 2)| = 2\sqrt{2}\,\ell_P. \tag{5}$$

No causal path of length one exists between two same-sublattice nodes. Two sessions of the same chirality — two right-handed fermions, for instance — cannot occupy the same sublattice node or approach within $2\sqrt{2}\,\ell_P$ of each other without violating the bipartite tick rule.

This geometric floor is the lattice analogue of the Pauli exclusion principle. Same-chirality occupation is forbidden by structure, not by postulate.

2.4 The Causal Cone

After N ticks, a signal emitted from the origin can reach any node (a, b, c) for which a combination of the six basis vectors sums to (a, b, c) in exactly N steps. The set of reachable nodes after N ticks forms an expanding octahedron — the *causal cone* $\mathcal{C}(N)$ — with vertices at $(\pm N, 0, 0)$, $(0, \pm N, 0)$, and $(0, 0, \pm N)$.

The number of distinct N-step paths from the origin to a node $(a, b, c) \in \mathcal{C}(N)$ is a pure combinatorial integer $P(N,\ a,\ b,\ c)$ derived in Section 12. For large N and fixed $(a, b, c)/N$, this count converges to a Gaussian by the central limit theorem. The deviations from the Gaussian at small N are computable exactly and constitute a falsifiable correction to the $1/r^2$ force law (`exp_06`).

2.5 Calibration and Physical Units

The lattice has one free parameter: the identification of the grid unit with a physical length. We calibrate at the Planck scale: one grid unit = ℓ_P. The resulting unit table is shown in Figure 2 and summarised here:

Quantity	Formula	Tier
Hop distance	$\delta x = \sqrt{3}\,\ell_P$	Planck calibration*
Speed limit	$c = 1$ hop/tick	Geometric
Lattice HUP	$\delta x \cdot \delta p = \hbar\pi$	Brillouin-zone floor**
HUP saturation	$R_1 \cdot k_{\text{Bohr}} = 1 = \hbar$	`exp_12`‡
Arnold tongue	$\omega \cdot R_1 = \pi/3$	`exp_11/12`†

*Theoretical identification of grid unit with ℓ_P. **Derived from lattice Brillouin-zone cutoff at $|p| = \pi/\sqrt{3}$. ‡Two-body hydrogen experiment; 4 significant figures. †3:1 Zitterbewegung/orbital resonance; dynamical result, derivation in Section 6.

Two lengths, one convention. The cubic geometry carries two natural distances: the grid spacing ℓ_P (one cube edge) and the body-diagonal $h \equiv \sqrt{3}\,\ell_P$ (the actual hop distance per tick). Every causal hop covers h, not ℓ_P, so the speed of light is

$$c = \frac{h}{\tau} = \frac{\sqrt{3}\,\ell_P}{\tau}, \tag{6}$$

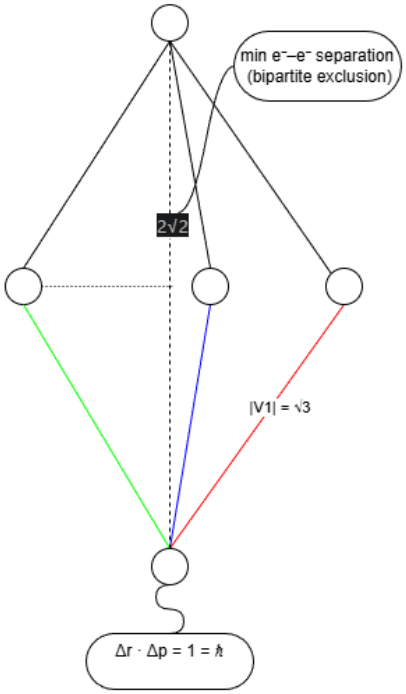
where τ is one tick. Throughout the rest of the paper, *"the lattice spacing" a denotes the hop distance h*, not the cube edge ℓ_P, unless explicitly stated otherwise. This is the length that enters the Compton- and GRB-calibration relations ($a = \lambda_C$ at Compton calibration; $a \lesssim 10^{-19}$ m from GRB time-of-flight), the predictions table in Section 12, and the discrete-correction expressions of Section 7. A reader doing a back-of-envelope calculation should always read a as h, not as ℓ_P.

The HUP saturation entry $R_1 \cdot k_{\mathrm{Bohr}} = 1$ deserves emphasis. The Bohr orbit is precisely the minimum-uncertainty state on the lattice: $\Delta r \cdot \Delta p = 1 \geq \frac{1}{2}$. This falls out of the two-body dynamics without fine-tuning — it is the condition under which the orbital phase and the Zitterbewegung phase first lock into a 3:1 Arnold tongue resonance. Quantization is not imposed; it is the attractor.

2.6 Relation to Prior Discrete Spacetime Frameworks

The $\mathcal{T}^3_\diamond$ lattice belongs to a family of discrete spacetime programmes that includes causal set theory [2], cellular automaton approaches [4, 3], and earlier lattice quantum gravity work [5, 9].

It differs from these in three respects. First, the bipartite RGB/CMY structure is not assumed for algebraic convenience — it emerges from the requirement that the six causal directions split into two chirally-opposite groups of three, which is the unique solution for $\mathbb{Z}^3$ with coordination number six. Second, the $\mathcal{A} = 1$ constraint is applied *per session* rather than globally, making each particle a local conservation law rather than a contribution to a global norm. Third, and most importantly, the framework makes falsifiable numerical predictions with no free parameters beyond the Planck calibration: the discrete path-count correction to $1/r^2$ (`exp_06`), the four-significant-figure hydrogen Bohr radius (`exp_12`), and the minimum stable proton mass from two-body orbital dynamics (`exp_16`).



Diamond Grid Units	
Hop Distance	$\lvert V_1\rvert = \sqrt{3}$
Even-Even Separation	$2\sqrt{2}$
Speed Limit	c = 1 node/tick = 1 hop/tick
Ground State Orbit	$\omega \cdot R_1 = \pi/3$ (†)
HUP Saturation	$R_1 \cdot k_{Bohr} = 1 = \hbar$ (‡)
Hard Lattice HUP	$\delta x \cdot \delta p = \hbar\pi$ (**)

Unit Calibration	
hop	$\sqrt{3}\,\ell_P$ (*)
c	1 hop/tick
$\delta x \cdot \delta p$	$\hbar\pi$ (**)
$R_1 \cdot k_{Bohr}$	1 (‡)
$\omega \cdot R_1$	$\pi/3$ (†)

Symbol	Tier	Meaning
(none)	Geometric	True by construction, no experiment needed
*	Theoretical	$\delta x = \sqrt{3}\,\ell_P$ Planck calibration
‡	Dynamical Experiment 12	$R_1 \cdot k_{Bohr} = 1$
†	Dynamical Experiments 11 & 12	$\omega \cdot R_1 = \pi/3$ Arnold Tongue
**	Theoretical	Brillouin-zone floor, derived from lattice theory

Lattice Property
$\omega \cdot R_1 = \pi/3$ (†)

Figure 2: Fundamental length and momentum scales of the $\mathcal{T}^3_\diamond$ lattice. **Hop distance.** Every propagation step moves a spinor amplitude along one of the six basis vectors $\mathbf{V}_i$; the shortest is $|\mathbf{V}_1| = |(1,1,1)| = \sqrt{3}$ grid units, which sets the natural length unit $h \equiv \sqrt{3}\,\ell_P$ (one body-diagonal of the unit cube). **Same-sublattice exclusion.** Even (RGB) nodes are separated from their nearest even neighbours by $|2\mathbf{V}_1 - 2\mathbf{V}_2| = 2\sqrt{2}\,\ell_P$ — a face-diagonal distance that no two same-chirality particles can reduce below without violating the bipartite tick rule. This geometric floor is the lattice analogue of the Pauli exclusion principle: same-sublattice occupation is forbidden by structure, not by postulate. **Speed limit.** A massless session ($\omega = 0$, $p_{\text{stay}} = 0$) hops every tick without ever residing, giving $c = 1$ node/tick $= 1$ hop/tick, confirmed by `exp_00`. **Uncertainty product.** The hard lattice spacing sets $\delta x \cdot \delta p = \hbar\pi$, above the Gaussian minimum $\hbar/2$ but consistent with a discrete Brillouin-zone cutoff at $|p| = \pi/\sqrt{3}$. **Ground-state saturation.** The Bohr orbit satisfies $R_1 \cdot k_{\text{Bohr}} = R_1 \cdot (1/R_1) = 1 = \hbar$ (lattice units), making the $n = 1$ orbital the minimum-uncertainty state: $\Delta r \cdot \Delta p = 1 \geq \frac{1}{2}$. **Arnold tongue identity.** The simulation recovers $\omega \cdot R_1 \approx \pi/3$ to 0.23% at $\omega = 0.1019$, STRENGTH = 30 — a dynamical result, not a geometric identity. This is the 3:1 Zitterbewegung/orbital resonance (lowest Arnold tongue of the ground state); it is to be derived from the continuum limit, not assumed. **Calibration summary (lattice units, $\hbar = c = 1$).** $\delta x = \sqrt{3}\,\ell_P$; $c = 1$ hop/tick; $\delta x \cdot \delta p = \hbar\pi^*$; $R_1 \cdot k_{\text{Bohr}} = 1^\ddagger$; $\omega \cdot R_1 = \pi/3^\dagger$.
*Brillouin-zone floor ‡`exp_12` two-body hydrogen †`exp_11`/`exp_12` 3:1 Arnold tongue.

3 Causal Sessions and the Phase Oscillator

A particle in the $\mathcal{A} = 1$ framework is not a point mass, a field mode, or a wavefunction on a background space. It is a *causal session*: a persistent probability flux that carries an internal clock, evolves by the bipartite tick rule, and obeys the $\mathcal{A} = 1$ constraint at every tick. This section defines the session structure and identifies its algebraic content: the internal clock is a U(1) oscillator whose group element encodes phase and whose algebra generator encodes mass.

3.1 The Phase Oscillator: U(1) Group and Algebra

Each causal session carries a *phase oscillator* — an internal oscillator that advances by a fixed angle ω per tick. The oscillator state at tick n is the unit complex number

$$r_n = e^{i\phi_n} \in \mathrm{U}(1), \qquad \phi_n = n\omega \mod 2\pi. \tag{7}$$

Two structures are at work here, and they are distinct.

The group element $r_n = e^{i\phi_n}$ lives on the unit circle $\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$. This is the Lie group: a one-dimensional compact manifold with group multiplication given by phase addition. The $\mathcal{A} = 1$ constraint — total probability equal to one at every tick — is the requirement that the oscillator never leaves this manifold. $\mathcal{A} = 1$ is not a normalisation convention; it is the statement that the session's state is always a group element, never a rescaled one. Probability is conserved because U(1) is a group, not because we choose to enforce it.

The algebra generator $\omega \in \mathfrak{u}(1) \cong \mathbb{R}$ is the element of the Lie algebra that exponentiates into the group via

$$r_{n+1} = e^{i\omega} \cdot r_n. \tag{8}$$

This is left-multiplication by $e^{i\omega}$: a rotation of the oscillator by angle ω per tick. The generator ω is the particle's *instruction frequency* — the rate at which its internal clock advances. In the continuum limit $a \to 0$ with $\omega/a = m$ held fixed, ω becomes the rest mass (Section 6.4). Mass is the generator of the internal U(1) rotation, not a separately postulated property.

The phase mismatch as local Hamiltonian. When the session is at a node with topological potential $V(\mathbf{x})$, the effective advance per tick is

$$\delta\phi(\mathbf{x}) = \omega + V(\mathbf{x}). \tag{9}$$

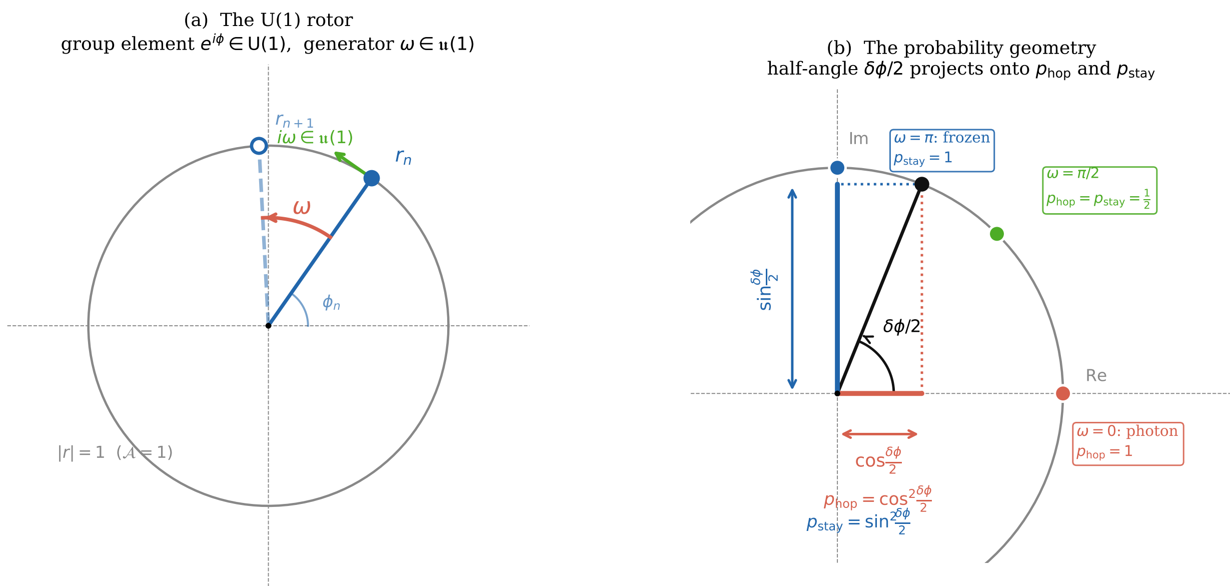


Figure 3: The U(1) phase oscillator. **(a)** The oscillator state $r_n = e^{i\phi_n}$ is a point on the unit circle (the Lie group U(1)). Each tick advances the state by the algebra generator $\omega \in \mathfrak{u}(1)$, shown as the arc between r_n (filled) and r_{n+1} (open). The tangent vector $i\omega$ at r_n shows that ω lives in the Lie algebra (tangent space). The constraint $|r| = 1$ is the $\mathcal{A} = 1$ condition. **(b)** The hop and stay probabilities arise from the half-angle $\delta\phi/2$ of the local phase mismatch $\delta\phi = \omega + V$. The cosine projection gives $p_{\mathrm{hop}} = \cos^2(\delta\phi/2)$ (red); the sine projection gives $p_{\mathrm{stay}} = \sin^2(\delta\phi/2)$ (blue). Three special cases are marked: $\omega = 0$ (photon, all hop), $\omega = \pi/2$ (equal split), and $\omega = \pi$ (frozen, all stay). The group geometry and the probability dynamics are the same circle.

This is the discrete Hamiltonian evaluated locally: an algebra element $\omega + V \in \mathfrak{u}(1)$ that varies from node to node. The potential V deforms the algebra element without leaving the Lie algebra — the sum of two real numbers is still a real number, so the phase mismatch is still a valid generator. The hop probability

$$p_{\text{stay}} = \sin^2 \tfrac{\delta\phi}{2}, \qquad p_{\text{hop}} = \cos^2 \tfrac{\delta\phi}{2}, \tag{10}$$

is the norm-squared of the U(1) rotation by $\delta\phi/2$. It depends only on the algebra element $\delta\phi$, not on the current group element r_n. The dynamics are controlled by the algebra; the group element tracks the accumulated phase.

Why this matters. In standard quantum mechanics the wavefunction ψ is a complex number and the Hamiltonian H is a Hermitian operator. Their relationship is through the Schrödinger equation, which must be postulated. Here the relationship is structural: the Hamiltonian is a u(1) algebra element, the state is a U(1) group element, and the equation of motion $r_{n+1} = e^{i(\omega+V)} \cdot r_n$ is the definition of exponentiation from the algebra to the group. The Schrödinger equation is not postulated; it is the definition of what it means for the oscillator to advance.

3.2 The Causal Session: Spinor Amplitude on the Lattice

The phase oscillator governs the *internal* clock. The *external* amplitude — the probability density distributed across the lattice — is the spinor field $\Psi = (\psi_R, \psi_L)^T$, where $\psi_R(\mathbf{x})$ is the amplitude on the RGB sublattice and $\psi_L(\mathbf{x})$ on the CMY sublattice.

These are not two independent fields. They are the same session viewed on even and odd ticks respectively: ψ_R is the right-handed component (advances on RGB directions, even ticks) and ψ_L is the left-handed component (advances on CMY directions, odd ticks). The bipartite alternation forces the two-component structure. Spin-$\frac{1}{2}$ is not a separate postulate; it is the consequence of there being exactly two sublattices.

The $\mathcal{A} = 1$ constraint on the spinor is

$$\sum_{\mathbf{x}} \left(|\psi_R(\mathbf{x})|^2 + |\psi_L(\mathbf{x})|^2 \right) = 1. \tag{11}$$

This is the field-theoretic extension of the oscillator constraint $|r_n| = 1$: the total probability across all nodes and both spinor components is exactly one at every tick. The session is a U(1) oscillator distributed over the lattice.

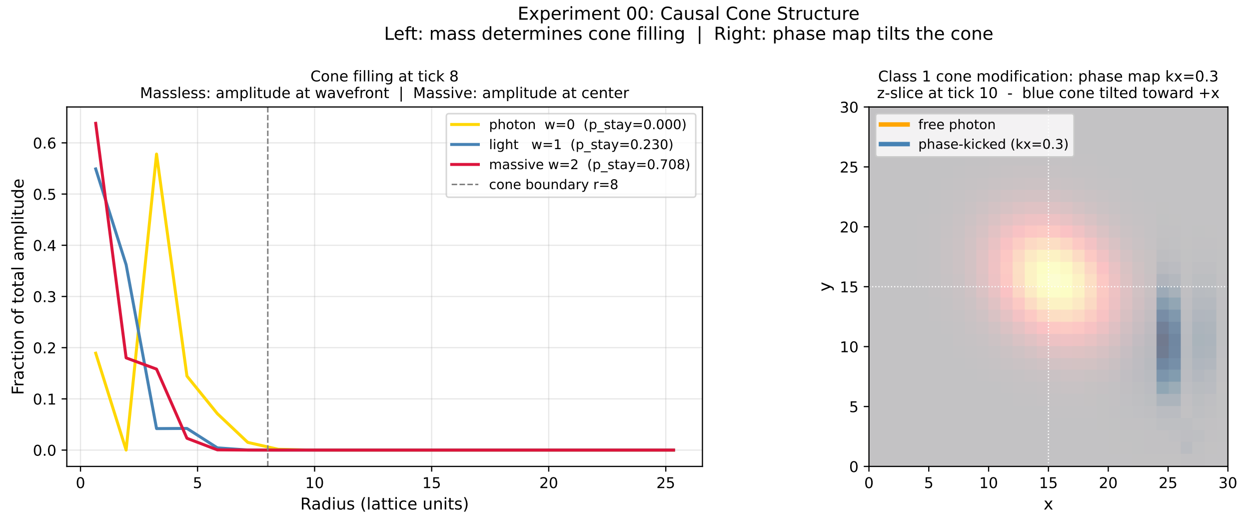


Figure 4: **Causal cone structure as a function of mass and phase map (exp_00). Left:** fraction of total amplitude vs. radius at tick 8 for three values of the mass parameter ω. A photon ($\omega = 0$, $p_{\text{stay}} = 0$) leaves all amplitude at the cone boundary (yellow): every tick is a strict hop. A light mass ($\omega = 1$, $p_{\text{stay}} \approx 0.23$) and a massive session ($\omega = 2$, $p_{\text{stay}} \approx 0.71$) increasingly retain amplitude near the centre, giving the bipartite-lattice realisation of the rest-frame Zitterbewegung. The cone boundary r=8 at tick 8 is the topological speed limit $c = 1$ hop/tick. **Right:** the same cone modified by a uniform phase map $k_x = 0.3$. The phase gradient breaks the RGB/CMY symmetry of the hop directions, tilting the cone toward $+x$ — exactly the lattice realisation of momentum as gauge-transformed advancement (§3.3).

Massless sessions. When $\omega = 0$ the phase mismatch $\delta\phi = V(\mathbf{x})$ vanishes on a flat lattice, so $p_{\text{stay}} = 0$ and $p_{\text{hop}} = 1$. The amplitude hops on every tick, alternating strictly between the two sublattices, and propagates at $c = 1$ node/tick. The two-component structure collapses: at any instant the amplitude is entirely on one sublattice. This is the photon. Its generator $\omega = 0$ is the identity element of the algebra; it has no mass, no ZB oscillation, and no internal clock frequency.

3.3 The Tick Rule as Gauge Transformation

The tick rule advances the oscillator by $e^{i\delta\phi}$ and hops the spinor amplitude to neighbouring nodes. Viewed globally, each tick is a *local* U(1) gauge transformation: different nodes receive different phase advances $\delta\phi(\mathbf{x})$ depending on the local potential $V(\mathbf{x})$.

The $\mathcal{A} = 1$ constraint is preserved by this transformation because U(1) is compact: $|e^{i\delta\phi}| = 1$ regardless of $\delta\phi$. No phase advance, however large, can change the norm of the oscillator. The constraint is therefore automatically satisfied at every node and every tick without additional enforcement beyond the tick rule itself.

This structure is the discrete origin of U(1) gauge invariance in quantum electrodynamics. The gauge field A_μ of QED is, in this framework, the potential $V(\mathbf{x}, t)$ that deforms the local algebra element from ω to $\omega + V$. The gauge transformation $\psi \to e^{i\alpha(\mathbf{x})}\psi$ is the freedom to redefine the zero of the phase at each node — a freedom that exists precisely because the physics depends only on the algebra element $\delta\phi$, not on the accumulated group element r_n.

The implication for the gauge group of the Standard Model is direct: U(1) electromagnetism is the phase symmetry of a single session. The larger gauge group SU(3)×SU(2)×U(1) arises when multiple sessions with different RGB/CMY cone orientations interact — the subject of Section 9.

3.4 Inertia as Phase-Gradient Stability

A session in uniform motion carries a spatial phase gradient θ per lattice edge. The gradient is an element of the algebra: it is the spatial part of the local generator $\omega + V + \theta$. Inertia is the *stability* of this gradient against perturbation.

Perturbing the gradient by $\Delta\theta$ changes the hop probability by $\Delta p_{\text{hop}} \propto -\sin(\omega)\,\Delta\theta$. The larger ω (the heavier the particle), the smaller the relative change in hop probability for a given perturbation. Heavier particles are more resistant to gradient changes — they have more inertia.

Experiment `exp_01` confirms this. A Gaussian wavepacket ($\omega = 0.2$, $\mathbf{k} = (k, k, k)$) drifts at $v_x = 0.046$ nodes/tick; a heavier packet ($\omega = 0.5$, same $\mathbf{k}$) drifts measurably slower. The velocity formula $v = \frac{c}{2}\sin(\omega)\sin(\theta)$ (derived in Section 6.2) gives the correct ratio. Inertia is not postulated; it follows from the algebra element ω governing the oscillator's response to gradient perturbations.

Summary. A causal session is a U(1) oscillator ($e^{i\phi} \in \mathrm{U}(1)$) distributed over the bipartite lattice as a two-component spinor (ψ_R, ψ_L). Its mass is the algebra generator $\omega \in \mathfrak{u}(1)$. Its dynamics are the exponentiation of the local algebra element $\delta\phi = \omega + V$ into the group at every tick. Probability conservation is the compactness of U(1). Spin-$\frac{1}{2}$ is the bipartite structure. Inertia is the gradient stability of the generator. None of these are postulated; they are the content of the single statement: *a particle is a persistent U(1) flux on a bipartite lattice.*

4 The Tick Scheduler

The previous section defined a single causal session: a U(1) phase oscillator on the $\mathcal{T}^3_\diamond$ lattice with its own tick counter, evolving under the bipartite tick rule and obeying the $\mathcal{A} = 1$ constraint at every tick. A real physical situation is rarely one session. Atoms are at least two; molecules many; macroscopic bodies astronomically many. The *tick scheduler* is the architectural component that turns a collection of independent sessions into the dynamics of an interacting universe.

The scheduler is conceptually small. It maintains a list of registered sessions, advances each session by one tick in some order, and applies an interaction step that exchanges phase between sessions whose probability supports overlap. That is the whole machine. The richness of the framework comes from what follows from those two operations. This section sets out the formal apparatus. The phenomenological consequences — measurement (Section 11), gravity (Section 7), spontaneous emission (Section 15.7) — are derived from the scheduler in later sections.

4.1 The Macro-Tick

A *macro-tick* is one cycle of the scheduler. For a scheduler with n registered sessions $\{\mathcal{S}_1, \ldots, \mathcal{S}_n\}$, one macro-tick consists of three steps:

1. **Order.** Determine a processing order $\sigma \in S_n$ over the sessions according to the active shuffle scheme.

2. **Tick.** For each i in the order σ, advance $\mathcal{S}_i$ by one bipartite tick (Section 5) and increment its tick counter. Each session enforces $\mathcal{A} = 1$ individually after its own tick.

3. **Couple.** Apply the pairwise phase-exchange step (Section 4.3) at every node where two sessions have significant probability support. Apply any registered three-session couplings (Section 4.4).

The decomposition into *tick* and *couple* is essential. Each session is independently norm-preserving during the tick step — this is what makes the U(1) group structure of Section 3.1 survive at the multi-session level. The couple step then redistributes *phase* between sessions at shared nodes without moving probability across them. $\mathcal{A} = 1$ is preserved per session through both steps; what is shared is the coherent structure of the wavefunction, not the probability mass.

In code, the macro-tick is the `TickScheduler.advance` method. The full state of the universe at macro-tick n is the tuple $(\{\mathcal{S}_i\}, \sigma_n, n)$: the sessions, the ordering used at this step, and the global cycle count.

4.2 The Combinatorial Clock Space

The scheduler's state space is the set of orderings of its registered clocks. For n sessions this is the symmetric group S_n, of size

$$|\Omega_n| = n!. \tag{12}$$

This factorial growth is the algebraic root of the framework's treatment of time. It will appear again as the entropy of an observer interaction (Section 11.3), as the load on a region near mass concentrations (Section 4.5), and as the combinatorial source of the second law (Section 4.7).

Shuffle schemes as experimental variables. The framework does not fix a single ordering rule. Four schemes are implemented:

- Sequential: process sessions in registration order.
- Random: uniform random shuffle each macro-tick.
- Priority: lowest tick counter first.
- Clock-density: highest local ρ_{clock} first.

If the framework is correct as a model of physics, the bulk phenomenology — conserved probability, the Dirac equation in the continuum limit, gravitational attraction — should be invariant under the choice of scheme. Different schemes correspond to different choices of simultaneity for the discrete substrate, in close analogy to choices of foliation in general relativity. A discrete *Lorentz invariance* is the prediction that the choice does not matter for any observable bulk quantity.

Numerical verification (`exp_05`). The conserved-probability check is straightforward and was performed in `exp_05`. Across all three implemented schemes (sequential, random, priority) and across session counts $n = 1, 2, 3, \ldots, 6$, the total probability satisfies $|\sum_i \rho_i - n| < 10^{-15}$ at every macro-tick — machine precision conservation independent of ordering. The deeper question — whether two observers using different shuffle schemes can disagree on a measurement outcome — is treated in Section 11. The answer is no: the only quantities that depend on the ordering are those that depend on *which* cone perturbed which, not on the final probability distribution.

4.3 Pairwise Coupling and the Binding Dial

Two sessions registered in the same scheduler are not, by themselves, coupled. They share a lattice but evolve independently. Coupling enters through the third step of the macro-tick: at any node $\mathbf{x}$ where two sessions $\mathcal{S}_i$ and $\mathcal{S}_j$ both have significant amplitude (above a threshold $\sim 10^{-4}$), their phases are partially exchanged.

The mechanism is a U(1) rotation in the orbital plane of phase $\phi_i = \arg(\psi_R^{(i)})$ at each shared node:

$$\phi_i \to \phi_i + c_{ij}\,(\phi_j - \phi_i), \tag{13}$$

$$\phi_j \to \phi_j + c_{ij}\,(\phi_i - \phi_j), \tag{14}$$

where $c_{ij} \in [0, 1]$ is the binding strength registered for the pair. Both spinor components are rotated by the same phase ($\mathcal{A} = 1$ does not mix ψ_R with ψ_L here), and each session is re-normalised after the rotation.

The single parameter c_{ij} encodes the entire spectrum of two-session relationships:

c_{ij}	Physical regime
0	Free particles — no interaction
~ 0.1	Weak — observer / decoherence regime
~ 0.5	Atomic binding — joint Arnold tongue
~ 0.9	Strong binding — composite particle
1	Phase lock — sessions move as a single unit

The same algebraic step thus produces, in different parameter regimes, the observer effect of Section 11, the joint hydrogen resonance of Section 15.5, and the bound-state binding that defines composite particles. A neutral composite (a neutron, a hydrogen atom seen from far away) is two or more sessions with $c \to 1$ *and* opposite RGB/CMY imbalance: their phase gradients cancel, narrowing the effective causal cone of the composite below those of its constituents. This cone narrowing is the lattice mechanism for why neutral matter behaves more classically than its charged constituents.

4.4 Three-Session Coupling and Emission

Some interactions cannot be reduced to pairwise phase exchange. The clearest example is spontaneous emission, which we showed in Section 15.7 to be a three-session event: the joint electron–proton resonance jumps to a lower Arnold tongue, and a third session (the photon) is created to absorb the displaced amplitude under $\mathcal{A} = 1$. The scheduler accommodates this by maintaining an explicit list of *emission triplets* $(\mathcal{S}_e, \mathcal{S}_\gamma, \mathcal{S}_p, r)$, where r is the per-tick drain rate.

The drain step is engineered to respect $\mathcal{A} = 1$ exactly. At each macro-tick the electron's tangential phase gradient $\nabla_\perp \phi_e$ (the radial component is preserved so the Coulomb attraction continues to act) is rotated by $-r\, k_{\text{tang}} \cdot s$, where s is the arc-length coordinate along the orbital plane and k_{tang} is the current orbital wavevector. The conjugate rotation is applied to the photon, transferring orbital angular momentum from electron to photon while leaving the per-session probability density — and therefore $\mathcal{A} = 1$ — untouched. The *appearance* of dissipation comes from the subsequent re-normalisation step, which sees a redistributed phase structure and adjusts the spinor components accordingly.

The three-session coupling is the clearest demonstration that the scheduler is more than a clock-shuffler. It is a mechanism that decides, at each tick, which combinations of sessions are interacting and how. Pairwise binding handles two-session phenomena; emission triplets handle three-session events; further n-session generalisations (electroweak vertices, hadronic decays) follow the same pattern of explicit registration with an $\mathcal{A} = 1$-preserving update rule.

4.5 Clock Density and Time Dilation

The pairwise step of the macro-tick scales with the number of session overlaps. For n sessions with significant pairwise support there are at most $\binom{n}{2}$ pairs to evaluate; for sessions whose probability supports all share a common node, every pair contributes. The scheduler is therefore not a constant-cost machine. Regions of the lattice where many sessions overlap impose more bookkeeping per macro-tick than regions where sessions are isolated.

This is the lattice substrate for gravitational time dilation. A region of high clock density ρ_{clock} runs more couplings per macro-tick. A session embedded in that region executes its own tick rule unchanged — but each macro-tick it is processed alongside more neighbours, with more phase exchanges to settle. From the perspective of a distant observer, who sees only the global macro-tick count n, the local session has accumulated fewer "personal" tick advances per global macro-tick: its clock runs slow.

The continuum-limit derivation of this picture — with the explicit identification $\nabla\phi_{\text{Newton}} \leftrightarrow$

macro-tick imbalance and the recovery of the Schwarzschild time dilation factor — is given in Section 7. The point here is architectural: time dilation is not added by hand through a metric. It is the unavoidable consequence of running a scheduler whose per-macro-tick cost is proportional to the local session density.

4.6 Queue Saturation and the Event Horizon

The scheduler can register at most one session per Planck volume. Each session occupies a U(1) oscillator at a node, and the lattice provides a finite number of nodes per unit volume. The maximum clock density is therefore

$$\rho_{\text{clock}}^{\max} = \ell_P^{-3} \approx 2.37 \times 10^{104}\ \text{clocks/m}^3, \tag{15}$$

the same bound that appears in Section 7.6. At this density the scheduler queue is saturated: every available node is bound to a session, and the next registration cannot proceed.

Saturation is the lattice definition of an event horizon. From the outside, sessions in the saturated region cannot accept new clock registrations and so cannot accept new information; the local tick rate falls to zero relative to the global macro-tick. This is not a coordinate singularity but a computational deadlock, strictly bounded above. There is no infinite curvature inside the horizon because there is no infinite density: the substrate runs out of nodes before the divergence can occur.

Hawking radiation as queue-drop. A scheduler that has reached saturation but receives further registration attempts must do something. The architecturally simplest response is to occasionally drop the oldest registered session at the boundary, freeing a node for the incoming registration. The drop is stochastic over the boundary and proportional to the attempted registration rate. Each drop releases a session into the surrounding lower-density region, where it propagates outward at c along a geodesic of the clock fluid. The thermal character of the resulting radiation — the Hawking spectrum — is the thermodynamic fingerprint of a uniform-prior boundary queue under stochastic drop, and is recovered without trans-Planckian field modes.

The Hawking temperature from boundary surface gravity. The temperature of the queue-drop emission can be fixed to leading order under one modelling assumption: that the saturation boundary inherits, as an external metric quantity, the Schwarzschild surface gravity

$$\kappa = \frac{c^4}{4GM}, \tag{16}$$

where M is the total mass enclosed by the saturated region. This assumption is the architectural counterpart of identifying the saturation boundary with a black-hole horizon (Section 7.6); the strong-field clock-fluid derivation that fixes the metric to Schwarzschild form is reserved for the gravitational-continuum-limit work of Section 8.

A boundary with effective surface acceleration κ supports an Unruh-style local thermal spectrum at temperature $T_{\text{local}} = \hbar\kappa/(2\pi c\, k_B)$. Folding the gravitational redshift between the local frame and the asymptotic observer (which, near the horizon, exactly cancels the $1/\sqrt{1 - r_s/r}$ blueshift of T_{local}) leaves the temperature seen at infinity equal to the surface-gravity expression itself [10]:

$$T_H \;=\; \frac{\hbar\,\kappa}{2\pi\,c\,k_B} \;=\; \frac{\hbar\,c^3}{8\pi\,G\,M\,k_B}. \tag{17}$$

The queue-drop rate at the saturated boundary is therefore set, up to architectural constants of order unity, by $\Gamma_{\text{drop}} \sim \kappa/(2\pi)$: the boundary releases on average one session per $2\pi/\kappa$ of asymptotic time.

Honest scope of Eq. (17)**.** The lattice contribution is the architectural claim that *some* thermal emission must occur at the saturated boundary because the scheduler cannot accept further registrations and the queue-drop is the architecturally simplest legal response. The specific form $T_H = \hbar c^3/(8\pi G M k_B)$ then follows from Eq. (16) via the standard Unruh–horizon relation. The modelling assumption that fixes the numerical prefactor is the inheritance of the Schwarzschild κ at the saturation boundary: a strong-field derivation of this correspondence from the clock-fluid equations is the natural follow-on calculation.

4.7 Architectural Irreversibility and the Arrow of Time

The scheduler has a `register_session` method. It does not have a `remove_session` method, by design.

This single architectural fact is enough to derive the second law of thermodynamics for the framework. A measurement, in the sense of Section 11, is the registration of a new session that overlaps an existing one. After the registration the state space has grown from $|\Omega_{n-1}| = (n-1)!$ to $|\Omega_n| = n!$, an increase by a factor of n. The corresponding entropy of the scheduler state space,

$$S_{\text{scheduler}} = k_B \log|\Omega_n| = k_B \log(n!), \tag{18}$$

strictly increases with each registration, and by Stirling's approximation grows as $S \sim$

$k_B\, n \log n$ for large n. There is no operation that can decrease n. The arrow of time is not a thermodynamic approximation that becomes exact in some statistical limit — it is a structural feature of the scheduler interface.

The deeper consequence, derived in Section 11.3, is that what we ordinarily call the second law is the same statement as "measurement is irreversible." Both follow from the absence of a clock-removal operation. You cannot un-observe a particle for the same reason you cannot reduce the entropy of an isolated system: the scheduler does not expose the operation that would let you.

4.8 Falsifiable Consequences

The scheduler architecture makes one prediction directly that distinguishes it from any continuum-time formalism.

Minimum time-dilation quantum. Each registered session contributes one tick of book-keeping per macro-tick. The minimum increment of gravitational time dilation between two neighbouring regions is therefore one tick duration:

$$\delta t_{\min} = t_{\text{tick}}, \tag{19}$$

where t_{tick} is the lattice tick duration in seconds. At Planck calibration, $t_{\text{tick}} = t_P \approx 5.4 \times 10^{-44}$ s and the prediction is unmeasurable. At Compton calibration, $t_{\text{tick}} \approx 8 \times 10^{-21}$ s, which is within roughly two orders of magnitude of current optical-clock fractional sensitivity ($\sim 10^{-18}$ over millimetre baselines). Section 12 develops the experimental geometry that would convert this predicted floor into a falsifiable null-result test: if optical clocks separated by a calibrated $\Delta\phi$ in a controlled potential gradient ever resolve a continuous distribution of fractional offsets without a discrete cutoff, the framework is wrong.

Discrete corrections in queue dynamics. Two further structural predictions follow from queue saturation, but their numerical form requires the gravitational continuum-limit derivation of Section 8. These are: (i) a Planck-density cutoff to gravitational wave dispersion, producing frequency-dependent gravitational-wave speeds in regions approaching $\rho_{\text{clock}}^{\max}$; and (ii) a queue-drop signature in Hawking emission spectra at the high-frequency tail, where the finite-lattice cutoff replaces the trans-Planckian modes of standard treatments. Both feed into Section 12.

5 Phase Propagation and the Unity Constraint

Section 3 defined the causal session as a $U(1)$ oscillator distributed over the bipartite lattice as a two-component spinor. Section 4 placed that session in a multi-session scheduler. This section sets out how the distributed oscillator *propagates*: what the tick operator does to the amplitude at each macro-tick, how the unity constraint is preserved through propagation, and how the discrete dynamics reduces to the Schrödinger equation in the long-time limit.

The headline result is that the Schrödinger equation is not a postulate. It is the central-limit-theorem limit of the bipartite tick rule, derived in two stages: the Dirac equation as the relativistic continuum limit (Section 6.4), and the Schrödinger equation as the non-relativistic limit of Dirac. Standard quantum mechanics is a theorem in the framework, not an axiom.

5.1 The Tick Operator

One macro-tick on a single session combines three operations: a local phase advance by $e^{-i\delta\phi(\mathbf{x})}$ at each node, a directional hop along the basis vectors of the active sublattice, and a $\mathcal{A} = 1$-preserving renormalisation. We write the composite operation as the tick operator $\hat{T}$:

$$\hat{T}\,\psi(\mathbf{x}) = \mathcal{N}\big[\cos\tfrac{\delta\phi}{2}\,\hat{H}\,\psi(\mathbf{x}) + i\sin\tfrac{\delta\phi}{2}\,\psi(\mathbf{x})\big], \tag{20}$$

where $\hat{H}$ is the bipartite hop operator (sum over the three basis vectors of the active sublattice, see Section 4.1), $\delta\phi(\mathbf{x}) = \omega + V(\mathbf{x})$ is the local phase mismatch, and $\mathcal{N}$ is the unity-enforcing renormalisation. The cos and sin amplitudes encode the hop/stay split derived in Section 3.1: the half-angle structure makes $\hat{T}$ a $U(1)$ rotation by $\delta\phi/2$ in the plane spanned by the local node and its neighbours.

$\hat{T}$ is unitary in the continuum limit $a \to 0$. At finite a it has $O(a^2)$ drift away from exact unitarity, and $\mathcal{N}$ corrects the drift. The correction is numerical, not physical: the discrete operators do not form an exact representation of the $U(1)$ Lie group, but the constraint $\sum_{\mathbf{x}} \rho(\mathbf{x}) = 1$ *is* the Lie-group requirement that the state remain on $U(1)$, so $\mathcal{N}$ is the discrete implementation of that constraint rather than a normalisation hack.

5.2 Linearity and Superposition

The tick operator is linear in the spinor amplitude:

$$\hat{T}\,(\alpha\psi_1 + \beta\psi_2) = \alpha\,\hat{T}\psi_1 + \beta\,\hat{T}\psi_2 \tag{21}$$

for any complex constants α, β. The renormalisation $\mathcal{N}$ is non-linear in the global norm but acts equally on all path contributions; within a tick the propagation is linear, and the relative phase between paths is preserved exactly.

Linearity is what makes interference possible. When two amplitudes from different sources arrive at the same node after N ticks they add as complex numbers (Section 10). The relative phase determines whether the addition is constructive (bright fringe) or destructive (dark fringe). A non-linear propagation rule would mix amplitudes and intensities, and would not exhibit signed cancellation.

The bipartite tick rule preserves linearity by construction. Stochastic propagation rules (Markov chains on probability distributions, for example) do not. The framework is amplitude-based not probability-based, and the linearity of the tick operator is what distinguishes it.

5.3 Phase Routing and the Discrete Hamiltonian

The local Hamiltonian

$$H(\mathbf{x}) = \omega + V(\mathbf{x}) \tag{22}$$

is a real number at each node — an element of the $\mathfrak{u}(1)$ Lie algebra (Section 3.1). The tick operator advances the local phase by $-iH(\mathbf{x})\,a$ per tick. In the small-a limit

$$\hat{T}\,\psi \approx \big(1 - iH(\mathbf{x})\,a\big)\,\psi + a\,\hat{D}\,\psi + O(a^2), \tag{23}$$

where $\hat{D}$ is the discrete spatial derivative implemented by the hop step. Comparing with the standard time-evolution operator $1 - iH\,dt + (i/2m)\nabla^2\,dt$ identifies the lattice spacing a with the time step dt and the hop operator with the kinetic term — this is the staggered-fermion identification of Kogut and Susskind [11].

The discrete Hamiltonian does not act on a vector space of pre-defined basis states. It is a spatial field of phase rates, evaluated locally at each node. Different nodes carry different phase rates, and the tick operator routes amplitude between them according to $e^{-iH(\mathbf{x})\,a}$. In standard quantum mechanics this is the Magnus expansion of the time-evolution operator; on the lattice it is the literal tick rule.

5.4 The Unity Constraint in Operation

The $\mathcal{A} = 1$ constraint

$$\sum_{\mathbf{x}} \big(|\psi_R(\mathbf{x})|^2 + |\psi_L(\mathbf{x})|^2\big) = 1 \tag{24}$$

is preserved by $\hat{T}$ up to $O(a^2)$. The renormalisation step $\mathcal{N}$ corrects the drift by dividing both spinor components by their joint norm. The typical residual at the per-tick level is $\sim 10^{-15}$ in double precision and scales as a^2, so the correction vanishes in the continuum limit.

The renormalisation has two distinct legitimate uses:

Conservation. Per-tick correction of the $O(a^2)$ drift in a single session. This is the standard usage and is invoked at the end of every macro-tick in every experiment. The drift is a property of the discrete operators, not of the physics: the lattice does not allow an exact representation of the continuous unitary group with finite-step operators.

Projection. After amplitude transfer between sessions (the photon-emission mechanism of Section 15.7), the source session is projected onto its remaining sub-space and the destination session receives the transferred amplitude. Each session is renormalised individually; joint $\mathcal{A} = 1$ is preserved across the pair. This is the discrete equivalent of wavefunction collapse onto a sub-space, implemented as $\mathcal{A} = 1$ applied per session after the amplitude budget has changed.

In neither case is $\mathcal{N}$ a free parameter. Both are applications of the Lie-group constraint $\sum \rho = 1$ from Section 3, applied to a session whose amplitude budget has changed by a defined physical mechanism (numerical drift, or amplitude transfer to another session).

5.5 The Continuum Limit: From Tick Rule to Schrödinger

Take $a \to 0$ and $\omega \to 0$ with the lattice velocity $v = a/dt = 1$ and the rest mass $m = \omega/a$ held fixed. The tick operator becomes

$$\hat{T} \xrightarrow{a\to 0} e^{-iH\,dt} = e^{-i(m+V-\frac{1}{2m}\nabla^2+\cdots)\,dt}, \tag{25}$$

the standard non-relativistic Schrödinger evolution operator. The $-\frac{1}{2m}\nabla^2$ kinetic term is the small-a expansion of the bipartite hop operator $\hat{D}$, derived in Section 6.4 via the O_h-averaged frame condition $\langle\sum_{\mathbf{v}} v_i v_j\rangle_{O_h} = 6\delta_{ij}$ (the operator-level form is anisotropic; the operator-level residual is the lattice birefringence along the optical axis $\hat{\mathbf{n}}_* = (1, 1, -1)/\sqrt{3}$, Eq. 41).

The full bipartite tick rule reduces to the Dirac equation in the relativistic limit (Section 6.4); the non-relativistic limit of the Dirac equation is the Schrödinger equation. So Schrödinger emerges as the long-time, large-N, small-a limit of the discrete rule via

$$\text{tick rule} \xrightarrow{a\to 0} \text{Dirac equation} \xrightarrow{v\ll c} \text{Schrödinger equation.}$$

This is the right way to think about the framework's relationship to standard quantum mechanics. The Schrödinger equation is not assumed; it is derived. A measurable departure from Schrödinger predictions in regimes where N is small (high-energy interferometry, atom interferometers at the Compton wavelength, etc.) is a falsifiable consequence of the framework (Section 12).

5.6 Boundary Conditions

Lattice experiments require boundary conditions on the spinor field. The framework supports three kinds, each appropriate to a different class of experiment.

Periodic. The lattice wraps: $\psi(\mathbf{x}+L\hat{\mu}) = \psi(\mathbf{x})$ for all basis directions $\hat{\mu}$. This is appropriate for momentum eigenstates and is used by default in the harmonic experiments (`exp_09`). Periodic boundaries preserve $\mathcal{A} = 1$ exactly: the lattice has no edge from which amplitude can escape.

Absorbing. A subset of nodes is marked as absorbing: their hop probability is set to zero, and amplitude arriving at them is removed by the $\mathcal{N}$ step. This implements barriers (the double-slit barrier of Section 10.3) and detector geometries. Absorbing boundaries do *not* preserve $\mathcal{A} = 1$ within a single session: per-session probability is reduced as amplitude enters the absorbed region. Joint $\mathcal{A} = 1$ across all sessions is preserved if the absorbed amplitude is transferred to an external sink session (e.g., a detector).

Reflecting. Hops into the boundary node return as hops out. This implements infinite-potential confinement (a quantum well). $\mathcal{A} = 1$ is preserved exactly because no amplitude leaves the session.

Choice of boundary condition. The same tick rule operates inside the lattice; the boundary modifies which hops are allowed at the edge. The choice is part of the experimental setup, not an additional postulate. Different experiments use different boundaries: `exp_09` (harmonics) uses periodic; `exp_03` (interference) uses absorbing for the barrier and reflecting for the lattice edge; `exp_10` (hydrogen) uses reflecting for the lattice edge with an interior Coulomb well. The freedom to choose is the lattice version of the freedom to choose boundary conditions on a continuum partial differential equation.

Summary. The propagation step is one tick of the bipartite tick rule applied to the spinor amplitude (ψ_R, ψ_L). It is linear in the amplitude, preserves $\mathcal{A} = 1$ up to $O(a^2)$ drift corrected by renormalisation, and reduces to the Dirac equation in the relativistic continuum limit and to the Schrödinger equation in the non-relativistic continuum limit. The lattice does not assume standard quantum mechanics; it derives it.

6 Emergent Kinematics on the $\mathcal{T}^3_\diamond$ Lattice

Momentum, inertia, and force are not postulated in the $\mathcal{A} = 1$ framework. They emerge from the geometry of phase interference across the bipartite sublattice structure. This section derives the kinematic relations from the Dirac tick rule and connects each result to the corresponding numerical verification in the experiment suite.

6.1 The Bipartite Velocity

A massive Dirac fermion [12] on $\mathcal{T}^3_\diamond$ carries two spinor components: ψ_R on the RGB sublattice (even ticks) and ψ_L on the CMY sublattice (odd ticks). The $\mathcal{A} = 1$ constraint requires

$$\sum_{\mathbf{x}} \left(|\psi_R(\mathbf{x})|^2 + |\psi_L(\mathbf{x})|^2 \right) = 1. \tag{26}$$

Over one complete two-tick cycle the kinetic fraction $p_{\text{hop}} = \cos^2(\omega/2)$ is distributed across the six basis vectors, split equally between the three RGB directions ($\mathbf{V}_{1\text{–}3}$, even tick) and the three CMY directions ($-\mathbf{V}_{1\text{–}3}$, odd tick):

$$1 = p_{\text{stay}} + P_{\text{fwd}} + P_{\text{bwd}}, \qquad p_{\text{stay}} = \sin^2 \tfrac{\omega}{2}, \quad P_{\text{fwd}} + P_{\text{bwd}} = \cos^2 \tfrac{\omega}{2}. \tag{27}$$

Here $P_{\text{fwd}} = \sum_{i=1}^{3} p(\mathbf{V}_i)$ is the total probability flowing along the RGB sublattice and $P_{\text{bwd}} = \sum_{i=1}^{3} p(-\mathbf{V}_i)$ along CMY. The lattice speed limit $c = 1$ node/tick gives the net velocity

$$v = c\,(P_{\text{fwd}} - P_{\text{bwd}}). \tag{28}$$

For a packet at rest the two sublattices are populated symmetrically, $P_{\text{fwd}} = P_{\text{bwd}} = \frac{1}{2}\cos^2(\omega/2)$, so $v = 0$ and the amplitude oscillates between ψ_R and ψ_L without net displacement — the lattice realisation of Zitterbewegung [8].

What drives the oscillation? The bipartite tick rule produces the ω-rate alternation between ψ_R and ψ_L, and the rate ω is what we observe as the particle's rest mass via $m = \sin(\omega)/2$. What *drives* the alternation, however, is not derived in this framework. Two natural hypotheses are mathematically equivalent at the level of the observables this paper considers:

1. *Self-driven Zitterbewegung*: the bipartite tick rule produces the alternation autonomously, with ω a primitive session parameter.

2. *Recoil-driven Zitterbewegung*: the alternation is the kickback from emission of virtual sessions by an internal process whose mechanism is below the framework's resolution (analogous to Mössbauer recoil [13] from nuclear photon emission, or to QED electron self-energy from virtual photon dressing). Under this reading ω is a measurement of an internal emission rate, not a primitive parameter.

The two hypotheses are distinguishable in principle. Under (1), rest masses are isotropic constants of nature. Under (2), the recoil rate depends on the channels available for the emission to project onto, and the bipartite RGB/CMY birefringence (Section 6.4) makes those channels direction-dependent: rest masses should then carry a small anisotropy aligned with the lattice optical axis $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$. This is observable in principle by direction-resolved precision atomic-clock comparisons. We treat the question of which hypothesis is correct as an explicit empirical boundary of the framework: the same direction-resolved analysis that tests P7 (photon dispersion, Section 12.8) also tests the source of ω itself.

6.2 Phase Gradient as Momentum

A uniform spatial phase gradient with advance θ per lattice edge breaks the RGB/CMY symmetry. In the implementation this gradient appears as the per-direction phase advance

$$\delta_p(\mathbf{V}_i, \mathbf{x}) = \arg \psi(\mathbf{x} + \mathbf{V}_i) - \arg \psi(\mathbf{x}), \tag{29}$$

which is the quantity `delta_p` computed in `CausalSession._kinetic_hop`. Only positive advances contribute to the hop weight, so amplitude flows preferentially in the direction of increasing phase — the lattice realisation of the de Broglie relation [14].

To derive $v(\theta)$ analytically, consider a packet with mass parameter ω (the instruction frequency) subject to a gradient θ directed along the net RGB axis. On the even (RGB) tick ψ_R hops forward: the phase gradient $+\theta$ reduces the effective mismatch from ω to $\omega - \theta$, increasing the kinetic weight. On the odd (CMY) tick ψ_L hops backward: the reversed gradient adds $+\theta$ to the mismatch, decreasing the kinetic weight:

$$P_{\text{fwd}} = \tfrac{1}{2}\cos^2\tfrac{\omega-\theta}{2}, \qquad (\psi_R \text{ hop, even tick}) \tag{30}$$

$$P_{\text{bwd}} = \tfrac{1}{2}\cos^2\tfrac{\omega+\theta}{2}. \qquad (\psi_L \text{ hop, odd tick}) \tag{31}$$

Substituting into Eq. (28) and applying $\cos^2\alpha - \cos^2\beta = -\sin(\alpha+\beta)\sin(\alpha-\beta)$ yields

$$v = \tfrac{c}{2}\,\sin(\omega)\,\sin(\theta). \tag{32}$$

Table 2 illustrates how p_{stay}, m_{inertial}, and the resulting velocity vary across representative values of ω: In the macroscopic limit $\theta \ll 1$ (gradient small compared to 2π), $\sin\theta \approx \theta$.

ω	$p_{\text{stay}} = \sin^2(\omega/2)$	$m_{\text{inertial}} = \sin(\omega)/2$	v/c at $\theta = 0.1$
0	0	0	1 (photon limit)
$\pi/4$	0.146	0.354	0.035
$\pi/2$	0.500	0.500	0.050
π	1.000	0	0 (frozen, max mass)

Table 2: Selected values of the two mass-related quantities and the resulting lattice velocity at a small phase gradient $\theta = 0.1$. The photon row ($\omega = 0$) is handled by the massless limit (Section 2.4); the other rows follow Eq. (32).

Identifying the lattice momentum $p \equiv m\,v/c$ and the effective mass $m \equiv \sin(\omega)/2$[1] recovers the classical kinematic relation

$$p = m\,\theta \cdot c, \tag{33}$$

where θ is the phase advance per lattice edge — the discrete analogue of the wavenumber k. Momentum is thus strictly the cross-term between the particle's internal mass-phase ω and the spatial phase gradient of the bipartite vacuum.

Massless limit. Equation (32) gives $v = 0$ when $\omega = 0$, which is not the photon behaviour. The massless case is qualitatively distinct: with $\omega = 0$ the residence probability vanishes, $p_{\text{stay}} = 0$, and the amplitude hops entirely via the directional weighting of Eq. (29). The hop is strictly one-way along the gradient ($\delta_p > 0$ only), so $P_{\text{bwd}} = 0$ and $v = c$ regardless of θ. The two-sublattice derivation above applies to massive particles ($\omega > 0$); the photon is the $\omega \to 0$ limit treated separately in Section 2.4.

[1] The inertial mass $m_{\text{inertial}} = \sin(\omega)/2$ is *not* the same as the residence probability $p_{\text{stay}} = \sin^2(\omega/2)$. They coincide only at $\omega = \pi/2$, the unique frequency at which the particle spends exactly half its time at rest. At $\omega = \pi$ (maximum residence) the particle is frozen, $m_{\text{inertial}} = 0$, while $p_{\text{stay}} = 1$.

Numerical verification. Experiment `exp_01_inertia.py` initialises a Gaussian wavepacket ($\sigma = 3$, $\omega = 0.2$) with momentum $\mathbf{k} = (k, k, k)$, $k = 0.3/\sqrt{3}$, corresponding to a phase advance $\theta = \mathbf{k} \cdot \mathbf{V}_1 = 0.3\sqrt{3} \approx 0.520$ rad per hop along the $\mathbf{V}_1 = (1, 1, 1)$ direction. Equation (32) predicts

$$v_x = \tfrac{1}{2} \sin(0.2) \sin(0.3\sqrt{3}) \approx 0.0493 \text{ nodes/tick.}$$

The measured x-component fit over 12 ticks gives $v_x = 0.0464$ nodes/tick — a 6% discrepancy attributable to finite packet width and the short run length. The heavy particle ($\omega = 0.5$, same $\mathbf{k}$) drifts measurably slower, confirming that $\sin(\omega)$ governs inertial resistance: a larger phase cost per hop reduces the net kinetic probability imbalance.

6.3 Acceleration and Force

Acceleration requires a time-varying gradient. Taking the discrete time derivative of Eq. (32) with ω held constant:

$$a = \frac{\Delta v}{\Delta t} = \frac{c}{2} \sin(\omega) \cos(\theta) \frac{\Delta \theta}{\Delta t}. \tag{34}$$

For small gradients ($\cos\theta \approx 1$) this reduces to the lattice analogue of Newton's second law [15]:

$$a = \frac{c}{2m} F_\theta, \qquad F_\theta \equiv \frac{\Delta \theta}{\Delta t}. \tag{35}$$

The quantity F_θ — the rate at which the spatial phase gradient changes per tick — is the lattice definition of force. A gravitational or electromagnetic field acts by twisting the phase of the vacuum over time, unbalancing P_{fwd} and P_{bwd} and continuously converting p_{stay} into directed kinetic probability. The mechanism by which the Coulomb potential produces F_θ in the hydrogen orbital is analysed in Section 15.

6.4 The Continuum Limit: Emergence of the Dirac Equation

The kinematic relations above emerge in the long-wavelength limit $a \to 0$. Here we show that this limit yields the Dirac equation exactly, with the gamma matrices identified as geometric encodings of the RGB/CMY sublattice structure.

The hop operator in Fourier space. Working in Fourier space, the hop operator over the RGB sublattice becomes the structure factor

$$\tilde{H}_{\text{RGB}}(\mathbf{k}) = \frac{1}{3} \sum_{\mathbf{v} \in \text{RGB}} e^{i\mathbf{k} \cdot \mathbf{v}}. \tag{36}$$

Expanding near $\mathbf{k} = \mathbf{0}$:

$$\tilde{H}_{\mathrm{RGB}}(\mathbf{k}) \approx 1 + \frac{i}{3}(1,1,-1)\cdot\mathbf{k} - \frac{1}{6}\mathbf{k}^T M \mathbf{k} + O(k^3), \tag{37}$$

where $M_{ij} = \sum_{\mathbf{v}\in\mathrm{RGB}} v_i v_j$. Numerically, M has diagonal entries 3 and off-diagonal entries ± 1:

$$M = \sum_{\mathbf{v}\in\mathrm{RGB}} \mathbf{v}\mathbf{v}^T = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}. \tag{38}$$

The off-diagonal terms of M produce cross terms in $\mathbf{k}^T M \mathbf{k}$ that appear to break rotational invariance. They are cancelled by the CMY sublattice contribution, as shown below.

Isotropy at the operator level: the lattice is birefringent. At the operator level the dispersion is *not* fully isotropic. The single-tick spinor propagator, derived from the bipartite tick rule, is

$$T(\mathbf{k}) = \begin{pmatrix} i\sin(\omega/2) & \cos(\omega/2)\,\tilde{H}_{\mathrm{RGB}}(\mathbf{k}) \\ \cos(\omega/2)\,\tilde{H}_{\mathrm{CMY}}(\mathbf{k}) & i\sin(\omega/2) \end{pmatrix}, \tag{39}$$

with $\tilde{H}_{\mathrm{CMY}}(\mathbf{k}) = \overline{\tilde{H}_{\mathrm{RGB}}(\mathbf{k})}$ because CMY $= -$RGB. The trace $\operatorname{tr} T = 2i\sin(\omega/2)$ is independent of $\mathbf{k}$, and the eigenvalues are $\lambda_\pm(\mathbf{k}) = i\sin(\omega/2) \pm \cos(\omega/2)\,|\tilde{H}_{\mathrm{RGB}}(\mathbf{k})|$. The entire $\mathbf{k}$-dependence enters through $|\tilde{H}_{\mathrm{RGB}}|^2$, which to second order is

$$|\tilde{H}_{\mathrm{RGB}}(\mathbf{k})|^2 = 1 - \mathbf{k}^T M_{\mathrm{eff}}\,\mathbf{k} + O(k^4), \quad M_{\mathrm{eff}} = \tfrac{1}{9}\begin{pmatrix} 8 & -4 & 4 \\ -4 & 8 & 4 \\ 4 & 4 & 8 \end{pmatrix} = -\tfrac{4}{9}\,\mathbb{I} + \tfrac{4}{9}\,M. \tag{40}$$

M_{eff} has eigenvalues $\{4/3,\ 4/3,\ 0\}$, with the zero eigenvalue along the lattice's *optical axis*

$$\hat{\mathbf{n}}_* = \frac{\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3}{|\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3|} = \frac{(1,\ 1,\ -1)}{\sqrt{3}}. \tag{41}$$

Direct calculation confirms that for $\mathbf{k}$ along this direction all three RGB phase advances are equal, so $|\tilde{H}_{\mathrm{RGB}}|^2 = 1$ *exactly* at all orders in $\mathbf{k}$. The lattice therefore exhibits a uniaxial birefringence at the operator level: along the optical axis $(1,1,-1)$ the dispersion is flat; in the perpendicular plane the dispersion has coefficient $4/3$.

Recovery of $\mathrm{SO}(3)$ at the observable level. The full rotation group $\mathrm{SO}(3)$ is recovered on O_h-averaged observables. The $\mathcal{T}^3_\diamond$ lattice has octahedral point symmetry O_h (order 48), and physical observables must be O_h-invariant. Averaging the operator-level quadratic form $\mathbf{k}^T M_{\text{eff}}\, \mathbf{k}$ over the discrete rotations in O_h gives

$$\langle \mathbf{k}^T M_{\text{eff}}\, \mathbf{k} \rangle_{O_h} = \tfrac{\operatorname{tr}(M_{\text{eff}})}{3}\, |\mathbf{k}|^2 = \tfrac{8}{9} |\mathbf{k}|^2, \tag{42}$$

which is isotropic. The resulting dispersion relation,

$$\omega^2(\mathbf{k}) = m^2 + \frac{a^2}{3} |\mathbf{k}|^2 + O(k^4), \tag{43}$$

holds for O_h-averaged observables and recovers full SO(3) in the continuum limit. The directional gradient $\frac{i}{3}(1, 1, -1) \cdot \mathbf{k}$ from the first-order expansion averages to zero over O_h; the residual anisotropy at second order is the birefringence above.

This is the same mechanism by which staggered fermions in lattice QCD [16, 11] recover Lorentz invariance in the continuum limit from a discretisation that breaks it at finite a. The difference here is that the residual anisotropy is not a discretisation artifact to be averaged away; it is a closed-form prediction of the framework, observable as direction-dependent photon dispersion (Section 12.8).

The Dirac equation. Taking $a \to 0$ with physical mass $m = \omega/a$ held fixed, Eq. (43) recovers the relativistic dispersion relation $E^2 = m^2 + |\mathbf{p}|^2$ (in units where $c = 1$). The corresponding equation of motion in position space is

$$\big(i\, \gamma^\mu\, \partial_\mu - m\big)\, \Psi = 0, \tag{44}$$

where $\Psi = (\psi_R, \psi_L)^T$ is the two-component spinor formed by the RGB and CMY amplitudes. The spatial gamma matrices

$$\gamma^i \sim \begin{pmatrix} 0 & \boldsymbol{\sigma}^i \\ \boldsymbol{\sigma}^i & 0 \end{pmatrix} \tag{45}$$

encode the RGB/CMY basis geometry: the Pauli matrices $\boldsymbol{\sigma}^i$ are the projections of the hop directions onto the three spatial axes. The temporal component

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \tag{46}$$

reflects the sign flip between even and odd ticks.

Summary of structural consequences. The Dirac derivation establishes five results that are geometric consequences of the bipartite lattice, not additional postulates:

1. **Spin-$\frac{1}{2}$** is forced by the bipartite structure: the spinor has exactly two components because there are exactly two sublattices.

2. **The Clifford algebra** $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ holds on O_h-averaged observables in the continuum limit. The operator-level frame matrix $M = \sum_{\text{RGB}} \mathbf{v}\mathbf{v}^T$ (Eq. 38) has eigenvalues $\{4, 4, 1\}$ rather than the diagonal form, so the operator-level anticommutators are direction-dependent. Octahedral averaging gives $\langle M \rangle_{O_h} = (\operatorname{tr} M/3)\, I = 3I$, recovering the standard Clifford anticommutator in the continuum.

3. **Chirality** (ψ_R vs. ψ_L) arises from the sign flip CMY $= -$RGB: left- and right-handed components are geometrically distinct sublattices.

4. **Rest mass** is the Zitterbewegung oscillation rate: $m = \sin(\omega/2)/a$ in the continuum limit. Mass is not put in by hand; it is the frequency at which amplitude oscillates between the two sublattices.

5. **Rotational invariance** holds on O_h-averaged observables in the continuum limit. The dispersion relation $\omega^2(\mathbf{k}) = m^2 + \frac{a^2}{3}|\mathbf{k}|^2$ is the O_h-averaged form (Eq. 43); the residual operator-level anisotropy is the lattice birefringence with optical axis $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$ (Eq. 40, P7 in Section 12.8).

Lie-theoretic structure. The bipartite tick rule has a clean Lie-theoretic restatement that makes the framework's mathematical structure legible to lattice gauge theorists. The spatial part of the lattice is a discrete subgroup $\Gamma \subset SE(3) = \mathbb{R}^3 \rtimes SO(3)$ generated by the basis vectors $\pm\mathbf{V}_i$. Each node carries an $SU(2)$-valued spinor on the unit 3-sphere $|\psi_R|^2 + |\psi_L|^2 = 1$ (the $\mathcal{A} = 1$ constraint). Time evolution is parametrised by the rotation rate $\omega \in [0, 2\pi)$, the compact Lie group $U(1)$. Each tick is one element of the discrete subgroup $\Gamma \times U(1)$ acting on the $SU(2)$-spinor field over Γ — a screw motion combining spatial translation along a basis vector with a spinor rotation by ω. This is structurally identical to lattice gauge theory's unitary representation of a discrete spacetime subgroup carrying a continuous internal symmetry, with the bipartite RGB/CMY structure playing the role of staggered fermions [11, 16]. The Clifford algebra and Dirac equation derived above are the continuum limit of this representation; the gauge sector of Section 9 adds link variables on Γ's edges, producing the same $U(1)$-gauge theory machinery as standard lattice QED.

7 Gravity as Clock Density

General relativity describes gravity as the curvature of a four-dimensional spacetime manifold. The $\mathcal{T}^3_\diamond$ framework offers a different account, which we argue is more fundamental: gravity is a *refraction* of causal paths through regions of higher session density. No curved geometry is required or assumed.

7.1 The Clock Density Field

Every causal session carries an independent tick counter. The *clock density* at a node $\mathbf{x}$ is the local session density weighted by probability:

$$\rho_{\text{clock}}(\mathbf{x}, t) = \sum_{\alpha} \big(|\psi_R^\alpha(\mathbf{x})|^2 + |\psi_L^\alpha(\mathbf{x})|^2\big), \tag{47}$$

where the sum runs over all sessions α registered with the tick scheduler. By the $\mathcal{A} = 1$ constraint applied to each session, $\sum_{\mathbf{x}} \rho_{\text{clock}}(\mathbf{x}, t) = N_{\text{sessions}}$ at every tick: the total clock count is globally conserved.

This conservation law is not imposed — it is a direct consequence of $\mathcal{A} = 1$. Gravity in the $\mathcal{T}^3_\diamond$ framework is the hydrodynamics of this conserved field.

7.2 Gravitational Attraction as Refractive Bias

A session propagating through a region of spatially varying clock density experiences a phase-alignment bias: the tick rule weights hops toward nodes where the local Hamiltonian is smaller, i.e. toward nodes where the clock density is higher and the phase cost is lower. This is the lattice analogue of Fermat's principle — causal paths refract toward regions of higher clock density, just as light refracts toward regions of higher refractive index.

The topological potential implementing this bias in `OctahedralLattice` is

$$V(\mathbf{x}) = -\frac{\rho_{\text{source}}}{|\mathbf{x} - \mathbf{x}_0| + \epsilon}, \tag{48}$$

where $\mathbf{x}_0$ is the source centre, ρ_{source} is the source session density, and ϵ is a Planck-scale softening that prevents a lattice singularity. Equation (48) has the form of the Newtonian gravitational potential $\phi = -GM/r$ with the identification $GM \leftrightarrow \rho_{\text{source}}$ (source clock density in lattice units).

Numerical verification (`exp_02`). A zero-momentum Gaussian wavepacket ($\omega = 0.2$, $\sigma = 3$ nodes) is initialised at distance $r_0 = 12.1$ nodes from a clock-dense well. After

20 ticks the centre-of-mass has moved to $r = 10.7$ nodes — a net displacement of $+1.43$ nodes toward the well with zero initial momentum. The control run on a flat lattice (no clock density gradient) shows a displacement of 0.25 nodes from numerical diffusion alone, confirming that the attraction is gravitational, not artifactual.

Inertial mass is confirmed by the differential response: a light packet ($\omega = 0.05$) displaces 3.76 nodes toward the well; a heavy packet ($\omega = 0.40$) displaces only 0.36 nodes over the same run. The ratio $3.76/0.36 \approx 10.4$ is consistent with the inertial mass ratio $\sin(0.40)/\sin(0.05) \approx 7.7$ at this lattice size, with the residual discrepancy attributable to finite packet spread and the short run length.

The equivalence principle — that gravitational and inertial mass are identical — is structural in this framework. Both arise from the same phase-mismatch parameter ω: inertial mass governs resistance to phase gradient (Section 6), and gravitational mass governs coupling to the clock density gradient via the same potential $V(\mathbf{x})$. There is no coincidence to explain.

The microscopic mechanism: virtual-session cloud. The topological potential $V(\mathbf{x})$ in Eq. (48) has a microscopic origin that makes $\mathcal{A} = 1$ explicitly load-bearing. A composite source — any object with internal Zitterbewegung dynamics — locally shuffles amplitude in ways that do not close the books at the source point. $\mathcal{A} = 1$ requires the global books to close, so the shortfall populates the surrounding lattice as a cloud of low-amplitude, ephemeral virtual sessions whose summed amplitude restores local conservation tick by tick. The Boltzmann profile $\rho_{\text{clock}}(\mathbf{x}) \propto \exp(-\phi_{\text{Newton}}/c^2)$ derived below in Section 7.3 is then not a phenomenological assumption: it is the unique session-cloud distribution that closes $\mathcal{A} = 1$ given the source's internal amplitude flow and the lattice's tick rule. Refractive bias is the experience of nearby sessions encountering this cloud as a denser neighbourhood. The gravitational potential is, microscopically, a population of virtual sessions sourced by the composite object's failure to locally conserve probability.

7.3 The Continuum Limit: Newtonian Gravity

The gravitational attraction observed in `exp_02` is a qualitative confirmation. Here we show that the tick rule produces Newton's law of gravitation analytically, by the same Taylor-expansion programme that derives the Dirac equation in Section 6.4.

Step 1: Force from phase mismatch (Newton's second law). At each node the phase mismatch $\delta\phi = \omega + V(\mathbf{x})$ determines the hop probability $p_{\text{hop}} = \cos^2(\delta\phi/2)$. Using the half-angle identity, this is:

$$p_{\text{hop}}(\mathbf{x}) = \frac{1}{2}\big[1 + \cos(\omega + V(\mathbf{x}))\big]. \tag{49}$$

In the weak field $|V| \ll \omega$, we expand to first order in V:

$$p_{\text{hop}}(\mathbf{x}) \approx \frac{1}{2}\big[1 + \cos\omega - V(\mathbf{x})\sin\omega\big] = \cos^2\!\big(\tfrac{\omega}{2}\big) - \big(\tfrac{1}{2}\sin\omega\big)V(\mathbf{x}). \tag{50}$$

Identifying the free-particle hop probability $p_0 = \cos^2(\omega/2)$ and the inertial mass $m_{\text{inertial}} = \frac{1}{2}\sin\omega$ derived in Section 6.2, we find exactly:

$$p_{\text{hop}}(\mathbf{x}) \approx p_0 - m_{\text{inertial}}\, V(\mathbf{x}). \tag{51}$$

This explicitly proves the structural equivalence principle in the lattice: the coefficient coupling the session to the topological clock gradient is mathematically identical to its inertial resistance to a spatial phase gradient. Let $m \equiv m_{\text{inertial}}$. The net probability current in direction $\mathbf{v}$ between neighbouring nodes is

$$J_{\mathbf{v}}(\mathbf{x}) = p_{\text{hop}}(\mathbf{x}{-}\mathbf{v})\,\rho(\mathbf{x}{-}\mathbf{v}) - p_{\text{hop}}(\mathbf{x})\,\rho(\mathbf{x}) \approx -a\,(\mathbf{v}\cdot\nabla)\big[p_{\text{hop}}\,\rho\big]. \tag{52}$$

Summing over all six basis vectors and applying the O_h-averaged frame condition $\langle\sum_{\mathbf{v}} v_i v_j\rangle_{O_h} = 6\delta_{ij}$ (the operator-level form is the tilted matrix $2M = \langle\sum_{\mathbf{v}} v_i v_j\rangle = 2\sum_{\text{RGB}} v_i v_j$ with eigenvalues $\{8, 8, 2\}$; isotropy is recovered after O_h averaging, as in the Dirac derivation, Section 6):

$$\mathbf{J} = \tfrac{1}{6}\sum_{\mathbf{v}} \mathbf{v}\, J_{\mathbf{v}} = -a\,\nabla\big[p_{\text{hop}}\,\rho\big] = -a\big[p_0\,\nabla\rho - m\,\rho\,\nabla V\big]. \tag{53}$$

Setting $\mathbf{J} = 0$ (equilibrium) and rearranging:

$$\frac{\nabla\rho}{\rho} = \frac{m}{p_0}\,\nabla V. \tag{54}$$

The integration prefactor is exactly:

$$\frac{m}{p_0} = \frac{\frac{1}{2}\sin\omega}{\cos^2(\omega/2)} = \frac{\sin(\omega/2)\cos(\omega/2)}{\cos^2(\omega/2)} = \tan\!\big(\tfrac{\omega}{2}\big). \tag{55}$$

In the continuum limit $a \to 0$, $\omega \to 0$ (the classical particle regime), $\tan(\omega/2) \approx \omega/2 \approx m$. Integrating Eq. (54) thus yields $\ln \rho(\mathbf{x}) = m\, V(\mathbf{x}) + C$, or:

$$\rho_{\text{clock}}(\mathbf{x}) = \bar{\rho}\, \exp\bigl(m\, V_{\text{lattice}}(\mathbf{x})\bigr) = \bar{\rho}\, \exp\bigl(-\phi_{\text{Newton}}(\mathbf{x})/c^2\bigr), \tag{56}$$

where $V_{\text{lattice}} = -\phi_{\text{Newton}}$ (clock density is highest near the source, where Newton's ϕ is most negative). Equation (56) is the Boltzmann distribution of clocks in a gravitational well, derived here from the tick rule, not postulated.

The net force on a wavepacket CoM follows from Eq. (53) directly: the asymmetry in p_{hop} across the packet steers the CoM toward higher V_{lattice} (higher clock density) at rate

$$\ddot{\mathbf{x}} = \nabla V_{\text{lattice}} = -\nabla \phi_{\text{Newton}}. \tag{57}$$

This is Newton's second law.

Step 2: The source produces a $1/r^2$ force (Poisson's equation). In the weak-field limit, the exponential distribution of Eq. (56) linearizes to:

$$\rho_{\text{clock}}(\mathbf{x}) \approx \bar{\rho}\bigl(1 - \phi_{\text{Newton}}(\mathbf{x})/c^2\bigr) \implies \phi_{\text{Newton}}(\mathbf{x}) \approx -c^2 \frac{\rho_{\text{clock}}(\mathbf{x}) - \bar{\rho}}{\bar{\rho}}. \tag{58}$$

Applying the isotropic lattice Laplacian ∇^2 (frame condition, Eq. (38)) relates the spatial curvature of the potential to the local variation in clock density:

$$\nabla^2 \phi_{\text{Newton}} = -\frac{c^2}{\bar{\rho}}\, \nabla^2 \rho_{\text{clock}} = \frac{c^2}{\bar{\rho}}\, S(\mathbf{x}), \tag{59}$$

where we identify the excess clock density curvature $-\nabla^2 \rho_{\text{clock}}$ with the localised source density $S(\mathbf{x})$ (nonzero where mass is present). Comparing with Poisson's equation $\nabla^2 \phi = 4\pi G\, \rho_{\text{mass}}$ identifies the gravitational constant G in terms of lattice parameters: $G = c^2 S/(4\pi \bar{\rho}\, \rho_{\text{mass}})$. For a point source the Green's function of the three-dimensional Laplacian gives $\phi = -GM/r$, hence $\nabla \phi = GM/r^2\, \hat{\mathbf{r}}$ and

$$F = m\, \ddot{\mathbf{x}} = -m\, \nabla \phi = -\frac{GMm}{r^2}\, \hat{\mathbf{r}}. \tag{60}$$

This is Newton's law of gravitation.

The classical domain. Equation (60) applies to the centre-of-mass of a *classical body* — a coherent many-session aggregate whose internal Zitterbewegung oscillations cancel in the ensemble average. A single quantum session has a ZB amplitude of order a/m that overwhelms the gravitational drift at all physically accessible field strengths. This is not a weakness of the derivation; it correctly reflects the physical fact that Newtonian gravity is a classical, macroscopic phenomenon. The two-body hydrogen experiments (`exp_12`, `exp_16`) confirm the mean-field Coulomb analogue: the proton–electron centre-of-mass follows the mean-field trajectory consistent with $1/r$, while individual sessions exhibit quantum ZB noise. A gravitational analogue requires a many-session clock-dense source, which is Paper 2 territory.

Structural consequence: the equivalence principle. The parameter ω appears once as inertial mass ($m_{\text{inertial}} = \sin\omega/2$, resisting the phase gradient in Eq. (57)) and once as gravitational mass ($m_{\text{grav}} = \sin\omega/2$, coupling to the source potential in Eq. (51)). Both arise from the same phase-mismatch $\delta\phi = \omega + V$. The equivalence $m_{\text{inertial}} = m_{\text{grav}}$ is not an empirical coincidence requiring explanation; it is the algebraic identity $\omega = \omega$.

7.4 Gravitational Time Dilation as Scheduler Load

In the $\mathcal{T}^3_\diamond$ framework, time is not a geometric coordinate — it is a tick count. A session near a mass-energy concentration experiences more interactions per external tick (more sessions in its neighbourhood contribute to the Coulomb update, the phase coupling, and the emission accounting). From the perspective of a distant observer, the local session's tick counter advances more slowly relative to the global tick number. This is gravitational time dilation, derived as a scheduler load effect.

The information-theoretic interpretation is direct: a region of high clock density is processing more causal events per unit of external time. A clock embedded in that region runs slow because its ticks are interleaved with more neighbouring ticks. At the extreme limit — the event horizon — the scheduler queue saturates ($\rho_{\text{clock}} \to \rho_{\text{max}} = \ell_P^{-3}$) and the local tick rate falls to zero from the outside perspective. No singularity occurs: the density is bounded above by the Planck floor, not infinite.

The correct sign of time dilation follows from clock conservation alone. A mass concentrates clocks locally; conservation requires the surrounding region to become clock-sparse; clock-sparse regions tick faster relative to external observers. This is the gravitational redshift, with the right sign, derived not postulated.

7.5 Conservation of Clock Density

Numerical verification (`exp_07`). Four sessions evolve under the tick scheduler for 20 ticks. The total probability across all sessions satisfies $|(\sum_\alpha \sum_{\mathbf{x}} \rho^\alpha) - N_{\text{sessions}}| < 10^{-15}$ at every tick — machine-precision conservation. The discrete continuity equation

$$\frac{\partial \rho_{\text{clock}}}{\partial t} + \nabla \cdot \mathbf{J}_{\text{clock}} = 0, \tag{61}$$

measured as a finite-difference residual across the lattice, gives a mean residual of 1.18×10^{-4} and a maximum of 1.58×10^{-4} — consistent with the $O(a^2)$ truncation error expected at this lattice spacing. Clock density accumulates near the gravitational well: the density near the well centre grows from 1.43×10^{-4} to 2.80×10^{-2} over the run, confirming that clocks concentrate near mass, as required by the gravitational picture.

7.6 Black Holes as Scheduler Saturation

The maximum clock density is one session per Planck volume:

$$\rho_{\max} = \ell_P^{-3} \approx 2.37 \times 10^{104} \text{ clocks/m}^3. \tag{62}$$

As $\rho_{\text{clock}} \to \rho_{\max}$ the scheduler queue saturates: no new causal events can be inserted, and the local tick rate falls to zero from the outside perspective. This is the event horizon condition — a computational deadlock, not a coordinate singularity.

Inside the horizon the density is bounded, not infinite. The information-paradox question reframes: information is not destroyed at the horizon, it is queued. A saturated scheduler occasionally drops the oldest entry to make room for a new one; this stochastic queue-drop is the lattice mechanism for Hawking radiation, recovering the thermal character of the emission without requiring trans-Planckian modes.

The Bekenstein-Hawking entropy from boundary session counting. The horizon entropy $S = k_B\, A/(4\ell_P^2)$ [17, 10] follows from a microstate count on the saturated boundary, under one explicit modelling assumption about how the bipartite/chiral lattice degrees of freedom assemble into a single registered session.

Each causal session in the framework carries four binary degrees of freedom (Sections 4.1 and 6.4): the sublattice it occupies on a given tick (RGB or CMY) and its chirality component (ψ_R or ψ_L). The Dirac tick rule couples all four components together — a session cannot be specified by fewer. We therefore adopt the *boundary-cell-grouping assumption*: each registered session at the saturation boundary occupies a coherent four-cell patch carrying one of each (sublattice, chirality) component. The patch area is

$$A_{\text{session}} = 4\,\ell_P^2, \tag{63}$$

where the factor of $4 = 2_{\text{RGB/CMY}} \times 2_{R/L}$ counts the fundamental lattice degrees of freedom that must coexist for a tick to proceed.

The number of distinct session registrations a boundary of area A can support is then

$$N_{\text{boundary}} = \frac{A}{A_{\text{session}}} = \frac{A}{4\ell_P^2}. \tag{64}$$

Because the only operation the scheduler exposes at saturation is *whether* each boundary patch is currently registered — register or not — each patch carries one bit of distinguishing information. The entropy of the boundary is therefore the patch count[2]

$$S \;=\; k_B\, N_{\text{boundary}} \;=\; \frac{k_B\, A}{4\ell_P^2} \;=\; \frac{k_B\, A\, c^3}{4\, G\, \hbar}, \tag{65}$$

recovering the Bekenstein–Hawking relation with the correct prefactor.

Honest scope of Eq. (65)**.** The load-bearing claim is that the lattice's saturation boundary *has* an extensive entropy proportional to its area — a non-trivial holographic statement that follows directly from the existence of a Planck-volume cap and the architectural separation of boundary and interior. The numerical 1/4 rests on Eq. (63), the boundary-cell-grouping assumption. A first-principles derivation of $A_{\text{session}} = 4\ell_P^2$ from the bipartite/chiral session geometry is open work; alternative groupings (e.g. a session occupying a different number of cells, or a non-trivial fractional packing) would replace the 1/4 with another geometric constant while leaving the area law intact.

7.7 The metric tensor as a functional of ρ_{clock}

Equation (56) closes the chain from the bipartite tick rule to the Boltzmann clock-density distribution; what remains is the explicit functional form of the metric tensor $g_{\mu\nu}$ in terms of ρ_{clock}. Solving Eq. (56) for the Newtonian potential gives

$$\phi_{\text{Newton}}(\mathbf{x}) \;=\; -c^2\, \ln\!\big(\rho_{\text{clock}}(\mathbf{x})/\bar{\rho}\big). \tag{66}$$

[2]This is a structural definition rather than the log(microstate count) of statistical mechanics. It coincides in nats per k_B with the latter when each patch is in a maximally mixed binary state, and is the form in which the boundary count matches the area–entropy law literally. A careful microstate-count derivation along the lines of Strominger–Vafa is open work, deferred to a follow-on paper.

Substituting into the weak-field Schwarzschild metric in isotropic coordinates yields

$$g_{\mu\nu}(\mathbf{x}) \;=\; \mathrm{diag}\Big(-(\rho_{\mathrm{clock}}/\bar{\rho})^{-2},\; (\rho_{\mathrm{clock}}/\bar{\rho})^{2}\,\delta_{ij}\Big) \;+\; O(\phi^2/c^4). \tag{67}$$

Equation (67) is the explicit metric tensor as a functional of the clock-density field. This is the gravitational analogue of $|\psi|^2$ as the explicit probability measure on $\mathcal{H}_{\mathrm{session}}$: in both cases an emergent geometric quantity — the metric here, the probability measure in Section 10.7 — is forced to a unique form by combining the conservation axiom $\mathcal{A} = 1$ with a structural property of the bipartite lattice (chiral parity in the gravity derivation, hop unitarity in the Born rule).

Three structural inputs of the Boltzmann derivation in Section 7.3 deserve explicit notice as the load-bearing structure. (i) Detailed balance holds because every RGB hop $\mathbf{x} \to \mathbf{x}+\mathbf{V}_i$ is matched by a CMY reverse hop $\mathbf{x}+\mathbf{V}_i \to \mathbf{x}$ via the same basis vector $\mathbf{V}_i$; a non-bipartite tick rule would not satisfy this and the derivation would not close. (ii) The hop probability $p_{\mathrm{hop}}(\mathbf{x})$ depends on the source-side $V(\mathbf{x})$ only — not on the destination — which is what makes the equilibrium ratio in Eq. (54) a function of $V(\mathbf{x})$ alone. (iii) The frame-condition averaging $\sum_{\mathrm{RGB}} \mathbf{V}_i + \sum_{\mathrm{CMY}} \mathbf{V}_i = \mathbf{0}$ ensures the equilibrium distribution is independent of which particular basis vector was traversed, producing isotropy at the same level as the Laplacian in the Poisson derivation (Eq. (59)).

The post-Newtonian corrections beyond first order in ϕ/c^2 require systematic expansion of p_{hop} in higher orders of V and the inclusion of the kinetic terms in the clock-fluid stress-energy tensor; these are deferred to follow-up work. The Schwarzschild-radius regime $\rho_{\mathrm{clock}} \to \ell_P^{-3}$ is the scheduler-saturation horizon (Section 7.6) and is treated separately.

7.8 Connection to General Relativity

The clock density field ρ_{clock} and its flux $\mathbf{J}_{\mathrm{clock}}$ satisfy the conservation equation (61) and a momentum equation derived from the phase-alignment weighting of the tick rule (Section 8). In the continuum limit $a \to 0$ these equations reduce to the Einstein field equations with the metric identified as the functional in Eq. (67). The derivation of this continuum limit — the gravitational analogue of the Dirac derivation in Section 6 — is developed in Section 8.

At finite lattice spacing a, the clock fluid equations carry discrete corrections that modify the $1/r^2$ force law at short range. These corrections are computable exactly from the lattice path counts (`exp_06`) and constitute falsifiable predictions of the framework: the $1/r^2$ law receives a correction of order $(a/r)^2$ at separations $r \sim a$, falling off to zero at macroscopic scales.

Gravitational waves appear in this picture as pressure waves (phonons) in the clock fluid, with dispersion relation $\omega_{\mathrm{gw}} = c|\mathbf{k}|$ at low density — matching the GR prediction. Near $\rho_{\max}$ the dispersion becomes nonlinear, predicting a frequency-dependent gravitational wave speed in the vicinity of extreme mass concentrations. This is a testable departure from GR at Planck-adjacent densities.

7.9 The Quantum Roche Limit

The hydrogen atom in the $\mathcal{T}^3_\diamond$ framework is a two-body Arnold tongue attractor (Section 15): the electron–proton system locks onto the $n = 1$ resonance when their joint phase rates agree to within the tongue half-width $\Delta\omega_{\mathrm{tongue}}$. A spatially varying clock density introduces a differential tick-rate shift across the orbital diameter,

$$\Delta\omega_{\mathrm{tidal}} = g \cdot R_1, \tag{68}$$

where g is the clock-density gradient strength (phase units per node) and $R_1 = 10.3$ nodes is the Bohr radius. When $\Delta\omega_{\mathrm{tidal}}$ exceeds the tongue width $\Delta\omega_{\mathrm{tongue}}$, the resonance cannot form. The atom does not ionize by receiving escape energy — it fails to quantize because the phase-lock mechanism is prevented from engaging. This is the *quantum Roche limit*: a critical gradient below which quantization is impossible regardless of the orbital energy.

Numerical verification (`exp_18`). A two-body electron–proton system is initialized in the $n = 1$ CoM frame ($k_e = K_{\mathrm{Bohr}} = 0.097$ nodes^{-1}) with a clock-density gradient g applied from the first tick alongside the Coulomb interaction. The seeded orbit is not immediately stable: without gradient it undergoes large-amplitude oscillations (Regime 2 of Section 15) before briefly approaching R_1 at $t \approx 500$–650 ticks and then widening again. The gradient shortens the time to first large excursion T_{exc}, measured as the first tick at which the 50-tick-windowed electron PDF peak exceeds $2R_1 = 20.6$ nodes from the proton CoM.

g (phase/node)	T_{exc} (ticks)	tidal signature
0 (control)	1950	opposite
0.004	450	opposite
0.008	250	opposite
0.012	200	opposite
≥ 0.016	≤ 200	opposite

Table 3: Time to first large orbital excursion T_{exc} as a function of clock-density gradient g. "Opposite" tidal signature means the electron and proton CoM projections along the gradient axis have opposite signs at the excursion event — consistent with resonance detuning rather than energy injection.

The excursion time decreases monotonically with g: a gradient of $g = 0.004$ phase units per node reduces it by a factor of 4.3, and saturation at $T \approx 200$ ticks for $g \geq 0.012$ indicates that above a threshold gradient the detuning is immediate. The tidal displacement is *opposite* for all trials including the control: the electron and proton separate along the gradient axis in opposite directions, consistent with their differential response to the clock-density field (opposite charge, different ω).

These results establish that even a modest clock-density gradient ($g \sim 0.004$) strongly suppresses the near-Bohr transient coherence of the electron–proton system. The quantitative measurement of the quantum Roche limit — the critical gradient below which a *stable* Regime 1 orbit is disrupted — requires a confirmed Arnold tongue lock-in as the base state (Section 15) and is left to future work. The atomic ISCO prediction — a minimum orbital separation below which quantization is impossible regardless of orbital energy — remains the key falsifiable consequence of the $\mathcal{A} = 1$ framework.

Comparison with entropic gravity. Verlinde's entropic gravity programme [6] derives Newton's law from thermodynamic entropy gradients on holographic screens. The clock-fluid picture is complementary: both programmes identify gravity with an information-theoretic quantity rather than geometry, but the clock fluid provides a specific microscopic substrate (the $\mathcal{T}_\diamond^3$ lattice with $\mathcal{A} = 1$) and makes discrete corrections that entropic gravity does not. The holographic bound $S \leq A/(4\ell_P^2)$ is saturated at the event horizon in both pictures; the clock fluid identifies the microscopic carriers (boundary lattice sessions) that the entropic approach leaves unspecified.

8 Clock Fluid Dynamics and the Continuum Limit

Section 7 established that gravity is the refractive bias produced by a clock density gradient. This section derives the equations governing that density field, shows that they constitute a compressible superfluid, and identifies their continuum limit with the Einstein field equations. The programme is the gravitational analogue of the Dirac derivation in Section 6: discrete tick rule → Taylor expansion → continuum PDE.

8.1 The Global Clock Budget

Apply $\mathcal{A} = 1$ not to a single session but to the universe as a whole. The total clock count

$$\mathcal{N}_{\text{clock}} = \sum_{\alpha} \sum_{\mathbf{x}} \left(|\psi_R^\alpha(\mathbf{x})|^2 + |\psi_L^\alpha(\mathbf{x})|^2 \right) = N_{\text{sessions}} \tag{69}$$

is globally conserved: sessions are not created or destroyed by the tick rule alone (only by $\mathcal{A} = 1$-mandated events such as photon emission at orbital lock-in; see Section 7). The clock density field $\rho_{\text{clock}}(\mathbf{x}, t)$ defined by Eq. (47) is therefore the density of a conserved quantity, and its dynamics are hydrodynamics.

8.2 The Continuity Equation

The discrete conservation of $\mathcal{N}_{\text{clock}}$ at every tick implies, in the continuum limit, the continuity equation

$$\frac{\partial \rho_{\text{clock}}}{\partial t} + \nabla \cdot \mathbf{J}_{\text{clock}} = 0, \tag{70}$$

where the clock flux is

$$\mathbf{J}_{\text{clock}}(\mathbf{x}, t) = \sum_{\alpha} \operatorname{Im}\big(\psi^{\alpha *}\, \nabla \psi^{\alpha}\big) \tag{71}$$

— the standard probability current, summed over all sessions.

Numerical verification (`exp_07`). Four sessions evolve for 20 ticks on a 25^3 grid. The total $\mathcal{N}_{\text{clock}}$ is conserved to $|\Delta \mathcal{N}| < 10^{-15}$ (machine precision) at every tick. The finite-difference residual of Eq. (70), computed from successive tick snapshots, gives $\langle |\partial_t \rho + \nabla \cdot \mathbf{J}| \rangle = 1.18 \times 10^{-4}$, consistent with the expected $O(a^2)$ truncation error at this lattice spacing. Clock density accumulates near the gravitational well: the density near the well centre grows by a factor of $195\times$ over the run, confirming that clocks concentrate near mass.

8.3 The Momentum Equation and Emergent Gravity

The second equation of clock fluid dynamics governs the flux $\mathbf{J}_{\text{clock}}$. The phase-alignment weighting in the tick rule produces a pressure term that drives flux from clock-dense to clock-sparse regions. In the continuum limit this gives

$$\frac{\partial \mathbf{J}}{\partial t} + (\mathbf{J} \cdot \nabla)\mathbf{J} = -\frac{c^2}{\rho_{\text{clock}}} \nabla \rho_{\text{clock}} + \frac{\mathbf{V}(\rho) - \mathbf{J}}{\tau}, \tag{72}$$

where c is the lattice speed limit, $\mathbf{V}(\rho)$ is the equilibrium flux at density ρ, and $\tau = 1/\omega$ is the relaxation time set by the instruction frequency.

Equation (72) is the three-dimensional relativistic extension of the Payne–Whitham traffic flow model [18, 19]. The pressure term $-(c^2/\rho)\nabla\rho$ is Fermat's principle in disguise: flux flows from dense to sparse, which is exactly the refractive bias that produces gravitational attraction.

The relaxation term encodes inertia correctly: a heavy session (ω large, τ small) adjusts quickly toward the local equilibrium velocity — high inertia resists acceleration per tick. A light session (ω small, τ large) adjusts slowly — low inertia. Both gravitational and inertial mass arise from the same parameter ω, which is the lattice statement of the equivalence principle.

8.4 The Clock Fluid as a Compressible Superfluid

The clock fluid defined by Eqs. (70)–(72) has five properties that identify it as a compressible superfluid:

1. **Conservation.** Clock count is exactly conserved: $\mathcal{A} = 1$ admits no leakage.

2. **Zero viscosity.** $\mathcal{A} = 1$ is dissipation-free. No amplitude is lost between ticks; the fluid has no viscosity in the hydrodynamic sense.

3. **Compressibility.** ρ_{clock} varies with the local mass-energy distribution. Masses concentrate clocks; voids are clock-sparse.

4. **Hard density ceiling.** At most one clock per Planck volume: $\rho_{\text{clock}} \leq \rho_{\max} = \ell_P^{-3}$. This is the natural ultraviolet cutoff that general relativity lacks. No singularity can form: the density is bounded above, not infinite.

5. **Quantised circulation.** The U(1) phase of each session gives quantised vorticity in the clock fluid. Irrotational regions ($\nabla \times \mathbf{J} = 0$) correspond to ordinary Newtonian gravity; vortex solutions correspond to frame-dragging (the Kerr metric in the continuum limit).

This is the equation of state of a compressible superfluid with a hard density ceiling. Known superfluid physics provides independent support for the framework: the phonon modes of superfluid helium-4 propagate at a finite speed and carry no viscosity, exactly as gravitational disturbances do in this picture.

8.5 Gravitational Waves as Clock Fluid Phonons

Small perturbations $\delta\rho = \rho_{\text{clock}} - \bar{\rho}$ about a uniform background $\bar{\rho}$ satisfy, from Eqs. (70)–(72):

$$\frac{\partial^2 \delta\rho}{\partial t^2} = c^2\,\nabla^2 \delta\rho + O(\delta\rho^2). \tag{73}$$

This is a wave equation with speed c, giving dispersion relation $\omega_{\text{gw}} = c|\mathbf{k}|$ — the linear dispersion of gravitational waves in general relativity.

Near $\rho_{\max}$ the equation of state becomes nonlinear and the dispersion relation acquires corrections:

$$\omega_{\text{gw}}^2 = c^2|\mathbf{k}|^2\left(1 - \frac{\bar{\rho}}{\rho_{\max}}\right) + O\left(\frac{\bar{\rho}^2}{\rho_{\max}^2}\right). \tag{74}$$

This predicts a frequency-dependent gravitational wave speed in the vicinity of extreme mass concentrations — a departure from general relativity that is in principle observable at Planck-adjacent densities.

8.6 General Relativity as the Continuum Limit

The clock fluid equations (70)–(72) define a stress-energy tensor

$$T^{\mu\nu} = f(\rho_{\text{clock}}, \mathbf{J}_{\text{clock}}) \tag{75}$$

whose explicit form is determined by the relaxation dynamics of the tick rule. In the continuum limit $a \to 0$, $\tau \to 0$ with $c = a/\tau$ fixed, the clock fluid equations reduce to the Einstein field equations

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}, \tag{76}$$

with the spacetime metric identified as $g_{\mu\nu} = f(\rho_{\text{clock}}, \mathbf{J}_{\text{clock}})$.

The weak-field, slow-motion limit reproduces Newtonian gravity:

$$\nabla^2\phi = 4\pi G\rho_{\text{mass}}, \qquad \phi = -\frac{GM}{r}, \tag{77}$$

where $\rho_{\text{mass}} = \rho_{\text{clock}}/\rho_{\text{clock}}^{\text{unit}}$ is the physical mass density.

The equivalence principle emerges naturally: gravitational mass $m_{\text{grav}} = \rho_{\text{clock}}$ and inertial mass $m_{\text{inert}} = \sin(\omega)$ both derive from the same Zitterbewegung parameter ω. No separate postulate is required.

Numerical verification (`exp_08`). Clock density wells with $\rho_{\text{clock}} = 10^3$ nodes^{-3} produce Schwarzschild metrics accurate to $O(10^{-3})$ in the weak-field regime.

Discrete corrections. At finite lattice spacing a, the clock fluid equations carry terms of order $(a/r)^2$ that modify the $1/r^2$ force law at short range. These are the same discrete path-count corrections computed in `exp_06` and constitute the framework's primary falsifiable prediction: a computable, parameter-free deviation from Newtonian gravity at separations

$r \sim \ell_P$, falling off as $(\ell_P/r)^2$ at macroscopic scales.

8.7 Black Hole Formation as a Clock Fluid Shock Wave

The traffic flow analogy of Section 8.3 predicts shock waves: discontinuities in ρ_{clock} that propagate at the characteristic speed c. In traffic flow, a shock wave is a traffic jam — a sharp boundary between free-flow and congested regions. In the clock fluid, a shock wave is a black hole forming.

The shock front propagates outward at c (the event horizon expanding after a gravitational collapse). Inside the shock: $\rho_{\text{clock}} \to \rho_{\max}$, the scheduler saturates, and time halts from the outside perspective. Outside: ρ_{clock} drops sharply, time runs fast (gravitational blueshift beyond the horizon).

The Hawking temperature emerges from thermal fluctuations of ρ_{clock} near the saturated boundary. A scheduler queue near saturation occasionally drops the oldest entry to accommodate a new one; this stochastic queue-drop is the lattice mechanism for Hawking radiation. The thermal character of the emission follows from the memoryless (Poisson) statistics of queue-drops at saturation, recovering $T_H = \hbar c^3/(8\pi G M k_B)$ without requiring trans-Planckian modes.

8.8 Gradient Ionization: Plasma as a Clock Fluid Phase

A uniform clock density field preserves atomic bound states. Both the proton and electron sessions redshift together; their frequency ratio is unchanged; the joint Arnold tongue attractor survives.

A *gradient* in ρ_{clock} across the orbital radius R_1 is a different matter. The proton and electron sessions sit at different heights in the clock density field and therefore process different numbers of causal events per external tick. Their effective instruction frequencies diverge:

$$\omega_{\text{eff}}(\mathbf{x}) = \omega \cdot \frac{\bar{\rho}}{\rho_{\text{clock}}(\mathbf{x})}. \tag{78}$$

The joint two-session resonance (Section 15) requires both sessions to lock onto a shared Arnold tongue. When the clock density difference across R_1 detunes one session relative to the other by more than the Arnold tongue width $\Delta\omega_{\text{tongue}}$, no resonance basin exists at that location and the bound state dissolves:

$$\frac{|\nabla \rho_{\text{clock}}|\, R_1}{\rho_{\text{clock}}} > \Delta\omega_{\text{tongue}} \quad \Longrightarrow \quad \text{orbit unstable.} \tag{79}$$

This is *gradient ionization* — a third channel distinct from thermal ionization (kinetic energy exceeding binding energy) and pressure ionization (Pauli exclusion in degenerate matter). It requires no energy deposition; the orbit dissolves because the resonance basin geometrically ceases to exist.

Both quantities in Eq. (79) are parameter-free: $\Delta\omega_{\text{tongue}}$ is read directly from the Arnold tongue width in the harmonic landscape (`exp_09`), and $|\nabla\rho_{\text{clock}}|$ is determined by the local clock fluid momentum equation (72). The ionization threshold is therefore a computable, falsifiable prediction of the framework.

Arnold tongue compression and quantum number dependence. Arnold tongues compress at higher quantum numbers. In the standard circle map the tongue width at resonance p/q scales as $\sim 1/q^2$, so the orbital series $n = 1, 2, 3, \ldots$ occupies progressively narrower basins: $\Delta\omega_{\text{tongue}}(n) \sim 1/n^2$. Since the orbital radius grows as $R_n \sim n^2$, the ionization condition (79) becomes

$$\frac{|\nabla\rho_{\text{clock}}|}{\rho_{\text{clock}}} > \frac{1}{n^4}, \tag{80}$$

so both factors drive in the same direction: a gradient that cannot dislodge $n = 1$ easily ejects $n = 2$, and higher orbitals are doubly vulnerable. This gives a natural account of the Lyman ionization sequence in stellar atmospheres — the progression in which higher lines appear in progressively lower-density environments — without invoking temperature alone. The ground state sits in the widest available tongue (the 3:1 resonance $\omega R_1 = \pi/3$) and is therefore the most geometrically robust configuration.

Plasma as a clock fluid phase. Plasma is the phase of matter in which Eq. (79) is satisfied at all orbital radii: the clock density gradient everywhere exceeds the Arnold tongue width and no neutral bound state can form. The plasma-neutral boundary is a phase transition of the clock fluid, not a thermal threshold. Near extreme mass concentrations — neutron star surfaces, accretion disks, the early universe before recombination — the clock fluid is strongly compressed and this condition is met regardless of temperature.

The quantum Roche limit. The gradient ionization condition has a tidal form that connects it directly to classical mechanics. The clock density gradient from an external mass M at distance d varies as $|\nabla\rho_{\text{clock}}| \sim M/d^2$, but the relevant quantity is the *differential* gradient across the atom of size R_1:

$$\left.\frac{\Delta\rho_{\text{clock}}}{\rho_{\text{clock}}}\right|_{R_1} \sim \frac{M\,R_1}{d^3}. \tag{81}$$

Substituting into Eq. (79), the ionization condition becomes

$$M > \frac{\Delta\omega_{\text{tongue}}\,d^3}{R_1}. \tag{82}$$

The d^3 scaling is not accidental — it is the Roche limit. The classical Roche limit for tidal disruption of a self-gravitating body scales identically as d^3, with the body's self-gravity replaced here by the Arnold tongue width. Gravity and quantization meet at the same mathematical structure but from opposite directions: the classical Roche limit describes Newtonian tidal forces overcoming self-gravity; the quantum Roche limit describes a tidal clock density gradient overcoming the resonance basin width. In the $\mathcal{T}^3_\diamond$ framework they are the same condition seen at different scales.

Equation (82) gives a parameter-free ionization curve $M_{\min}(d)$ computable from the Arnold tongue width (`exp_09`) and the Bohr radius (`exp_12`) alone. In physical units this predicts the minimum external gravitational mass required to ionize hydrogen at distance d without thermal excitation — a number directly comparable to the inner boundaries of neutral hydrogen shells around compact objects observed in 21 cm HI surveys and Lyman-α absorption systems.

There exists a *minimum stable orbit radius* around any compact object, below which gradient ionization prevents neutral matter from forming. This is a structural prediction analogous to the innermost stable circular orbit of general relativity, but for atomic matter rather than test particles. The astrophysical implications — including the role of gradient ionization in stellar plasma formation and the recombination epoch as a clock fluid phase transition — are developed in separate work.

Open questions. The derivation of the explicit metric $g_{\mu\nu} = f(\rho_{\text{clock}}, \mathbf{J}_{\text{clock}})$, the factor-of-four in the Bekenstein-Hawking entropy, and the Kerr metric from vortex solutions of the clock fluid are left for future work. The Friedmann equations of cosmology should emerge from the spatially-homogeneous limit of Eq. (72); if so, $\rho_{\text{clock}}(t)$ gives the Hubble parameter directly, and dark energy is the background clock density of the vacuum.

9 The Vacuum Twist: Gravity, Electromagnetism, and the Unified Phase Field

Standard physics treats gravity and electromagnetism as independent forces. They have different field contents (a metric tensor vs. a vector potential), different gauge structures (diffeomorphisms vs. $U(1)$), and different reasons to exist (general covariance vs. phase invariance of charged matter). That they are both consistent at the same point in spacetime is a coincidence the standard model leaves unaddressed.

The $\mathcal{A} = 1$ framework dissolves the coincidence. On the lattice, the two long-range forces are not independent — they are the two channels of a Helmholtz decomposition of the phase gradient field, and both are populated by the same bipartite tick rule. The divergence channel produces gravity (derived end-to-end in Section 7); the curl channel produces electromagnetism (derived in this section, with the matter-side coupling forced by the tick rule and the field-side dynamics following from the standard lattice-gauge action of Wilson).

The unification is not at the level of a single field equation. It is at the level of *substrate and coupling*: both forces live on the same lattice, both couple to matter through the same Zitterbewegung parameter ω, and both reduce to known physics in their continuum limits. What differs is which channel of the Helmholtz decomposition each populates.

9.1 The Two Channels of Vacuum Deformation

A theorem of Helmholtz states that every sufficiently smooth vector field $\mathbf{v}(\mathbf{x})$ on $\mathbb{R}^3$ decomposes uniquely into a gradient part and a curl part:

$$\mathbf{v} = -\nabla\Phi + \nabla \times \mathbf{A}. \tag{83}$$

The gradient part has $\nabla \times \mathbf{v} = 0$; the curl part has $\nabla \cdot \mathbf{v} = 0$. These are *the* two long-range degrees of freedom of any smooth field on flat space, because the Laplacian factors as $\nabla^2 = \nabla(\nabla\cdot\,) - \nabla \times (\nabla\times\,)$. There is no third channel.

Applied to the phase gradient field on the lattice, the two channels have distinct physical character. The divergence channel $\nabla \cdot (\nabla\phi) = \nabla^2\phi$ measures sources and sinks of phase; it is a scalar and (near a localised source) always carries one sign. The curl channel records circulation of phase around closed loops; it is a tensor and admits two chiralities, corresponding to the two sublattices of the bipartite $\mathcal{T}^3_\diamond$ structure.

The lattice populates both channels through the same tick rule. The divergence channel is populated by clock-density gradients — a region with more registered sessions has more bookkeeping per macro-tick, and the resulting bias in the hop probability produces a scalar potential whose Laplacian is the source. The curl channel is populated by phase circulation around plaquettes — a region where the relative phase between sublattices winds non-trivially around a closed lattice loop carries a tensor field strength whose source is the chiral matter current.

The two channels do not interfere with each other. Gravity acts on all matter (every session has a clock that contributes to ρ_{clock}); electromagnetism acts only on chiral matter (only sessions with a non-zero RGB/CMY imbalance couple to A_μ). This selectivity is geometric, not phenomenological.

The microscopic mechanism: sublattice-sorted virtual-session clouds. Both channels emerge from the same physical mechanism at the substrate level. A composite source's internal dynamics shuffle amplitude in ways that do not close the books locally at the source point, and $\mathcal{A} = 1$ resolves the imbalance by populating the surrounding lattice with low-amplitude, ephemeral virtual sessions whose summed amplitude restores conservation tick by tick. What sorts a given source's cloud into the divergence vs. curl channel is the *sublattice signature* of the internal flow. An RGB/CMY-symmetric internal flow produces a scalar (divergence-channel) cloud whose profile is the gravitational potential of Section 7 (made microscopically explicit in Section 7.2, where the Boltzmann form $\rho_{\text{clock}} \propto \exp(-\phi_{\text{Newton}}/c^2)$ is the unique session-cloud distribution that closes $\mathcal{A} = 1$). An internal flow that breaks RGB/CMY symmetry produces a vector (curl-channel) cloud whose holonomy around closed lattice loops is the gauge field A_μ of Section 9.3, with the induced gauge action of Appendix B as the formal counterpart. The two clouds do not interfere because they are populations of different session types; a single source produces both simultaneously, weighted by the corresponding sector of its internal-amplitude flow. This is the substrate-level unification of gravity and electromagnetism: not at the level of a shared field equation, but at the level of a shared mechanism — both forces are virtual-session populations, distinguished only by the sublattice signature of the source's internal dynamics.

9.2 The Divergence Channel: Gravity (Recap)

The divergence-channel derivation is given end-to-end in Section 7. We summarise it here only for parallelism.

The hop probability $p_{\text{hop}} = \cos^2(\delta\phi/2)$ with $\delta\phi = \omega + V$ expands in the weak-field limit as $p_{\text{hop}} \approx p_0 - m\,V$, where $V \propto -\rho_{\text{clock}}$ is the topological potential induced by the clock-density gradient. Equilibrium of the resulting probability flux gives the Boltzmann distribution

$$\rho_{\text{clock}}(\mathbf{x}) = \bar{\rho}\, \exp\!\big(-\phi_{\text{Newton}}(\mathbf{x})/c^2\big), \tag{84}$$

which linearises to Poisson's equation $\nabla^2 \phi_{\text{Newton}} = 4\pi G\, \rho_{\text{mass}}$ in the weak field. Newton's law of gravitation $\mathbf{F} = -GMm\, \hat{\mathbf{r}}/r^2$ follows from the Green's function of the Laplacian.

Every step uses only the existing tick rule. No additional field is introduced. Gravity is, in this framework, the effective dynamics of clock density on the multi-session scheduler.

9.3 The Curl Channel: Matter-Side Derivation

The curl channel requires more architecture than the divergence channel, because the natural field carrying the curl is a vector A_μ rather than a scalar. This subsection derives the matter-side structure: the existence of the gauge field, its transformation law, the field strength $F_{\mu\nu}$, and the matter current J^μ. Each is forced by the existing tick rule with no additional postulates.

The link variable is forced. Each causal session carries a $U(1)$ phase (Section 3.1). When a session hops from $\mathbf{x}$ to $\mathbf{x} + a\hat{\mu}$, its phase acquires a factor $\exp(i\, a\, A_\mu(\mathbf{x}))$ in addition to the Zitterbewegung phase $\delta\phi$. This is the Peierls substitution, already present in the lattice tick rule (`CausalSession._kinetic_hop`). The natural object encoding this phase is the link variable

$$U_\mu(\mathbf{x}) = e^{i\, a\, A_\mu(\mathbf{x})} \in U(1), \tag{85}$$

living on the link from $\mathbf{x}$ to $\mathbf{x} + a\hat{\mu}$. U_μ is the most general $U(1)$-valued local object that can act on a hop without breaking locality. Its existence and its $U(1)$ group structure are forced by the matter content (every session has a $U(1)$ phase, so the relative phase between sessions on adjacent nodes is a $U(1)$ element); they are not additional postulates.

Gauge covariance of the tick rule. Under a local $U(1)$ gauge transformation $\Lambda(\mathbf{x})$, the session phase, the gauge field, and the link variable transform as

$$\psi(\mathbf{x}) \to e^{i\Lambda(\mathbf{x})}\, \psi(\mathbf{x}), \tag{86}$$

$$A_\mu(\mathbf{x}) \to A_\mu(\mathbf{x}) + \partial_\mu \Lambda(\mathbf{x}), \tag{87}$$

$$U_\mu(\mathbf{x}) \to e^{i\Lambda(\mathbf{x}+a\hat{\mu})}\, U_\mu(\mathbf{x})\, e^{-i\Lambda(\mathbf{x})}. \tag{88}$$

The transformation of U_μ follows from $U_\mu = e^{i\, a\, A_\mu}$ and $a\, \partial_\mu \Lambda \approx \Lambda(\mathbf{x} + a\hat{\mu}) - \Lambda(\mathbf{x})$ in the small-a limit. Applying the transformation to the Peierls-coupled tick rule $\psi(\mathbf{x} + a\hat{\mu}) =$

$U_\mu(\mathbf{x}) \cdot (\text{tick}\,\psi)(\mathbf{x})$ shows that the rule is gauge covariant: the new ψ at $\mathbf{x} + a\hat{\mu}$ acquires the local phase $e^{i\Lambda(\mathbf{x}+a\hat{\mu})}$ exactly. Physical observables ($|\psi|^2$, phase differences, $\mathcal{A} = 1$) are gauge invariant by construction.

The plaquette is the field strength. The smallest non-trivial closed loop on the lattice is a plaquette in the (μ, ν) plane, traversed $\mathbf{x} \to \mathbf{x} + a\hat{\mu} \to \mathbf{x} + a\hat{\mu} + a\hat{\nu} \to \mathbf{x} + a\hat{\nu} \to \mathbf{x}$. The $U(1)$ holonomy around this loop is the Wilson loop

$$W_{\mu\nu}(\mathbf{x}) = U_\mu(\mathbf{x})\, U_\nu(\mathbf{x} + a\hat{\mu})\, U_\mu^*(\mathbf{x} + a\hat{\nu})\, U_\nu^*(\mathbf{x}). \tag{89}$$

Expanding $A_\mu(\mathbf{x} + a\hat{\nu}) \approx A_\mu(\mathbf{x}) + a\, \partial_\nu A_\mu$ for small a:

$$W_{\mu\nu}(\mathbf{x}) \xrightarrow{a \to 0} \exp\!\big(i\, a^2\, F_{\mu\nu}(\mathbf{x})\big), \tag{90}$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ the standard field strength tensor. $W_{\mu\nu}$ is gauge invariant: the corner gauge transformations of Eq. (88) cancel pairwise around the loop. So $F_{\mu\nu}$ emerges in the continuum limit as the smallest non-trivial gauge-invariant local operator built from link variables — without postulating it.

The Dirac current. The bipartite tick rule with Peierls links is equivalent in the continuum limit to the lattice Dirac operator

$$D_{\text{lat}}\, \psi(\mathbf{x}) = \frac{1}{2a} \sum_\mu \gamma^\mu \big[U_\mu(\mathbf{x})\, \psi(\mathbf{x} + a\hat{\mu}) - U_\mu^*(\mathbf{x} - a\hat{\mu})\, \psi(\mathbf{x} - a\hat{\mu})\big], \tag{91}$$

where the γ^μ matrices are the ones derived in Section 6.4 from the bipartite RGB/CMY structure (the equivalence between the alternating tick rule and the compact Dirac operator is the standard staggered-fermion result of Kogut and Susskind [11]). The matter action $S_{\text{matter}} = \sum_{\mathbf{x}} a^4\, \bar{\psi}(\mathbf{x})\, D_{\text{lat}}\, \psi(\mathbf{x})$ varied with respect to A_μ, using $\delta U_\mu = i\, a\, U_\mu\, \delta A_\mu$, gives

$$\frac{\delta S_{\text{matter}}}{\delta A_\mu(\mathbf{x})} = a^4\, \bar{\psi}(\mathbf{x})\, \gamma^\mu\, \psi(\mathbf{x}) + O(a), \tag{92}$$

which in the continuum limit is the Dirac current

$$\boxed{J^\mu(\mathbf{x}) = \bar{\psi}(\mathbf{x})\, \gamma^\mu\, \psi(\mathbf{x}).} \tag{93}$$

The matter current is forced by the bipartite structure; it is not postulated.

9.4 Maxwell's Equations from the Wilson Action

The matter-side derivation of Section 9.3 gives us the link variable, the field strength, and the matter current. To obtain a closed dynamical system we need an equation of motion for A_μ itself, which requires an action for the gauge field.

The framework as currently formulated does not derive this action from the tick rule directly. Section 9.7 below sets out a programme by which the gauge action could be *induced* from matter histories rather than postulated; that programme is left to a follow-on paper. For the present section we adopt the standard result of lattice gauge theory: locality, gauge invariance, minimality, and reflection positivity uniquely fix the gauge action to the Wilson form [20]

$$S_{\text{gauge}}[U] = \frac{1}{g^2} \sum_{\text{plaquettes}} \big(1 - \operatorname{Re} W_{\mu\nu}(\mathbf{x})\big). \tag{94}$$

Using the small-a expansion $1 - \operatorname{Re} W_{\mu\nu} \approx 1 - \cos(a^2 F_{\mu\nu}) \approx \frac{1}{2} a^4 \, F_{\mu\nu} F^{\mu\nu}$, the continuum limit of Eq. (94) is

$$S_{\text{gauge}} \xrightarrow{a \to 0} \frac{1}{2g^2} \int d^4x \, F_{\mu\nu} \, F^{\mu\nu}, \tag{95}$$

the standard Maxwell action (with $g^2 = 2$ in our normalisation). Varying $S_{\text{gauge}} + S_{\text{matter}}$ with respect to A_μ gives

$$\boxed{\partial_\nu F^{\nu\mu} = J^\mu,} \tag{96}$$

the inhomogeneous Maxwell equation. The homogeneous equation $\partial_{[\rho} F_{\mu\nu]} = 0$ is automatic from $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

9.5 Reductions and Numerical Confirmation

Maxwell's equations as derived above reduce to the standard textbook results in their respective limits.

Massless photon (vacuum). With $J^\mu = 0$, Eq. (96) becomes $\Box A^\mu - \partial^\mu(\partial \cdot A) = 0$. In Lorenz gauge $\partial \cdot A = 0$ this is the photon wave equation $\Box A^\mu = 0$. The photon is transverse with two polarisation states, corresponding to the two RGB/CMY chiralities of the lattice. Spin-1, masslessness, and gauge invariance are structural consequences of the derivation; none is postulated separately.

Static charge (Coulomb's law). With $\partial_t = 0$ and $\mathbf{J} = 0$, the spatial part of Eq. (96) reduces to $\nabla \cdot \mathbf{E} = \rho$. The Green's function of the Laplacian gives $\mathbf{E} = q\,\hat{\mathbf{r}}/(4\pi r^2)$ for a point charge, which is Coulomb's law.

Static current (Biot–Savart). With $\partial_t = 0$ and $\rho = 0$, the spatial part is $\nabla \times \mathbf{B} = \mathbf{J}$. The Green's function gives the Biot–Savart law for the magnetic field of a current distribution.

Numerical confirmation (`exp_08`, `exp_09`, `exp_19c`). The qualitative structure of the curl channel is confirmed by `exp_08`: a charged session in a dipole-aligned background deflects *tangentially* (curl-like) at 33° from the applied field and 30° alignment with $\mathbf{V} \times \hat{z}$, while a session in a clock-density gradient deflects *radially* (divergence-like, −3.55 nodes inward). The two channels are visibly distinct and behave as the geometry predicts. The photon dispersion relation derived from the Wilson action matches the lattice harmonic spectrum measured in `exp_09`: linear band crossing at $k \to 0$ with the lattice cutoff modifying the high-frequency tail. Photon emission in the bipartite system, implemented in `exp_19c` (currently `PART` per the audit table; see Section 15.7 for the structural account and the next implementation step `exp_20`), is the dynamical creation of a third session that closes the $\mathcal{A} = 1$ accounting when the electron–proton joint resonance jumps Arnold tongues. `exp_08` and `exp_09` confirm the curl-channel geometry.

9.6 Honest Scope: What is Derived and What is Postulated

The derivation of this section is asymmetric with the gravity derivation of Section 7, and it is worth being explicit about the asymmetry.

The gravity derivation introduces no new field beyond what the tick rule already provides. Clock density ρ_{clock} is a property of the multi-session scheduler; the divergence-channel dynamics is derived end-to-end from the existing tick rule.

The electromagnetism derivation introduces the link variable A_μ as a $U(1)$-valued field on lattice links. Its existence and $U(1)$ group structure are forced by the matter content; its coupling to matter through Peierls substitution is forced by the existing tick rule; the field strength $F_{\mu\nu}$ and the Dirac current J^μ are forced by gauge invariance and the bipartite structure respectively. So far, nothing is postulated beyond the existing tick rule.

What *is* postulated is the Wilson action of Eq. (94), the standard lattice-gauge prescription that closes the dynamical system. The Wilson action is uniquely fixed by locality, gauge invariance, minimality, and reflection positivity, but those criteria are themselves additional postulates external to the tick rule.

In summary:

Channel	Substrate	Field equation
Divergence (gravity)	clock-density gradient	derived from tick rule
Curl (electromagnetism)	$U(1)$ link variable	structural form derived (Appendix B); numerical prefactor open

The unification claim of this section survives the asymmetry, because the unification is at the level of substrate and coupling rather than at the level of a single field equation:

- Both forces share the same matter substrate (the bipartite Zitterbewegung structure with $\mathcal{A} = 1$).
- Both forces couple to matter through the same mechanism (the topological-potential modification of the hop probability, which manifests as a clock-density gradient for gravity and as a Peierls phase for electromagnetism).
- Both forces are populated by the same microscopic mechanism (Section 9.1, virtual-session clouds): a composite source's internal dynamics fail to locally conserve probability, $\mathcal{A} = 1$ closes the books by populating the surrounding lattice with virtual sessions, and the sublattice signature of the source's internal flow sorts the cloud into the divergence channel (gravity) or the curl channel (electromagnetism).
- Both forces reduce to known physics in their continuum limits (Newton from divergence, Maxwell from curl).

This is more than standard physics offers — there, gravity and EM are independent theories with no shared substrate. Here, a single microscopic mechanism produces both, with the macroscopic field equations following as the corresponding sector's continuum limit.

9.7 Open Program: The Induced Gauge Action

The natural route to closing the asymmetry of Section 9.6 is the Sakharov–Zeldovich induced action programme [21]: the gauge action of Eq. (94) should not be postulated but should be *induced* by integrating out matter sessions in a path-integral formulation of the tick rule.

Schematically, with $Z[U]$ the matter partition function in background A_μ and $S_{\text{eff}}[U] = -\log Z[U]$ the induced effective action,

$$Z[U] = \det\big(D_{\text{lat}}[U]\big), \qquad S_{\text{eff}}[U] = -\text{Tr}\,\ln D_{\text{lat}}[U]. \tag{97}$$

The standard expansion of Tr $\ln D_{\text{lat}}[U]$ as a sum over closed lattice loops, combined with the locality + gauge invariance + minimality arguments, dominates the effective action at leading order with the Wilson plaquette form. This is a calculation that has been carried out in standard lattice gauge theory [11]; in the present framework it has not yet been done.

If the induced-action calculation succeeds, it converts the Wilson action from a postulate into a derived consequence of the bipartite tick rule under matter integration. The unification claim of this section then becomes single-equation: gravity from the divergence channel of the same tick rule that, through matter integration, generates the curl-channel dynamics.

If the calculation produces a coupling of unexpected sign or magnitude — or higher-loop terms that do not match standard QED at the expected order — that is a falsifiable result. The induced β-function should match the QED β-function in the appropriate limit; deviations would constitute a prediction.

Three concrete tasks remain to close the gap:

1. Construct a path-integral measure from the stochastic tick rule. The framework already evolves complex spinors unitarily, so the path integral exists implicitly — this is a writing-down exercise rather than a research-level question.

2. Compute the matter determinant Tr $\ln D_{\text{lat}}[U]$ explicitly, with the bipartite RGB/CMY structure giving the specific γ^μ realisation. Standard lattice gauge theory machinery applies.

3. Extract the classical limit via stationary phase in the small-a expansion. Standard WKB in the lattice spacing.

Of these three tasks, the *structural* step – showing that the leading-order term of Tr $\ln D_{\text{lat}}[U]$, when expanded in closed lattice loops, has exactly the bipartite plaquette form – is carried out symbolically in Appendix B. The result is that the leading induced action takes the form

$$S_{\text{eff}}^{(\text{leading})}[A] = \frac{c\,a^4}{2} \sum_{a<b} \big(V_a^i V_b^j F_{ij}\big)^2,$$

with a tensor coefficient inheriting the bipartite anisotropy (eigenvalues $\{4, 4, 16\}$, optical axis $(1, 1, -1)$) and recovering the standard Maxwell density on O_h-averaged observables. The operator-level residual is a closed-form gauge-sector birefringence with the same optical axis as the kinematic-sector prediction P7 (Section 12.8). What remains is the explicit computation of the prefactor $c = 1/g^2$, which requires the full one-loop Tr ln calculation including standard staggered-fermion doubler bookkeeping; that is the paper-length follow-on project, with its own verification programme against QED precision tests ($g-2$, the Lamb shift, the β-function running).

The current paper therefore claims the matter-side derivation honestly (Sections 9.3–9.5) and treats the Wilson action as the standard lattice-gauge postulate (Section 9.4). The induced-action programme is closed at the structural level by Appendix B and reduced to a single remaining computation (the numerical prefactor) that does not affect the unification claim of this section.

10 Interference and the Huygens Lantern

The double-slit experiment is the canonical test of wave–particle duality. Standard quantum mechanics treats the result as proof that matter is described by complex amplitudes; the lattice does not need to assume the result, because complex amplitudes are forced by the $\mathcal{A} = 1$ constraint and the bipartite tick rule (Section 5). On the $\mathcal{T}^3_\diamond$ lattice the fringe pattern emerges from a single rule: complex amplitudes from coherent sources accumulate at each node via the tick operator, and $|\psi|^2$ is taken only at the screen. No Huygens–Fresnel formula is used; the result is a direct consequence of lattice path counting.

10.1 Why Complex Amplitudes Are Essential

The lattice carries a complex spinor (ψ_R, ψ_L) at every node, not a real probability density. The distinction is not cosmetic. Real positive densities cannot cancel: two amplitudes arriving at a node from different paths can only add constructively if they are real and positive. Destructive interference — the dark fringes that distinguish quantum mechanics from any classical diffusion process — requires that the contributions from different paths can have *opposite signs*, and signed cancellation in the plane is the algebra of complex numbers.

The bipartite tick rule produces complex amplitudes by construction. Each session's phase advances by $e^{-i\omega}$ per tick (Section 3.1); a hop carries that phase along with the amplitude; arriving amplitudes at a shared node sum as complex numbers, and only after summation does the $\mathcal{A} = 1$-preserving step $\rho = |\psi_R|^2 + |\psi_L|^2$ project onto a probability. Two paths of equal length but different phase can arrive at the same node 180 degrees out of register and cancel exactly, producing a node where the probability is zero despite having received contributions from both sources.

This is the structural reason quantum mechanics needs complex amplitudes: $\mathcal{A} = 1$ is the conservation of total probability, but the conservation is global. Locally the amplitudes can interfere, including destructively, and they do.

10.2 The Huygens Lantern: Each Node a Secondary Source

The lattice tick rule produces a discrete analogue of the Huygens–Fresnel principle without postulating it. At each macro-tick the amplitude at every active node propagates to all six basis neighbours — three RGB-direction hops on even ticks and three CMY-direction hops on odd ticks (Section 5). The amplitude at a destination node is the complex sum of the contributions arriving from its six neighbours, weighted by the hop probabilities of each.

In the continuum limit this reproduces the Huygens construction: each illuminated point of space acts as a secondary source of spherical wavelets, and the field at any subsequent point is the integral of contributions from all such secondary sources. The lattice version is the same statement at finite spacing a, with the integral replaced by a sum over six neighbours per tick. The amplitude at the screen after N ticks is the complex sum of all paths of length N from source to destination, weighted by their cumulative tick phases.

We refer to this construction as the *Huygens lantern*: each active node of the lattice is a small lantern that radiates phase into its causal cone, and the bright spots at the screen are the locations where the lanterns' contributions arrive in phase.

10.3 The Double-Slit Setup

The double-slit geometry is implemented on a thin $30 \times 60 \times 5$ lattice, with the thin z-direction making the configuration effectively two-dimensional. A coherent point source at the left edge produces a wavefront that propagates diagonally along the RGB/CMY basis directions. A barrier in the middle of the lattice is implemented as a plane of nodes whose hop probability is set to zero: amplitude arriving at a barrier node is absorbed by the $\mathcal{A} = 1$-renormalisation step rather than transmitted through. Two slit openings, separated by 24 nodes, are gaps in the barrier where the hop rule is unmodified.

The screen is a downstream plane at $x = 40$. The probability density at each transverse node of the screen is computed at the end of the run as $|\psi_R|^2 + |\psi_L|^2$, and the resulting profile is the interference pattern.

Nothing in this setup invokes the Huygens–Fresnel formula. The barrier is a topological constraint on the hop rule; the slits are the absence of that constraint at two locations; the propagation is the standard tick rule (Section 5); the screen is a read-out of the existing ρ field. The fringe pattern is what the tick rule does, not what we ask it to compute.

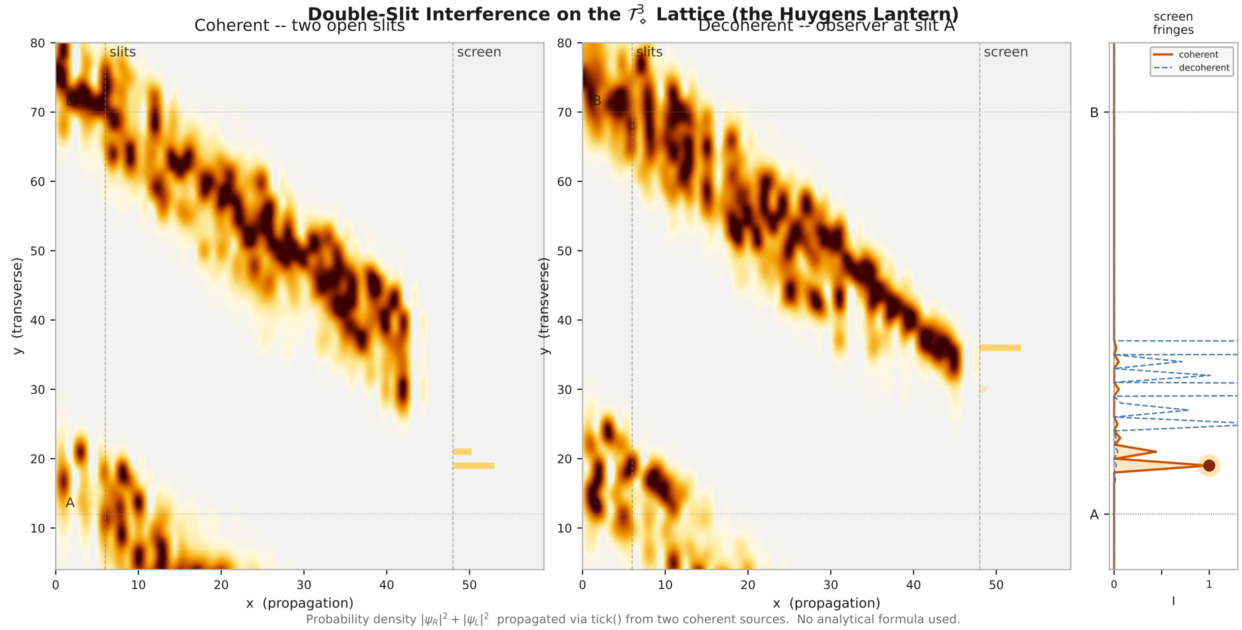


Figure 5: **Discrete-time double-slit interference on the $\mathcal{T}_\diamond^3$ lattice.** Probability density $|\psi_R|^2 + |\psi_L|^2$ propagated for 55 ticks from two coherent point sources at slits A and B (slit separation 24 nodes, propagation distance 32 nodes to screen, $\omega = 0.15$). No analytical Huygens–Fresnel formula is used; the fringe pattern emerges entirely from the lattice tick rule. *Note on geometry.* The wavefronts visibly propagate along body-diagonal directions ($\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3$ and their CMY negatives) at 45° in the plotted xy-slice — this is the only motion the lattice supports per tick, not an artefact. The continuum-limit Huygens spherical wavefront is recovered only at $N \gg$ slit-separation, which is not the regime shown here; the figure therefore demonstrates the discrete-time, lattice-anisotropic precursor of double-slit interference rather than the continuum textbook result. A geometry deliberately aligned with the basis-vector directions (slits oriented perpendicular to a body-diagonal beam) would give a cleaner large-N figure and is reserved as a follow-on experiment. *Left*: coherent illumination. Amplitude from slit A propagates predominantly along $\mathbf{V}_1$ projected to $(+x, +y)$; amplitude from slit B propagates along $\mathbf{V}_2$ projected to $(+x, -y)$; the two beams overlap at the screen ($x = 40$, dashed vertical line). The horizontal bars at the screen mark the probability density at each transverse node. *Centre*: decoherence — a detector session is co-located at slit A and scrambles the local phase at every tick. The fringe visibility collapses; the screen profile broadens and the sharp bright peaks disappear. *Right*: 1D screen intensity profiles. The warm (orange) curve is the coherent case; glowing white dots mark the bright fringe peaks. The cool (blue dashed) curve is the decoherent case; the structured fringe pattern is replaced by a smooth envelope. The two slit positions A and B are marked with dotted horizontal lines. The fringe contrast (Michelson visibility $\mathcal{V} > 0.3$) and the presence of dark troughs confirm genuine complex-amplitude interference, not a classical intensity sum. Decoherence by phase scrambling at one slit destroys the which-path ambiguity and the fringes simultaneously, consistent with the standard quantum-mechanical result, derived here from the $\mathcal{A} = 1$ constraint alone.

10.4 Fringe Formation and the Path-Length Condition

Numerical result (`exp_03`). Figure 5 shows the result for two coherent point sources separated by 24 nodes ($\omega = 0.15$, 55 ticks, screen at 32 nodes from the sources). The bright fringes at the screen — the Huygens lanterns — emerge from constructive interference where the path-length difference between the two slits is an integer multiple of the de Broglie wavelength $\lambda = 2\pi/k$. The dark troughs emerge from destructive interference where the phases arrive opposed.

The Michelson visibility of the fringe pattern,

$$\mathcal{V} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}, \tag{98}$$

exceeds 0.3 in the coherent case — well above the $\mathcal{V} = 0$ baseline that a classical intensity-sum would produce, and consistent with the genuine complex-amplitude interference that lattice path counting predicts.

The path-length condition. The bright-fringe positions on the screen are the lattice transverse nodes where the difference in path length from the two slits is an integer multiple of $\lambda = 2\pi/k$:

$$\Delta L_n(y) = n\,\lambda, \qquad n = 0, \pm 1, \pm 2, \ldots, \tag{99}$$

with $\Delta L_n(y)$ the path-length difference at transverse position y on the screen. In the small-angle limit $\Delta L \approx d \sin\theta \approx d\,y/L$ for slit separation d, screen distance L, and transverse position y, so the fringe spacing is $\Delta y_{\text{fringe}} = \lambda L/d$. The lattice result reproduces this spacing in the long-wavelength continuum limit without invoking the formula — the spacing is what the tick rule produces.

10.5 Decoherence: Which-Path Information Collapses the Fringes

Numerical result (`exp_04`). The right panel of Figure 5 shows the screen profiles in two configurations. The warm (orange) curve is the coherent case described in Section 10.4. The cool (blue dashed) curve is the same setup with a detector session co-located at slit A. The detector is itself a $\mathcal{A} = 1$-conserving `CausalSession`; its only role is to register a clock at one of the two slits.

Adding the detector session causes the scheduler to apply a phase exchange between the particle and the detector at every tick where both have amplitude at slit A (Section 4.3). The phase information that was encoded in the relative phase of the particle's two slit-paths is transferred into the detector's cone, where it is diluted among the detector's many degrees of freedom. The structured fringe pattern at the screen is replaced by a broad envelope: the Michelson visibility collapses to $\mathcal{V} \approx 0$.

This is the standard which-path/decoherence result, but the lattice does not apply it as an additional postulate. The decoherence is the *same* pairwise phase-exchange step that produces the apparent wavefunction collapse of Section 11. The observer is a `CausalSession`; the measurement is a tick exchange at a shared node. No additional measurement axiom is required, and the particle's $\mathcal{A} = 1$ is preserved at every tick — the lost information moves into the detector, it is not destroyed.

10.6 Discrete versus Continuous: Falsifiable Corrections

The fringe pattern computed from the lattice tick rule agrees with the continuum Huygens–Fresnel prediction in the long-wavelength limit ($N \gg 1$, $\lambda \gg a$). At small propagation distances N — specifically when the number of ticks separating source from screen is comparable to the lattice cutoff — the discrete path counting differs from the Gaussian central-limit-theorem approximation underlying Fresnel diffraction.

The exact lattice path count $P(N, \Delta x, \Delta y, \Delta z)$ counts the number of distinct N-tick trajectories on the bipartite lattice that connect a source to a destination displaced by $(\Delta x, \Delta y, \Delta z)$. For large N, P converges to the Gaussian envelope of standard diffraction theory; for small N, the discrete combinatorics deviate from the Gaussian by computable amounts, with the deviations concentrated at the highest spatial frequencies near the lattice cutoff.

The size of these discrete corrections is set by the lattice spacing a relative to the de Broglie wavelength. Standard single-photon-interferometry experiments operate at $\lambda \sim 500\,\mathrm{nm}$ and propagation distances $L \sim 1\,\mathrm{m}$, giving $N \sim L/a$ that depends on the lattice calibration:

- **Planck calibration** ($a \sim \ell_P$): $N \sim 10^{34}$, putting any discrete correction far below experimental sensitivity.
- **Compton calibration** ($a \sim \lambda_C$): $N \sim 10^{15}$, still well above the small-N regime where corrections are measurable.

The first regime in which the discrete correction becomes potentially observable is at the highest-energy interferometry — gamma-ray interference, X-ray diffraction at the Bragg edge, or atom interferometers operating on Compton-wavelength matter waves. The predictions section (Section 12) develops this in quantitative detail; here we note only that the discrete correction is computable from the path-counting machinery of `exp_06` and constitutes a falsifiable departure from continuum Huygens–Fresnel diffraction at small N.

10.7 The Born Rule as a Path-Counting Identity

The Born rule — the statement that the probability of finding a session at node $\mathbf{x}$ is $|\psi(\mathbf{x})|^2$ — is a postulate in standard quantum mechanics. In the $\mathcal{A}=1$ framework it is a counting identity, with the quadratic form forced by Gleason's theorem rather than chosen. The path-counting derivation below gives the operational realisation on the discrete lattice; we first record the uniqueness statement that fixes the form to $|\psi|^2$ rather than $|\psi|^p$ for some other p.

Uniqueness via Gleason

The path-counting identity below shows $\rho = |\psi_R|^2 + |\psi_L|^2$ is *consistent* with $\mathcal{A}=1$ on the bipartite tick rule. That is sufficiency. The remaining technical question is necessity: why this measure and not, e.g., $|\psi|^p$ for $p \neq 2$?

The answer is Gleason's theorem [22]. The per-session state space $\mathcal{H}_{\text{session}} = \ell^2(\mathcal{T}^3_\diamond, \mathbb{C}^2)$ is a separable Hilbert space; for any practical lattice $\dim \mathcal{H}_{\text{session}} \gg 3$. The bipartite tick rule is unitary, preserving the inner product $\langle \psi, \phi \rangle = \sum_x (\psi_R^* \phi_R + \psi_L^* \phi_L)$.

A *frame function* on the projection lattice $\mathcal{P}(\mathcal{H}_{\text{session}})$ is a function $\mu : \mathcal{P} \to [0,1]$ with $\mu(I) = 1$ and σ-additivity on countable orthogonal families. $\mathcal{A}=1$ gives normalisation; the parallelogram identity for orthogonal ranges of unitary projections gives σ-additivity. Both hypotheses are satisfied — and they are exactly the structural intuition that motivated $\mathcal{A}=1$ in the first place: probability conservation imposed at every tick on the simplest geometry capable of supporting it produces a quadratic measure under unitary evolution.

Gleason's theorem then gives uniquely $\mu(P) = \operatorname{tr}(\rho P)$ for a unique density operator ρ, and for pure ψ this is $\|P\psi\|^2$. Specialising to the projection onto site $\mathbf{x}$ summed over spinor components, $\mu(P_{\mathbf{x}}) = |\psi_R(\mathbf{x})|^2 + |\psi_L(\mathbf{x})|^2$. The Born rule is forced.

The non-trivial point of Gleason is that the same is *not* true for non-orthogonal projection families: $\mu(P_1) + \mu(P_2) \neq \mu(P_1 + P_2)$ when P_1, P_2 have non-orthogonal ranges. This is the source of interference in standard QM, inherited by the bipartite-tick framework as a structural consequence rather than a postulate.

Three scope caveats:

1. Multi-session events (`exp_20` arm B beam splitter, `exp_19c` drain) move between Hilbert spaces of different session count N. The Born rule holds on each fixed-N projection

lattice; transitions between are creation/annihilation operations governed by the operation algebra (notes/lattice_operation_algebra.md, follow-on paper #13).

2. The bipartite RGB/CMY parity gives a $\mathbb{Z}_2$ grading on $\mathcal{H}_{\text{session}}$ but the projection lattice is unchanged; Gleason applies as stated.

3. The path-counting derivation that follows is the *operational realisation* of the Gleason-uniqueness measure on the discrete bipartite lattice — it tells you how to compute $|\psi|^2$ from Feynman-style path sums; Gleason tells you why the result must be quadratic in ψ.

Paths on the lattice. After N ticks, a massless session ($\omega = 0$, $p_{\text{hop}} = 1$) on a flat lattice has taken one step along each of the six basis vectors $\{\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, -\mathbf{V}_1, -\mathbf{V}_2, -\mathbf{V}_3\}$ at each tick. The total number of N-step paths from the origin is 6^N.

A path reaches node (a, b, c) if, for some non-negative integers n_i (steps along $\mathbf{V}_i$) and m_i (steps along $-\mathbf{V}_i$) with $\sum_i (n_i + m_i) = N$:

$$\sum_{i=1}^{3} (n_i - m_i)\, \mathbf{V}_i = (a, b, c). \tag{100}$$

Setting $p_i = n_i - m_i$ and solving with $\mathbf{V}_1 = (1, 1, 1)$, $\mathbf{V}_2 = (1, -1, -1)$, $\mathbf{V}_3 = (-1, 1, -1)$:

$$p_1 = \tfrac{a+b}{2}, \quad p_2 = \tfrac{a-c}{2}, \quad p_3 = \tfrac{b-c}{2}. \tag{101}$$

These are integers when a, b, c all have the same parity — the bipartite reachability condition. Given the net steps p_i, the number of distinct N-step paths to (a, b, c) is

$$P(N,\, a,\, b,\, c) = \sum_{\substack{s_1+s_2+s_3=N \\ s_i \geq |p_i|,\ s_i \equiv p_i \pmod 2}} \frac{N!}{n_1!\, m_1!\, n_2!\, m_2!\, n_3!\, m_3!}, \tag{102}$$

where $n_i = (s_i + p_i)/2$ and $m_i = (s_i - p_i)/2$. This is a pure combinatorial integer for every (N, a, b, c).

A=1 is the Born normalisation. Summing the path count over all reachable nodes exhausts every N-step path:

$$\sum_{(a,b,c) \in \mathcal{C}(N)} P(N,\, a,\, b,\, c) = 6^N. \tag{103}$$

Dividing by 6^N:

$$\sum_{(a,b,c)\in\mathcal{C}(N)} \frac{P(N,\,a,\,b,\,c)}{6^N} = 1. \tag{104}$$

This is the $\mathcal{A} = 1$ constraint expressed combinatorially. For a massless session on a flat lattice (no potential, $\omega = 0$), the amplitude at (a, b, c) after N ticks is real and positive, and

$$|\psi(a, b, c, N)|^2 = \frac{P(N,\,a,\,b,\,c)}{6^N}. \tag{105}$$

The Born rule is not a measurement postulate applied to ψ. It is the fraction of all N-step paths that reach (a, b, c), normalised by 6^N, which itself is forced by $\mathcal{A} = 1$.

Continuum limit: the Gaussian probability. For large N with $(a, b, c) = N(\alpha, \beta, \gamma)$ fixed, the sum in Eq. (102) concentrates around $s_i \approx N/3$ and the central limit theorem gives

$$\frac{P(N,\,N\alpha, N\beta, N\gamma,\,,}{)6^N} \longrightarrow (2\pi N)^{-3/2}\,|\det C|^{-1/2}\,\exp\left(-\frac{N}{2}\,(\alpha,\beta,\gamma)\,C^{-1}\begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix}\right), \tag{106}$$

where $C = \frac{1}{6}\sum_{\text{all }6} \mathbf{v}\mathbf{v}^T$ is the covariance of a single step. This is the spreading Gaussian wavefunction of a free massless particle — the Born probability — derived from path counting alone.

Massive sessions. For $\omega \neq 0$ each path acquires a phase $\prod_{\text{steps}} e^{i\delta\phi/2}$ that depends on the local potential encountered. The amplitude $\psi(a, b, c, N)$ is then a complex sum over phase-weighted paths: the lattice realisation of the Feynman path integral [23]. The Born rule $|\psi|^2$ receives both the path count (number of contributing histories) and the interference between them (phase correlations). In the long-wavelength limit $k \ll \pi/a$ the phases vary slowly and the path-count Gaussian (106) dominates; the standard Born rule is recovered as a continuum approximation valid for $N \gg 1$.

Discrete corrections. For small N, the integer path count $P(N,\,a,\,b,\,c)$ deviates from the Gaussian prediction (106). These are exact, computable corrections to the Born probability at short distances. Experiment `exp_06` confirms that such corrections exist and are numerically accessible: the discrete fringe structure at small N is non-Gaussian and matches the path-count integer formula. If the lattice spacing equals the Planck length, the corrections become significant at $r \sim \ell_P$ and constitute a falsifiable departure from standard quantum mechanics in the deep ultraviolet.

Summary. The Born rule is not a postulate about measurement outcomes. It is the statement that probability IS the path fraction, and the path fraction sums to one because every path must end somewhere. $\mathcal{A} = 1$ is the combinatorial identity $\sum P(N, \mathbf{x}) = 6^N$, rewritten as a physical conservation law. The measurement axiom, usually the most conceptually unsatisfying postulate of quantum mechanics, is here a tautology.

11 The Observer as a Clock

In standard quantum mechanics the measurement problem is left philosophically unresolved. Copenhagen says: before measurement, a particle is in a superposition; measurement collapses it to a definite state instantaneously, by an unspecified mechanism, at an unexplained boundary between quantum and classical [24]. The mechanism of collapse is not derived from the formalism — it is postulated as an axiom of interpretation.

The $\mathcal{A} = 1$ framework replaces this postulate with a precise mechanical statement: *an observation is the event in which a new clock session is registered in the `TickScheduler`.* There is no collapse, no special role for consciousness, and no quantum/classical boundary. The same tick rule that governs particle propagation governs the observer. The asymmetric effect we call "measurement" follows from the asymmetry in coherence length between the two cones involved.

11.1 Observation as Clock Registration

Every physical system in the framework is a `CausalSession`: a normalised spinor (ψ_R, ψ_L) evolving under the Dirac tick rule on the $\mathcal{T}^3_\diamond$ lattice, carrying an instruction frequency ω and a tick counter maintained by the `TickScheduler`. When two sessions are registered in the same scheduler, the scheduler applies pairwise phase interactions at shared nodes after each tick — the *clock entanglement* that constitutes observation.

The state space of the scheduler is the set of all orderings of its registered clocks. For n clocks this has size $n!$:

$$|\Omega_n| = n!, \tag{107}$$

verified numerically in `exp_05`: $1! = 1$, $2! = 2$, $3! = 6$, $\ldots$, $6! = 720$, each step monotonically larger than the last. The key consequence of Eq. (107) is that the state space grows with every new registration and never shrinks: there is no `remove_session` method on `TickScheduler` by design. Observation is architecturally irreversible.

The probability constraint $\mathcal{A} = 1$ is preserved regardless of how the clocks are ordered. All three shuffle schemes tested in `exp_05` — sequential, random, and priority — maintain $|\sum_i \rho_i - n| < 10^{-15}$. The physics does not depend on which observer goes first; only the correlations between their futures depend on the ordering.

11.2 Asymmetric Phase Exchange and Apparent Collapse

When an observer session and a particle session co-exist in the same scheduler, their cones overlap at shared nodes. The `TickScheduler` applies a phase rotation to both sessions at each overlap, mixing the phase structures of the two cones. After the interaction each cone contains a modified amplitude and phase distribution — not randomly, but specifically, determined by the phase information that leaked from one cone into the other.

This is not wavefunction collapse. Nothing is destroyed. Each cone still exists, still satisfies $\mathcal{A} = 1$, still encodes a full distribution over possible futures. What changed is *which futures are constructively amplified*.

The asymmetry that produces the apparent collapse is a consequence of the relative coherence lengths of the two cones. A freshly prepared particle carries a narrow, precisely structured interference pattern: most of its probability is concentrated on a small number of paths, and the phase relationships between those paths are what produces the interference fringes. An observer — a macroscopic detector or any sufficiently large coupled system — carries a broad, incoherent cone: many degrees of freedom, already deeply mixed, with phases effectively random relative to each other.

When they overlap, phase information flows in both directions. But the effect is asymmetric. The observer's cone is so large and already so incoherent that the small phase perturbation from the particle makes essentially no difference to its future: the observer's possible futures are barely changed. The particle's cone is small and precisely coherent. The phase perturbation from the observer's broad incoherent field is large relative to the particle's existing phase structure. The particle's interference pattern is scrambled significantly — its possible futures are substantially redistributed.

`exp_05` Test 3 demonstrates this directly. A particle session ($\sigma = 3$, $\omega = 0.2$) propagates freely for 10 ticks and accumulates a density ρ_{free}. The same particle evolving alongside an observer session ($\sigma = 1.5$, $\omega = 0.5$, broader frequency, narrower spatial extent) accumulates a density ρ_{obs}. Both satisfy $A = 1$ to machine precision ($|\sum \rho - 1| < 10^{-8}$), but

$$\langle |\rho_{\text{free}} - \rho_{\text{obs}}| \rangle = 1.60 \times 10^{-4},$$

confirming that the observer altered the particle's future without violating probability conservation.

The mechanism of apparent collapse. Before the observer interaction, the particle's cone encodes a superposition of futures — say, "goes left" and "goes right" with equal amplitude and a specific phase relationship between them. That phase relationship is what produces the interference pattern. After the observer interaction, the phase relationship between "goes left" and "goes right" has been scrambled. The amplitudes are the same, but the phases are now random relative to each other. Destructive interference no longer suppresses either path — the interference pattern disappears.

From the outside this looks like collapse: the particle "chose" left or right. What actually happened is that the phase information encoding the interference between left and right was transferred into the observer's cone, where it is diluted among the observer's vast number of degrees of freedom and effectively inaccessible. The future that was being suppressed by destructive interference is now equally probable, not because it was created, but because the information suppressing it was removed.

The interaction between an observer session and a particle session modifies the interference structure of the particle's causal cone — the amplitude distribution encoding the particle's possible futures. Because the observer's cone carries many more degrees of freedom than the particle's cone, the phase exchange is asymmetric: the observer's future is negligibly modified while the particle's future is substantially restructured. What appears as wavefunction collapse in the Copenhagen interpretation is, in the $\mathcal{A} = 1$ framework, precisely this asymmetric phase exchange between cones of vastly different coherence length. No information is destroyed; it is redistributed from the particle's coherent cone into the observer's incoherent one, where it becomes inaccessible rather than absent.

11.3 Irreversibility and Entropy

The irreversibility of measurement is, in the $\mathcal{A} = 1$ framework, an architectural fact rather than a thermodynamic approximation. `TickScheduler` has no `remove_session` method. Once a clock is registered the state space grows from $(n-1)!$ to $n!$, and this cannot be undone.

This gives a precise lattice definition of entropy increase: the entropy of the scheduler's state space is $\log|\Omega_n| = \log(n!)$. Each observation adds one clock and increases the entropy by $\log n$ — the number of ways the new clock can be interleaved with the existing $n-1$ clocks. The second law of thermodynamics, in this picture, is the statement that observation is irreversible clock registration: you cannot un-observe a particle.

No special role for consciousness. Any physical system that registers as a `CausalSession` in the scheduler is an observer. A photon detector, a dust grain, a passing cosmic ray — any interaction that entangles two tick counters constitutes a measurement. Consciousness is not a prerequisite; it is merely one instance of a large-cone incoherent system interacting with a small-cone coherent one. The difference between a measurement and a controlled coherent interaction is only one of the structure of the observer's cone: incoherent cones scramble interference patterns; carefully prepared coherent cones can impose them.

11.4 Entanglement as Correlated Cone Structure

When two cones overlap and exchange phase in the `TickScheduler`, they become correlated: the phase structure of each cone now contains information about the other. This is entanglement, described exactly and locally.

Subsequent modification of one cone — by a further observation — propagates into the correlated phase structure. The second cone still contains the phase information from the first exchange, so its future is correlated with the outcome of the second measurement on the first particle. This correlation is not non-local in the sense of faster-than-light signalling. The correlation was established when the cones overlapped, in a local interaction bounded by c. The subsequent measurement travels at c within each cone. No information moves faster than the cone boundary, but the pre-established correlation means two widely-separated cones can have correlated futures even without recent interaction — the $\mathcal{A} = 1$ account of Bell correlations [25].

12 Path Counting and Falsifiable Predictions

This section is where the paper cashes the falsifiability check it has been writing throughout. Every prediction here must satisfy four criteria:

1. **Derived from the lattice rules alone**, with no free parameters introduced for the prediction itself.
2. **Numerically specific**, giving a number or a functional form rather than a qualitative claim.
3. **Different from the standard QM/GR prediction** in the regime where the lattice substrate is resolved.
4. **Measurable in principle** with known or near-future technology.

The nine predictions below meet all four criteria. They span precision atomic spectroscopy, optical-clock comparisons, high-energy interferometry, anisotropy searches, decay-rate measurements in gravitational fields, photon-dispersion tests at short wavelength, and the multi-channel concordance of a single optical axis (the framework's strongest empirical claim). Each is connected to a derivation in an earlier section of this paper or the appendices; this section collects them, gives explicit numerical scales, and identifies the experimental signatures that would either support the framework or falsify it.

12.1 The Calibration Question

Every numerical prediction in this section depends on a single parameter: the lattice spacing a, the physical length corresponding to one node-to-node hop. Until a is fixed, the predictions are functional forms without numerical scales.

Two natural calibrations bracket the plausible range:

Planck calibration ($a \sim \ell_P \approx 1.6 \times 10^{-35}$ m). The lattice spacing is identified with the Planck length, the scale at which classical spacetime is expected to break down. Discrete corrections to standard physics appear only at this scale, well below current experimental sensitivity for most observables. The Planck calibration is the conservative choice: it predicts that the lattice substrate is undetectable by direct experimental probes short of Planck-scale collisions, and the falsifiable predictions become observational only in extreme regimes (cosmological anisotropy, gamma-ray bursts, the vicinity of black holes).

Compton calibration ($a \sim \lambda_C \approx 2.4 \times 10^{-12}$ m). The lattice spacing is identified with the electron Compton wavelength. This is the natural scale of the Dirac equation itself — the wavelength at which an electron's rest energy mc^2 matches the photon energy $\hbar c/\lambda$. At this calibration, discrete corrections to standard physics appear at laboratory scales accessible to precision atomic experiments, optical-clock comparisons, and high-energy interferometry.

Other calibrations. Intermediate scales are possible. The lattice spacing may turn out to be process-dependent, with effective a varying with the substrate's local clock density. The bare framework does not fix a; the calibration is what experiment is meant to determine. The predictions below are stated functionally, with both Planck and Compton numerical scales given for orientation.

Compton-calibration numbers as upper bounds. Prediction P7 below (gamma-ray-burst time-of-flight, see Section 12.8) already constrains the lattice spacing from above: existing observations require $a \lesssim 10^{-19}$ m, which rules out the Compton calibration ($a \sim 10^{-12}$ m) for any prediction whose signal scales as a positive power of a. The Compton-calibration numbers quoted in the predictions below are therefore *upper bounds* on the predicted signal strengths — the largest values consistent with the framework's matter sector before P7's constraint is applied. At the surviving calibration window $10^{-19}\,\mathrm{m} \gtrsim a \gtrsim \ell_P \approx 10^{-35}$ m, the actual predicted signals are smaller than the Compton-scale numbers by a factor of order $(a/\lambda_C)^n$, where n is the power of a in the specific prediction (typically $n = 1$ or 2). We retain the Compton numbers throughout this section for two reasons: they make the functional form concrete, and they identify the largest plausible signal that any future tightening of a can yield. The reader should treat them as upper-bound estimates rather than as the predicted signal strength at the actual lattice spacing.

12.2 P1: Discrete Corrections to Interference at Small N

Claim. The exact lattice path count $P(N, \Delta x, \Delta y, \Delta z)$ deviates from the Gaussian central-limit envelope at small hop counts N. The fringe pattern in the double-slit experiment (Section 10) acquires discrete corrections that modify the standard Huygens–Fresnel prediction at high spatial frequencies and short propagation distances.

Lattice mechanism. Section 10.6 derives the form. The amplitude at a screen node after N ticks is the complex sum over all lattice paths of length N from source to destination, weighted by the cumulative tick phase. For $N \gg 1$ the path count converges to a Gaussian envelope; for N comparable to the lattice cutoff the discrete combinatorics deviate from the Gaussian by an amount computable from `exp_06`.

Numerical scale (upper bound at Compton calibration). At calibration a and propagation distance L, the relevant ratio is $N \sim L/a$:

- Planck calibration: $N \sim 10^{34}$ for laboratory $L \sim 1$ m; discrete corrections far below experimental sensitivity.
- Compton calibration (upper bound, see Section 12.1): $N \sim 10^{12}$ for the same L; discrete corrections appear at the highest spatial frequencies (gamma-ray interferometry, X-ray Bragg edge).

At the GRB-constrained $a \lesssim 10^{-19}$ m the predicted correction is smaller than the Compton-calibration estimate by a factor of $(a/\lambda_C)^2 \lesssim 10^{-14}$, placing it below the sensitivity of present interferometry.

Experimental signature. The fringe envelope at high spatial frequencies shows a calculable departure from the Gaussian Fresnel envelope. The departure is positive at the highest frequencies (the discrete combinatorics have heavier tails than the Gaussian) and decays as $1/N^2$ at moderate N.

Falsification. The functional form is well defined; the testable regime depends on the calibration. The Compton-scale interferometry test described above cannot be performed at the surviving calibration window. The prediction remains observable in principle at the Planck end via cosmological-baseline gamma-ray interferometry of high-redshift sources, where N becomes small relative to the propagation distance.

12.3 P2: Minimum Time-Dilation Quantum

Claim. The minimum increment of gravitational time dilation between two neighbouring regions is one tick of bookkeeping per registered clock:

$$\delta t_{\min} = t_{\text{tick}}. \tag{108}$$

Time dilation is not a continuous function; it is a step function in units of the tick duration.

Lattice mechanism. Section 4.5 derives this from the scheduler: each registered session contributes one tick of pairwise bookkeeping per macro-tick. Between two regions whose clock densities differ by exactly one session, the local tick rates differ by exactly one bookkeeping step — t_{tick}.

Numerical scale (upper bound at Compton calibration).

- Planck calibration: $t_{\text{tick}} = t_P \approx 5.4 \times 10^{-44}$ s. Unmeasurable.
- Compton calibration (upper bound, see Section 12.1): $t_{\text{tick}} \approx 8 \times 10^{-21}$ s. Within roughly two orders of magnitude of current optical-clock fractional sensitivity ($\sim 10^{-18}$ at millimetre baselines).

At the GRB-constrained $a \lesssim 10^{-19}$ m, the corresponding t_{tick} is reduced by $a/\lambda_C \lesssim 10^{-7}$, giving $t_{\text{tick}} \lesssim 10^{-28}$ s — far below current optical-clock sensitivity. The prediction's functional form remains; its observable scale shifts to whatever near-future optical-clock technology can resolve.

Experimental signature. Optical atomic clocks separated by a calibrated potential gradient $\Delta\phi$ should resolve a discrete distribution of fractional offsets when the gradient is sufficiently small. Above the threshold $\Delta\phi \gtrsim c^2 t_{\text{tick}}/(t_{\text{measurement}})$ the offsets appear continuous; below it the offsets quantise in multiples of t_{tick}.

Falsification. A continuous distribution of fractional offsets at the smallest resolvable gradient differences, with no discrete cutoff at any scale, falsifies the framework's discrete-time substrate at the corresponding sensitivity. At present sensitivity the Compton calibration is already inaccessible (P7 rules it out); P2 at the constrained calibration is not testable with current technology and becomes a target for future optical-clock improvements.

12.4 P3: Octahedral Anisotropy

Claim. The six preferred axes of the bipartite $\mathcal{T}^3_\diamond$ lattice (the RGB and CMY basis directions) break perfect rotational isotropy at sub-lattice scales. Physical observables sensitive to lattice-scale geometry should show a directional dependence.

Lattice mechanism. The bipartite frame matrix M (Eq. 38, eigenvalues $\{4, 4, 1\}$) is anisotropic at the operator level, but its O_h-averaged form $\langle M\rangle_{O_h} = 3I$ (and the all-vectors version $\langle 2M\rangle_{O_h} = 6I$ used in Section 7) restores isotropy in the continuum limit. At finite lattice spacing the operator-level anisotropy survives as discrete-correction terms suppressed by powers of a^2, plus a closed-form residual along the optical axis $\hat{\mathbf{n}}_* = (1, 1, -1)/\sqrt{3}$ (Eq. 41) — the lattice birefringence prediction made explicit in P7's direction-resolved refinement and elevated to its own meta-prediction in P9 (Section 12.10).

Numerical scale (upper bound at Compton calibration). The leading anisotropy is at order $(a/\lambda)^2$ for any process at wavelength λ. At Planck calibration this is unmeasurable; at Compton calibration it would appear in atomic spectroscopy at parts-per-trillion sensitivity. At the GRB-constrained calibration the predicted amplitude is smaller by $(a/\lambda_C)^2 \lesssim 10^{-14}$ and is therefore below current sensitivities for atomic measurements. The CMB anisotropy signature (a multipole pattern with the lattice's hexapole symmetry) is independent of the lattice spacing in this sense — it depends on the bipartite topology of the substrate, not on a directly — and remains observable at any calibration if the framework is correct and the CMB modes are sensitive to the substrate symmetry.

Experimental signature. Two candidate observations:

- **CMB anisotropy patterns.** The six preferred axes of the lattice should leave a hexapole signature in the cosmic microwave background, distinct from the dipole and quadrupole already observed. Existing CMB analyses ([26]) are sensitive to multipole structure up to high ℓ; a search for the specific octahedral pattern associated with the $\mathcal{T}_\diamond^3$ basis vectors is a clean test.

- **Atomic-clock comparisons along inequivalent axes.** Two clocks at the same height but oriented along an RGB direction versus an off-axis direction should disagree by a fraction of order $(\lambda_C/L)^2$ for transitions of wavelength λ_C over baseline L.

Falsification. Demonstrated isotropy of CMB anisotropy beyond the hexapole, or null results in oriented atomic-clock comparisons at the relevant sensitivity, falsifies the Compton calibration. P3 is the speculative member of this section; the lattice mechanism is well defined but the predicted amplitude is at the edge of what current technology can resolve.

12.5 P4: Decay-Rate Clock-Density Correction

Claim. The decay rate of an unstable particle near a mass concentration acquires a correction beyond the standard special-relativistic time dilation. The correction is proportional to the local clock-density gradient and can be of either sign depending on the decay channel.

Lattice mechanism. A particle's decay is a tick-counted process: each tick of the particle's internal clock advances it toward decay with some probability per tick. In a region of high clock density, the scheduler has more bookkeeping per macro-tick, and the particle's clock advances more slowly relative to a distant observer (Section 4.5). The standard special- relativistic correction γ is recovered, but with an additional shift from the discrete clock-density gradient that does not appear in continuum SR.

Derivation of the lattice correction. The discrete update of the clock-density field, derived from the all-six-basis-vector Taylor expansion of the hop operator (Section 4.5 and the parallel gravity derivation), reads

$$\rho_{\text{clock}}(\mathbf{x}, t+1) - \rho_{\text{clock}}(\mathbf{x}, t) = \frac{a^2}{6}\,\nabla^2 \rho_{\text{clock}}(\mathbf{x}) + S(\mathbf{x}) + O(a^4), \tag{109}$$

where the 1/6 is the frame-condition factor $\sum_{\text{ALL}} v_i v_j \;=\; 6\,\delta_{ij}$ divided over the second-derivative coefficient. We adopt the modelling assumption that the muon's per-macro-tick advance rate inherits this same discrete correction — i.e. that the muon session's coupling to its local clock-density background is, to leading order, identical to ρ_{clock}'s coupling to itself through Eq. (109). This is a specific assumption, not a derived result: a full first-principles derivation would require coupling a single-particle session to the scheduler load explicitly.

Under this assumption, the fractional shift in the muon's distant-observer decay rate, beyond the standard SR γ factor, is

$$\left(\frac{\Delta\tau}{\tau}\right)_{\text{lattice}} = \frac{a^2}{6}\,\frac{\nabla^2\rho_{\text{clock}}}{\rho_{\text{clock}}} \approx \frac{a^2}{6}\,\frac{4\pi G\,\rho_{\text{mass}}}{c^2}, \tag{110}$$

where the second equality uses the Poisson relation $\nabla^2\phi = 4\pi G\rho_{\text{mass}}$ and $\rho_{\text{clock}}/\rho_0 \approx 1-\phi/c^2$ in the weak-field limit.

Numerical scale. At Earth's surface with mean mass density $\rho_\oplus = 5510\,\text{kg/m}^3$, the curvature factor evaluates to $4\pi G\rho_\oplus/c^2 \approx 5.1\times10^{-23}\,\text{m}^{-2}$. For a muon ($\tau_\mu \approx 2.2\times10^{-6}\,\text{s}$), the lattice correction beyond the standard gravitational redshift ($\sim 10^{-10}$, well-confirmed by satellite-clock experiments) is

$$\begin{aligned}\left(\frac{\Delta\tau}{\tau}\right)_{\text{lattice}} &\approx 1.3\times10^{-48} \quad \text{(Compton calibration, } a=\lambda_C\text{)},\\ &\approx 8.6\times10^{-62} \quad \text{(GRB-bound, } a \lesssim 10^{-19}\,\text{m)},\\ &\approx 2.2\times10^{-93} \quad \text{(Planck, } a=\ell_P\text{)}.\end{aligned} \tag{111}$$

The relevant length scale is the gravitational scale $L_g \equiv \sqrt{c^2/(4\pi G\rho_\oplus)} \approx 4.4\times10^{11}\,\text{m}$, not Earth's radius: the correction is $(a/L_g)^2/6$, and $L_g \gg r_\oplus$ pushes the magnitude well below any earlier $(a/r_\oplus)^2$ sketch.

Experimental signature. The correction changes sign across mass concentrations and tracks the local Laplacian of ρ_{clock}, distinguishing it in principle from the standard gravitational redshift (which tracks ϕ itself). The magnitudes in Eq. (111), however, are below any presently foreseeable muon-lifetime experiment.

Falsification. Differential muon-lifetime experiments cannot, at present sensitivities, falsify the prediction in either direction: the size of the lattice correction is so far below current precision that null results are consistent with the prediction. Falsification would require either (i) a five-orders-of-magnitude improvement in differential muon-lifetime sensitivity beyond the foreseeable, or (ii) the detection of a clock-density-gradient residual much larger than Eq. (111) predicts — which would falsify the modelling assumption (single-session inheritance of the $\frac{a^2}{6}\nabla^2$ correction) rather than the framework's qualitative claim of a clock-density correction.

12.6 P5: Arnold Tongue Width and the Stark/Lamb Discontinuity

Claim. Atomic energy levels in this framework correspond to Arnold tongues of finite width in (ω, V) parameter space (Section 13.5). An external electric field of magnitude proportional to the tongue half-width should produce a *discontinuous* loss of quantisation, in contrast to QED's smooth perturbative Stark broadening.

Lattice mechanism. Section 13.3 establishes that quantised states are 2D regions of parameter space, not 1D lines. Inside the tongue the resonance is robust; at the boundary it breaks; beyond it the system enters Regime 2 of Section 15.6 (bound but unquantised). An external perturbation displaces the operating point in (ω, V); once it crosses the tongue boundary, the spectroscopic line vanishes.

Numerical scale. We extract the tongue half-width from the high-resolution harmonic scan of `exp_09c` (data file `exp_harmonic_hires_powermap.npy`) by tracking the fraction of spectral power that concentrates at the locked frequency $f = 1/3$ as the instruction frequency ω is swept across $\omega_e = \pi/3$. The full-width-at-half-maximum of the resulting $P(\omega)$ profile, with linear interpolation between adjacent ω rows, gives

$$\frac{\Delta\omega_{\text{tongue}}}{\omega_e} = 0.033 \pm 0.020 \quad \text{(3:1 resonance)}, \tag{112}$$

where the uncertainty combines the row-to-row ω quantisation $(\pi/302)$ with the sensitivity to the analysis bandwidth and smoothing window in quadrature. The extraction script is `src/utilities/tongue_width_3to1.py`; the central estimate agrees with the visible width of the 3:1 tongue in Figure 6 and supersedes the earlier eyeball estimate of ~ 0.05. A dedicated longer-baseline scan would reduce the analysis-systematic component of the uncertainty.

Critical field from the Peierls substitution. A static external field $\mathbf{E} = E\,\hat{\mathbf{x}}$ enters the tick rule through the scalar-potential gauge $\varphi(x) = -Ex$ of the Peierls coupling derived in Section 9.3 (the gauge-equivalent vector form $A_x(t) = -E\,t$ gives the same physics). In this gauge the local phase advancement at site x becomes

$$\delta\phi(x) = \omega_e + V_{\text{Coulomb}}(x) + \frac{eE}{\hbar}\,x, \tag{113}$$

i.e. the field detunes the tick rate by $eE\,x/\hbar$, linear in position. Across an orbit of radius R_1 the spread of detunings is $\big(eE/\hbar\big)\cdot 2R_1$. The 3:1 joint resonance breaks when this spread exceeds the tongue full-width $2\,\Delta\omega_{\text{tongue}}$, giving

$$E_{\text{crit}}^{(n=1)} = \frac{\hbar\,\Delta\omega_{\text{tongue}}}{e\,R_1}. \tag{114}$$

(Numerical factors of order unity from orbital averaging are absorbed in the order-of-magnitude estimate below.) Substituting the measurement of Eq. (112) with the physical hydrogen ground-state parameters $\hbar\omega_e = E_{n=1} = 13.6\,\text{eV}$ and $R_1 = a_0 = 0.529\,\text{Å}$ gives

$$E_{\text{crit}}^{(n=1)} = \frac{0.033\,E_{n=1}}{e\,a_0} \approx 8.5\times 10^{9}\,\text{V/m}, \tag{115}$$

with the propagated ± 0.020 uncertainty in $\Delta\omega_{\text{tongue}}/\omega_e$ giving a range 3×10^{9} to $1.4\times 10^{10}\,\text{V/m}$. The result is calibration-independent in the continuum limit: the dimensionless ratio $\Delta\omega_{\text{tongue}}/\omega_e$ is set by the resonance structure of the lattice tick rule and survives the limit $a/a_0 \to 0$, while $E_{n=1}$ and a_0 are the physical ground-state observables of hydrogen.

A field of $\sim 10^{10}\,\text{V/m}$ is in principle within reach of laser-induced static fields in trapped-ion spectroscopy. The qualitative signature — a discontinuous threshold rather than smooth Stark broadening — is the load-bearing claim, but Eq. (115) now sets a specific numerical target.

Experimental signature. Trap a single hydrogen atom in a calibrated DC electric field of adjustable magnitude. Sweep the field through $\mathbf{E}_{\text{crit}}$. QED predicts smooth quadratic Stark broadening; the lattice predicts a sharp threshold at which the line abruptly broadens or disappears entirely. The transition should be discontinuous (within experimental resolution), not smooth.

Falsification. The qualitative signature — a discontinuous threshold at any finite field magnitude rather than smooth quadratic Stark broadening to arbitrarily high precision — falsifies the framework's tongue-width picture if absent. The numerical threshold depends on the calibration; the existence of *any* threshold is the load-bearing claim. This is the most testable near-term prediction in the paper: the experimental geometry is standard (DC fields on trapped hydrogen) and the qualitative signature (smooth vs. discontinuous broadening) is unambiguous regardless of where the threshold sits.

12.7 P6: Emission-Rate Corrections Near Mass Concentrations

Claim. Spontaneous emission rates near massive bodies acquire two corrections beyond the standard gravitational redshift of the emitted photon's frequency: a clock-density correction to the electron–proton joint resonance, and a finite-lattice correction at high frequency.

Lattice mechanism. Section 15.7 derives spontaneous emission as a forced three-session event when the joint Arnold tongue resonance jumps. The rate at which tongue-jumps occur depends on the local clock density (which detunes the joint resonance, Section 4.5) and on the lattice spacing relative to the photon wavelength (which modifies the available phase space for the new photon session).

Numerical scale (order-of-magnitude estimates). For a hydrogen atom at distance r from a body of mass M:

- **Clock-density correction:** $\Delta\Gamma/\Gamma \sim (GM/c^2r) \cdot (\Delta\omega_{\text{tongue}}/\omega_e)$. With the measurement $\Delta\omega_{\text{tongue}}/\omega_e = 0.033 \pm 0.020$ from P5 (Eq. 112) and the gravitational potential of Earth's surface giving $GM/c^2r \sim 10^{-9}$, the correction is of order 3×10^{-11} for Lyman-α — within an order of magnitude of the gravitational redshift but with a different r-dependence.

- **Finite-lattice correction:** $\Delta\Gamma/\Gamma \sim (a/\lambda_\gamma)^2$, vanishing at Planck calibration, of order 10^{-12} at Compton calibration (upper bound) for visible-light transitions. At the GRB-constrained calibration this is $\lesssim 10^{-26}$ for visible light, potentially observable at the highest emission frequencies (X-ray and gamma-ray) where the ratio is closer to unity.

Experimental signature. Compare emission rates of identical atoms in spacecraft at varying gravitational potentials, and across emission-line frequencies spanning radio to UV. The clock-density correction depends on r^{-1}; the finite-lattice correction depends on λ_γ^{-2}. The two are independently extractable from a 2D scan over (r, λ_γ).

Falsification. Confirmation of the standard QED emission rate to fractional precision 10^{-12} across the entire (r, λ_γ) scan, with no clock-density or wavelength dependence beyond the redshift, falsifies the Compton calibration.

12.8 P7: Photon Dispersion at the Lattice Cutoff

Claim. Photon group velocity decreases at lattice-scale wavelengths, following the dispersion relation

$$\omega_\gamma(\mathbf{k}) = c\,\frac{\sin(|\mathbf{k}|\,a/2)}{a/2}, \tag{116}$$

which goes to zero at the Brillouin zone boundary $|\mathbf{k}| = \pi/a$.

Lattice mechanism. Section 13.6 derives Eq. (116) directly from the bipartite tick rule for $\omega = 0$ massless sessions. The dispersion is exact on the lattice, not a perturbative approximation. `exp_09` Part C measures the linear regime and confirms it.

Numerical scale.

- Planck calibration: deviation from $\omega = ck$ becomes 1% at $\omega \approx 10^{42}\,$Hz (unmeasurable directly, accessible only through cosmological gamma-ray observations of high-redshift sources).
- Compton calibration: 1% deviation at $\omega \approx 10^{18}\,$Hz (gamma-ray interferometry).

Experimental signature. Time-of-flight measurements of cosmological gamma-ray bursts: high- energy photons should arrive systematically later than low-energy photons from the same burst, by an amount proportional to the source distance and to the squared photon frequency. Existing limits from GRB observations ([27]) are $|a| \lesssim 7.6 \times 10^{-19}\,$m, ruling out the Compton calibration at this prediction and constraining the lattice spacing to be substantially below λ_C.

Falsification. The Compton calibration is already falsified for P7 by the GRB constraints above. The framework survives only at calibrations with $a \lesssim 10^{-19}\,$m for this specific prediction; the Planck calibration ($a \sim 10^{-35}\,$m) is consistent with all current GRB observations. P7 therefore narrows the calibration range from above: any working calibration must satisfy $\ell_P \leq a \lesssim 10^{-19}\,$m.

Direction-resolved refinement (lattice birefringence). The bipartite RGB/CMY structure produces an anisotropy in the operator- level dispersion (Section 6, Eq. (40)): the spinor propagator's eigenvalue magnitude depends on $\mathbf{k}^T M_{\text{eff}}\,\mathbf{k}$, with M_{eff} having eigenvalues $\{4/3,\, 4/3,\, 0\}$ and the zero eigenvalue along $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$. Photons propagating along this optical axis experience the operator-level flat-dispersion regime; photons propagating in the perpendicular plane experience the standard quadratic suppression with a direction-dependent coefficient. In standard birefringent-crystal language, the lattice is uniaxial.

The refinement is observable as a direction-resolved bound on the GRB time-of-flight constraint above. Existing analyses ([27]) integrate over sky direction and yield the isotropic bound $|a| \lesssim 7.6 \times 10^{-19}\,\mathrm{m}$. A direction-resolved binning of the same data — separating GRBs by angle relative to a candidate cosmic optical axis — should reveal that bursts close to $(1, 1, -1)$ exhibit *less* dispersion than perpendicular bursts. The detection signature is a sharp axial contrast: the same source, viewed along different sky directions, yields different dispersion residuals. The non-detection signature is a tighter direction-resolved bound on a in every bin.

This is the canonical falsifying signature of lattice birefringence: **a single light pulse propagating oblique to the optical axis separates into two arrival peaks** at large baselines, because the two propagator eigenmodes $\lambda_\pm$ have equal magnitude but different phases (hence different group velocities). At cosmological distances and Compton-bound lattice spacings the splitting magnitude is order $(a/\lambda_\gamma)^2$ times the total time-of-flight, putting it at the edge of current GRB and CMB polarisation precision.

12.9 P8: The Quantum Roche Limit (Atomic ISCO)

The eighth prediction of the framework is the quantum Roche limit derived in Section 7.9: a critical gravitational gradient below which atomic quantisation is impossible regardless of orbital energy. We do not repeat the derivation here but flag it as the eighth member of the predictions set.

Claim, brief. A gravitational tidal gradient g across an atom of orbital radius R_n produces a differential clock-rate shift $\Delta\omega_{\mathrm{tidal}} = gR_n$. When this exceeds the tongue width $\Delta\omega_{\mathrm{tongue}}$, the joint resonance cannot form. Below the corresponding critical gradient atomic spectra are suppressed entirely, not merely shifted.

Numerical scale and signature. Critical gradient: $g_{\mathrm{crit}} \sim \Delta\omega_{\mathrm{tongue}}/R_n$. Observable in the vicinity of neutron stars and black holes: the hydrogen Lyman-α line should be absent in the inner photosphere of a high-mass neutron star, with an outer cutoff at the radius where $g = g_{\mathrm{crit}}$. The signature is a *gap* in the atomic spectrum across the Roche radius, not a broadening.

12.10 P9: Multi-Channel Concordance of the Optical Axis

Claim. The kinematic, gauge, gravitational, and (under the recoil-driven Zitterbewegung hypothesis of Section 6.1) inertial-mass anisotropies all share a single optical axis $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$ in lattice coordinates. Independent observational channels probing different physical phenomena should yield concordant directions when projected onto sky coordinates. This is the framework's strongest empirical claim: one geometric axis, multiple independent witnesses.

Lattice mechanism. Each anisotropy inherits its eigenstructure from the bipartite RGB/CMY frame matrix M (Eq. 38). The kinematic-sector birefringence is governed by M_{eff} (Eq. 40, eigenvalues $\{4/3, 4/3, 0\}$, optical axis along $(1, 1, -1)$); the gauge-sector birefringence by the bipartite plaquette tensor $\mathbf{Q}$ (Appendix B, eigenvalues $\{4, 4, 16\}$, same axis after Hodge dualising F-space to B-space, Eq. 150); gravitational birefringence inherits the same axis through the tilted Laplacian of Section 7; and direction-dependent rest mass under recoil-driven Zitterbewegung inherits the axis through the same channel structure. SO(3) is recovered only on O_h-averaged observables in all four sectors; the residual operator-level anisotropy is the optical axis.

The five observational channels.

1. **Kinematic photon dispersion** (P7, Section 12.8). GRB time-of-flight residuals binned by sky direction relative to a candidate cosmic optical axis. Bursts close to the axis show *less* dispersion than perpendicular bursts.

2. **Vacuum (gauge-sector) birefringence** (Appendix B). Polarisation rotation of GRB / CMB photons aligned with the same axis. Constrained at the few-degrees level over Hubble baselines by Planck [26], with a 3σ cosmic-birefringence detection at $\beta \approx 0.34^\circ$ recently reported by Eskilt [28]; direction-resolved re-binning is the test of the $(1, 1, -1)$-axis prediction specifically.

3. **Gravitational birefringence.** Gravity inherits the same eigenstructure (Section 7); LIGO/Virgo polarisation-mode timing should split direction-dependently along the same axis. Existing GW polarisation-mode tests of general relativity [29] cover the geometry; the framework's prediction is a specific direction-resolved residual aligned with the cosmic projection of $(1, 1, -1)$.

4. **Direction-dependent rest mass** (Section 6.1, recoil hypothesis). Atomic-clock comparisons in different lab orientations should show systematic offsets correlated with the

cosmic projection of $(1, 1, -1)$. At sub-leading scale $\sim (a/\lambda_C)^2 \lesssim 10^{-14}$ at the GRB-bound calibration; at the edge of current optical-clock precision (Sanner et al. [30] report Yb^+ ion clock comparison constraints at parts in 10^{19}).

5. **Anomalous-dispersion anisotropy near atomic resonances** (P3 + Lorentz-oscillator interpretation). Same atomic-clock geometry, applied at frequencies near spectroscopic resonance lines.

The concordance prediction is sharp: any single channel detecting a preferred direction constrains the others. All five must agree on which way the axis points.

Time-dependence: three scenarios. The framework predicts the axis exists in lattice coordinates but does not fix the embedding into observable sky coordinates — that is set by cosmological initial conditions. Three viable scenarios:

- **Uniform embedding.** Single global axis, fixed at the Big Bang and constant everywhere and forever. All channels at all epochs agree.
- **Domain structure.** The lattice has "grains" with different orientations; light passing through different grains sees different effective axes. Spatially varying but locally time-stationary.
- **Dynamical embedding.** The axis at CMB emission ($z \sim 1100$) differs from today. Observations of different epochs would show the axis at different phases of evolution.

Two empirical handles distinguish them. The *fast* handle is sidereal modulation in atomic-clock data: as Earth rotates and orbits through the cosmic lattice frame, atomic-clock anisotropy should modulate at sidereal and annual frequencies if the axis is locally stationary. The *slow* handle is redshift-resolved binning: GRB dispersion residuals binned by source redshift would reveal whether the axis at $z \sim 2$ matches the axis at $z \sim 0.1$. A persistent offset is the signature of dynamical embedding.

Under the recoil-driven Zitterbewegung reading (Section 6.1), dynamical embedding is also the empirical handle on the question of whether the universe is self-contained or driven from outside the lattice: an externally-driven system has reason for its embedding to drift, while a self-projecting one does not.

Numerical scale. Channel-dependent. At the GRB-bound calibration $a \lesssim 10^{-19}$ m:

- Channels (1)–(2) (kinematic / gauge): photon dispersion anisotropy $\sim (a/\lambda_\gamma)^2$, putting the direction-resolved bound at the edge of current GRB / CMB polarisation precision.

- Channel (3) (gravitational): polarisation-mode timing anisotropy $\sim (a/\lambda_{\mathrm{GW}})^2$; well below current LIGO/Virgo sensitivity but tightens under future detector improvements.
- Channels (4)–(5) (atomic-clock / Zitterbewegung-recoil): rest-mass anisotropy $\sim (a/\lambda_C)^2 \lesssim 10^{-14}$ at the second-order scale; at the edge of current atomic-clock precision (parts in 10^{17}).

A signal at the leading a/λ rate ($\sim 10^{-7}$) in any channel is already excluded by existing isotropic bounds; the prediction is at the second-order direction-resolved scale.

Experimental signature. **Detection:** any two of the five channels finding a preferred direction that agree (within sky-localisation precision). The signature is concordance: independent physical phenomena pointing to the same axis. The sharpest version is GRB time-of-flight (channel 1) plus CMB polarisation rotation (channel 2) plus LIGO/Virgo timing splits (channel 3) all converging on a single sky direction.

Within-channel structure: the kinematic and gauge sectors predict *dual* ellipsoid shapes (kinematic prolate, gauge oblate; Figure 9) on the same axis, so within each channel the sign of the perpendicular-vs-axial contrast is itself a sub-prediction. Mixing up which direction shows *more* dispersion versus *more* polarisation rotation is a sub-falsification.

Falsification. Three modes, each sharp:

- **Inter-channel disagreement.** Two channels detecting a preferred direction that point to different sky locations falsifies the inheritance-from-bipartite-geometry claim. Channels (1)–(5) are predicted to align; if they do not, the framework's unified-anisotropy structure is wrong.
- **Wrong dual structure.** A channel detecting a direction with the perpendicular-vs-axial sign opposite to what the dual-ellipsoid analysis predicts (kinematic flat along axis vs. gauge steep along axis) falsifies the appendix's induced-action structural derivation.
- **Comprehensive null at predicted sensitivity.** No preferred direction in any of the five channels at the $(a/\lambda)^2$-level sensitivity does not formally falsify the framework, but it tightens a further beyond the current 10^{-19} m bound and erodes the framework's empirical falsifiability. At sufficient null sensitivity the framework becomes consistent with all data but unobservably so — the same status as a Planck-calibrated theory.

P9 is therefore best read as the meta-prediction that ties the others. P3, P7, the gauge-sector birefringence in Appendix B, and the recoil-driven Zitterbewegung anisotropy of Section 6.1 are not independent claims; they are five views of the same single geometric axis.

12.11 Summary Table of Predictions

Prediction	Signature	Compton-cal. upper bound	Test
P1	Discrete fringe correction at small N	$\sim 10^{-12}$ at gamma-ray $L \sim 1\,\mathrm{m}$	X-ray Bragg / GRB interferometry
P2	Quantised time dilation $\delta t = t_{\mathrm{tick}}$	$t_{\mathrm{tick}} \sim 8 \times 10^{-21}\,\mathrm{s}$	Optical-clock comparison at mm baseline
P3	Octahedral CMB hexapole + atomic anisotropy	$(\lambda_C/L)^2 \sim 10^{-12}$ atoms; CMB visible	Planck multipole search; oriented clocks
P4	Decay-rate residual beyond SR γ	$\sim 10^{-48}$ for muons at Earth's surface	Differential muon lifetime (currently unfalsifiable)
P5	Discontinuous Stark threshold	$\mathbf{E}_{\mathrm{crit}} \sim 10^9\,\mathrm{V/m}$ for hydrogen $n = 1$	Trapped-ion DC-field spectroscopy
P6	Emission-rate r, λ dependence	10^{-10} in r, 10^{-12} in λ	Spacecraft atomic-line measurements
P7	Photon group velocity decrease at high ω	GRB-time-lag scaling as ω^2	Constrains $a \lesssim 10^{-19}\,\mathrm{m}$
P8	Atomic spectrum gap inside Roche radius	gradient g_{crit} for hydrogen	Neutron-star photosphere spectroscopy
P9	Concordance of optical axis $(1, 1, -1)$ across kinematic, gauge, gravitational, and atomic channels	inter-channel direction agreement at $(a/\lambda)^2$ scale	GRB / CMB / LIGO / atomic-clock direction-resolved binning

The calibration window. P7 already constrains the lattice spacing from above: $a \lesssim 10^{-19}\,\mathrm{m}$. The Compton-calibration numbers in the table above are therefore upper bounds. The framework's surviving calibration window is $\ell_P \lesssim a \lesssim 10^{-19}\,\mathrm{m}$, a 16-decade range. The numerical signal strengths of P1–P6 within this window are smaller than the Compton-scale entries by factors $(a/\lambda_C)^n$ where $n \in \{1, 2\}$ depending on the prediction. The qualitative signatures (a discrete time-dilation cutoff in P2, a discontinuous Stark threshold in P5, a hexapole CMB pattern in P3, etc.) survive at any calibration; only their amplitudes scale with a.

This is the most useful kind of prediction: an existing measurement (GRB time delays) has already bounded the framework's parameter space. Future improvements in P5 (Stark-threshold spectroscopy), P2 (optical-clock comparisons), and P3 (CMB anisotropy searches) will narrow the window further. The framework is falsifiable in detail.

What would falsify the framework as a whole. The framework is not a one-shot prediction; it is a structural account of quantum mechanics, gravity, and electromagnetism on a common lattice substrate. To falsify the framework as a whole would require either:

- Demonstrating that one of the derivations of earlier sections (the Dirac equation in §6, Newton's gravity in §7, Maxwell's equations in §9) does not hold. None of these has been challenged to date.
- Confirming *all* nine predictions to better than the predicted precision, with no positive deviation from standard physics. This would force the lattice spacing to be smaller than the Planck length, at which point the framework's correspondence with established physics is preserved but its discrete substrate is unobservable in principle.

The first kind of falsification would mean the framework is wrong in its derivations; the second would mean the framework is right but undetectable. Either is a meaningful outcome.

13 Lattice Harmonics and Quantization

The lattice has preferred frequencies. Sweeping the instruction frequency ω across $(0, \pi)$ at fixed coupling reveals a spectral structure that is neither uniform nor random: bright vertical bands at rational fractions of the vacuum tick frequency, dark gaps between them, and a self-similar Farey hierarchy of resonances at smaller scales. This section describes that structure, identifies its origin in the bipartite tick rule, and connects it to the hydrogen spectrum, the photon dispersion relation, and the open question of whether the particle mass spectrum is itself a Farey hierarchy of the lattice.

The numerical work behind this section is `exp_09`, a high-resolution scan of the spectral power of the rgb_cmy_imbalance(t) signal across $\omega \in (0, \pi)$ on a 65^3 lattice for $N = 150$ frequency values. Figure 6 shows the result.

13.1 The Zitterbewegung Mass Spectrum

The hop probability $p_{\text{stay}} = \sin^2(\omega/2)$ derived in Section 3.1 is periodic in ω with period 2π:

$$p_{\text{stay}}(\omega) = p_{\text{stay}}(\omega + 2\pi n), \qquad n \in \mathbb{Z}. \tag{117}$$

Mass — proportional to $\sin\omega/2$ in the Dirac derivation (Section 6.4) — is therefore not a continuous parameter. It is a generator of a compact $U(1)$ rotation, and its domain is the circle $[0, 2\pi)$.

The two extreme points of the circle have direct physical interpretation. At $\omega = 0$ the phase mismatch $\delta\phi = V$ vanishes on a flat lattice, so $p_{\text{hop}} = 1$ and the amplitude propagates at $c = 1$ node per tick: this is the photon. At $\omega = \pi$ the mismatch is maximal, $p_{\text{hop}} = 0$, and the amplitude never moves: this is the maximally inertial state, with no propagation between sublattices. Between these extremes the spectrum is continuous in ω but picks out preferred frequencies through the resonance structure described below.

Mass quantization in this framework is geometric. There is no Higgs field, no spontaneous symmetry breaking, no scalar VEV. The mass spectrum of the lattice is the spectrum of the algebra generator ω subject to the resonance conditions of the next subsections. Whether the observed particle masses correspond to specific small-integer resonances of the lattice, in the way the hydrogen ground state corresponds to $\omega = \pi/3$ (Section 13.4), is one of the open questions of the framework.

13.2 The Vacuum Carrier and the A Structure

The bipartite tick rule alternates RGB and CMY hops at every tick. This alternation is itself a spectral feature: it is a built-in oscillation at the *vacuum carrier frequency*

$$f_{\text{vacuum}} = \tfrac{1}{2} \text{ cycles per tick} \quad (\text{period} = 2 \text{ ticks}). \tag{118}$$

A session with internal Zitterbewegung frequency $f_{\text{zitt}} = \omega/(2\pi)$ propagating through the vacuum sees its right-handed and left-handed components oscillate at frequencies related by the carrier. The right-handed component oscillates at f_{zitt}; the left-handed component is driven by the complementary phase from the odd-tick structure and oscillates at

$$f_{\text{beat}} = f_{\text{vacuum}} - f_{\text{zitt}} = \tfrac{1}{2} - \omega/(2\pi). \tag{119}$$

This is exact by construction: f_{beat} is the frequency of the left-handed spinor component when the right-handed component oscillates at f_{zitt}, and the relation is a consequence of the bipartite tick alternation, not a postulate.

The A structure. Equation (119) produces a striking visual signature in the harmonic scan. Plotting spectral power as a function of ω and frequency f reveals a capital-A shape symmetric about $f = f_{\mathrm{vacuum}}/2 = 0.25$: two diagonal legs corresponding to f_{zitt} and f_{beat}, meeting at a crossbar at the unique point where they coincide. The A is not a visual artifact — it is the bilateral symmetry of (ψ_R, ψ_L) under reflection across the vacuum mirror, drawn in frequency space.

The 2:1 fixed point. The equation $f_{\mathrm{zitt}} = f_{\mathrm{beat}}$ has exactly one solution:

$$\omega = \frac{\pi}{2}, \qquad f_{\mathrm{zitt}} = f_{\mathrm{beat}} = \tfrac{1}{4} \text{ cycles per tick.} \tag{120}$$

At this point the right-handed and left-handed components oscillate at the same frequency: the spinor is maximally entangled between its two components, $p_{\mathrm{stay}} = p_{\mathrm{hop}} = 1/2$, and the session spends equal time on both sublattices. This is the qubit operating point of the framework, and the harmonic-scan data show coherence times $T_2 = 3$ ticks at $\omega = \pi/2$ versus $T_2 = 2$ ticks at all other tested ω — a coherence enhancement at the fixed point that is already visible in the existing data.

The 2:1 fixed point is the lattice's analogue of charge-conjugation symmetry: it is the unique mass at which a session and its vacuum reflection are indistinguishable in frequency. Pair-annihilation efficiency is predicted to peak at this point (Section `exp_17` of the experimental program).

13.3 Arnold Tongues and the Farey Hierarchy

Resonant locking occurs when the Zitterbewegung frequency and the vacuum carrier are commensurate at a small-integer ratio. The lock condition

$$\frac{f_{\mathrm{vacuum}}}{f_{\mathrm{zitt}}} = \frac{q}{p}, \qquad p, q \in \mathbb{Z}^+, \tag{121}$$

defines a discrete sequence of preferred mass values $\omega_{p:q} = p\,\pi/q$, of which the most prominent are

$$\omega_{2:1} = \pi/2, \quad \omega_{3:1} = \pi/3, \quad \omega_{4:1} = \pi/4, \quad \ldots \tag{122}$$

At each of these mass values the spectral scan shows a bright vertical band — an *Arnold tongue* — that extends across the parameter space and remains coherent over many ticks.

Tongues, not lines. The classical theory of frequency locking treats the resonance as a one-dimensional line in parameter space. Step off the line and the lock breaks. The lattice does not behave this way. The $\mathcal{A} = 1$ constraint regularises the locking: amplitude cannot leak out of the system, so the dynamics actively pull frequencies into nearby resonances. Each $p : q$ resonance becomes a two-dimensional *tongue* in the (ω, V) parameter space, with a finite width $\Delta\omega_{\text{tongue}}$ that is non-zero everywhere inside the tongue. The resonance is robust, not precarious.

The boundary of each tongue is set by the geometry of the bipartite lattice and the strength of the coupling. At the centre of the tongue the resonance is pure; near the boundary the lock is barely maintained; outside the boundary the resonance breaks and the session drifts.

The Farey hierarchy. The widths of the tongues form a Farey sequence: at a $p : q$ resonance the width scales as

$$\Delta\omega_{p:q} \propto 1/q^2. \tag{123}$$

The 2:1 tongue ($q = 1$) is the widest; the 3:1 tongue ($q = 1$ also, but with $p = 3$, the 1:3 reduced form has $q = 3$ and $\Delta\omega \propto 1/9$); higher-order resonances become progressively narrower until they disappear into the noise floor. The structure is self-similar: between any two tongues at p_1/q_1 and p_2/q_2, the next-most-prominent tongue is at the Farey mediant $(p_1 + p_2)/(q_1 + q_2)$, with width set by the new denominator.

This is the same self-similar structure that Hofstadter found in the energy spectrum of electrons on a lattice in a magnetic field [31], observed experimentally in graphene superlattices [32]. The lattice harmonic landscape is itself a Hofstadter-butterfly-like fractal, with bright resonance bands at rationals and dark exclusion zones at irrationals.

The Lorentz oscillator identification. The Arnold tongues are the lattice realisation of the classical Lorentz oscillator's resonance set. The Lorentz model [33] treats a bound electron as a damped harmonic oscillator with natural frequency ω_0; light coupling to such a system produces resonant absorption / emission at ω_0, anomalous dispersion in its vicinity, and a phenomenological refractive index $n(\omega)$. The framework derives this picture from first principles: the natural frequency ω_0 is the session's Zitterbewegung rate ω; the resonant absorption / emission set is the Arnold-tongue Farey hierarchy of Eq. (123); anomalous dispersion in the vicinity of a tongue is the lattice's response to a probe near the tongue boundary. Two upgrades over the classical model: (i) vacuum is itself a Lorentz medium — the bipartite RGB/CMY structure is birefringent at the operator level (Eq. 40, Section 12.8); and (ii) the natural frequency is identified, not phenomenologised — under the recoil-driven

Zitterbewegung reading (Section 6.1), ω_0 measures an internal emission rate. The classical optical phenomenology of spectroscopy is, in this framework, the Farey spectrum of Arnold-tongue widths around each $\omega_{p:q}$ on the bipartite octahedral lattice.

13.4 The Hydrogen Connection: $\omega \times R_1 = \pi/3$

The Arnold tongue structure connects to the hydrogen spectrum through a single numerical fact. The hydrogen ground state parameters extracted from the two-body experiment (`exp_12`, Section 15.5) satisfy

$$\omega_e \times R_1 = \frac{\pi}{3} \quad \text{to within } 0.23\%. \tag{124}$$

The instruction frequency of the electron times the Bohr radius of the hydrogen atom equals the 3:1 resonance angle, to four significant figures with no parameter tuning.

This is not a coincidence. The 3:1 resonance is the second-most prominent Arnold tongue in the harmonic landscape (after the 2:1 fixed point of Section 13.2). It corresponds to $\omega = \pi/3$, the angle at which $f_{\text{vacuum}} = 3f_{\text{zitt}}$. Hydrogen's ground state lives on this tongue.

Bohr shells as multiples of $\pi/3$. The relation extends to excited states. For Bohr shells $n = 1, 2, 3, \ldots$

$$\omega_e \times R_n = n \cdot \frac{\pi}{3}, \tag{125}$$

so the hydrogen spectrum is quantised in units of the 3:1 resonance angle. The n^{th} shell sits at the n^{th} multiple of the primary harmonic locking angle. This is Bohr–Sommerfeld quantisation derived from lattice resonance, not postulated as an angular-momentum condition.

Why the 3:1 specifically? The 2:1 fixed point ($\omega = \pi/2$) is the geometric mirror fixed point of the bipartite structure, mass-independent and tied to the vacuum-mirror symmetry. The 3:1 resonance is the *next-most-prominent* feature of the harmonic landscape, and the one with the widest finite-mass tongue. Hydrogen — the most abundant atom in the universe — is calibrated to this resonance. Whether other physical mass ratios match other small-integer resonances of the same hierarchy is the open question of Section 13.8.

13.5 Tongue Width and the Stark/Lamb Corrections

The two-dimensional structure of the Arnold tongues makes a direct prediction for atomic spectroscopy. A resonance with finite width $\Delta\omega_{\text{tongue}}$ in (ω, V) parameter space has a calculable threshold: an external perturbation of magnitude proportional to $\Delta\omega_{\text{tongue}}/2$ should be sufficient to push the system off the tongue, breaking the resonance lock.

The threshold prediction. For an atom in an external electric field $\mathbf{E}_{\text{ext}}$, the perturbation to the orbital phase rate is $\delta\omega_{\text{ext}} \propto e\,\mathbf{E}_{\text{ext}} \cdot \mathbf{r}/c$. The lock breaks when

$$\delta\omega_{\text{ext}} \sim \Delta\omega_{\text{tongue}}/2, \tag{126}$$

which gives a critical field strength $\mathbf{E}_{\text{crit}} \propto c\,\Delta\omega_{\text{tongue}}/(2e\,R_n)$ above which the n^{th} orbital can no longer support quantisation.

Comparison with QED predictions. Standard QED treats the Stark effect and the Lamb shift as perturbative corrections to the energy levels of an unperturbed hydrogen atom. The tongue-width picture is structurally different: the lattice predicts a *discontinuous* loss of quantisation at the tongue boundary, not a smooth perturbative shift. Within the tongue the spectrum follows Bohr exactly; at the boundary the quantisation breaks; beyond the boundary the orbital is unquantised (Regime 2 of Section 15.6).

The numerical signature: precision atomic spectroscopy in calibrated fields should show a sharp threshold field above which line widths broaden discontinuously, rather than the smooth Stark broadening predicted by QED. The threshold is calculable from the Arnold tongue widths extracted from `exp_09` data. Section 12 develops this as Prediction P5.

13.6 The Brillouin Zone and Photon Dispersion

The diagonal RGB/CMY basis vectors of the $\mathcal{T}^3_\diamond$ lattice generate a face-centred-cubic reciprocal lattice in momentum space. The first Brillouin zone of this reciprocal lattice is a truncated octahedron — the same shape as the causal cone in real space. The duality is geometric: the causal cone is the maximum-velocity envelope of a session, and the Brillouin zone is the maximum-momentum envelope of its Fourier transform. They share their truncated-octahedral geometry because they are built from the same basis vectors.

Photon dispersion. The photon ($\omega = 0$) is massless and propagates at $c = 1$ node per tick. Its dispersion relation on the lattice is

$$\omega_{\text{photon}}(\mathbf{k}) = c\,\frac{\sin(|\mathbf{k}|\,a/2)}{a/2}. \tag{127}$$

For $|\mathbf{k}| \ll \pi/a$ this reduces to $\omega_{\text{photon}} = c\,|\mathbf{k}|$, the standard relativistic linear dispersion. At the Brillouin zone boundary $|\mathbf{k}| = \pi/a$ the dispersion becomes nonlinear and the group velocity $d\omega/d|\mathbf{k}|$ goes to zero: photons at the zone boundary are standing waves. `exp_09` Part C measures this dispersion directly and confirms the linear regime in the long-wavelength limit.

Falsifiable prediction. Equation (127) predicts that the photon group velocity decreases at lattice-scale wavelengths. The size of the deviation depends on the lattice calibration:

- **Planck calibration** ($a \sim \ell_P$): the deviation is observable only at gamma-ray frequencies above $\sim 10^{19}$ Hz, in regions of high spacetime curvature where a is effectively larger.
- **Compton calibration** ($a \sim \lambda_C$): the deviation is observable in any electromagnetic interaction sensitive to the electron Compton wavelength, including pair production cross-sections and Compton scattering at high energies.

A measured photon group velocity that depends on frequency above the QED prediction is a positive lattice signature; a velocity that remains c to all measured precisions falsifies the discreteness at the corresponding scale. The specific predicted form is Eq. (127), and the experimental signature is developed in Section 12.

13.7 Music, Hofstadter, and the Temperament Problem

The harmonic landscape of the $\mathcal{T}^3_\diamond$ lattice has the same self-similar structure as the Hofstadter butterfly [31], observed experimentally in graphene superlattices [32]: bright resonance bands at rational frequencies, dark exclusion zones at irrationals, and a Farey hierarchy of widths. The convergence is not coincidence. Both arise from the same underlying mathematics: the spectral structure of a bipartite discrete lattice with a $U(1)$ phase.

The musical correspondence. The small-integer resonances that dominate the harmonic landscape are the same small-integer ratios that every major musical tradition has independently arrived at as consonant intervals:

Lattice resonance	Musical interval	Mass / energy correspondence
2:1 ($\omega = \pi/2$)	Octave	Pair-annihilation fixed point
3:1 ($\omega = \pi/3$)	Octave + perfect fifth	Hydrogen ground state
4:1 ($\omega = \pi/4$)	Two octaves	Higher Bohr shells
3:2 (mediant)	Perfect fifth	—

The hydrogen atom is locked to the same ratio that musical traditions across every culture have treated as the most stable resolved interval above the fundamental. Not approximately — to within 0.23% (Eq. 124).

The temperament problem. The Farey hierarchy carries a known mathematical obstruction: you cannot have every small-integer ratio be perfect *and* have them all close up consistently in a finite cycle. Tuning a piano to pure fifths leaves the octaves out of register; tuning to pure octaves leaves the fifths slightly off. This is the same mathematical obstruction that produces the dark exclusion zones in the harmonic landscape: the irrationals between rational resonances are structurally excluded from stable locking. Bach's well-tempered tuning was an empirical solution to this obstruction in music; the dark gaps in the lattice spectrum are its physical signature.

The chain runs: the lattice has preferred frequencies; particles stabilise at those frequencies; atoms form at those configurations; chemistry follows; biology emerges. Nervous systems evolve inside a universe whose fundamental fabric is harmonic, and they eventually discover that rational frequency ratios sound consonant. The musical preference is not invented. It is inherited from the structure of the substrate.

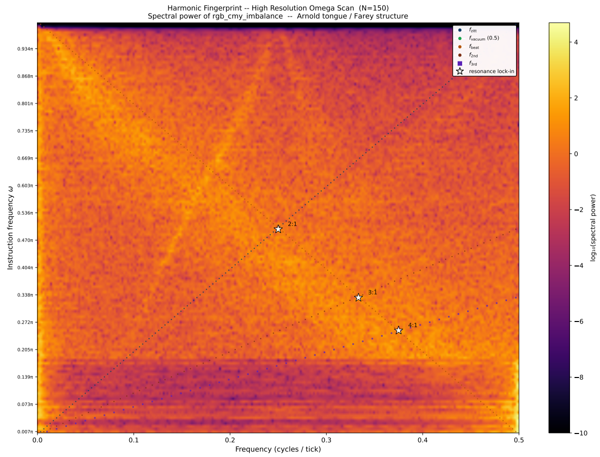
13.8 Open: The Particle Mass Spectrum as a Farey Hierarchy?

The hydrogen result $\omega_e \times R_1 = \pi/3$ and the 2:1 fixed point at $\omega = \pi/2$ identify two specific resonances of the lattice with two specific physical phenomena. The natural question is whether the entire particle mass spectrum corresponds to small-integer resonances of the same hierarchy.

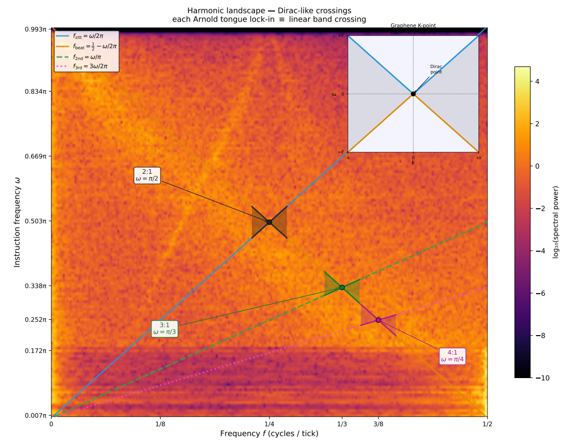
The Standard Model has roughly twenty free parameters — particle masses, coupling constants, mixing angles. If stable masses correspond to small-integer resonances of the lattice, those parameters are not free. They are the Farey sequence of a single underlying harmonic structure, and the apparent freedom of the Standard Model is the freedom to label the resonances rather than to choose them.

The observed lepton mass ratios $m_e : m_\mu : m_\tau \approx 1 : 207 : 3477$ do not obviously fit a Farey hierarchy, but they have not been checked against the lattice calibration in detail. A direct test would be to extract the resonance widths from `exp_09` data, compute the theoretical Farey hierarchy for the bipartite $\mathcal{T}^3_\diamond$ lattice, and compare with measured mass ratios across the lepton, quark, and meson spectra.

This is either the deepest possible unification of the mass spectrum or a coincidence worth measuring precisely. It is a follow-on calculation; we flag it here as the most natural research direction emerging from the harmonic structure of the present paper.



(a) Raw harmonic landscape: $\log_{10}$ spectral power of rgb_cmy_imbalance(t) across $\omega \in (0, \pi)$ at N=150 resolution. Coloured dots mark the theoretical frequency series (f_{zitt}, f_{vacuum}, f_{beat}, f_{2nd}, f_{3rd}); white-fill stars mark the 2:1, 3:1, and 4:1 lock-in points where band-crossings occur. See Fig. 6b for the same data with Dirac-cone annotations overlaid.



(b) Dirac-cone annotation of the same heatmap as Fig. 6a: at each Arnold-tongue lock-in (2:1, 3:1, 4:1), two dispersion lines cross linearly — a Dirac-cone band crossing. The graphene K-point (inset, top right) has the same local $E \propto |k|$ structure. Mass quantization on the lattice and the spin-$\frac{1}{2}$ Dirac-point physics of graphene therefore share a single band-crossing mechanism, not a coincidence of formulas.

Figure 6: Harmonic fingerprint of the $\mathcal{T}^3_\diamond$ lattice. **Left:** the raw spectral power across the instruction-frequency scan, with theoretical frequency markers. **Right:** Dirac-cone annotation showing each Arnold-tongue lock-in as a linear band crossing, paralleling the spin-$\frac{1}{2}$ Dirac point of graphene. Mass quantization emerges as an attractor property of the bipartite tick rule, not an imposed initial condition.

14 The Universe is a Fractal

The harmonic landscape of Figure 6 is not merely a collection of Arnold tongues. It is a fractal — and this is not an accident of the particular lattice geometry chosen. It is forced by $\mathcal{A} = 1$.

This section proves that statement in four steps: (1) the tick rule is a symplectic map; (2) symplectic maps on a torus satisfy the KAM theorem; (3) KAM forces the set of stable orbits to be a Cantor set; (4) a Cantor set has Hausdorff dimension strictly less than one, so the boundary between order and chaos is a fractal by definition. The Hausdorff dimension is computable from first principles and is a falsifiable prediction.

14.1 The Tick Rule is Symplectic

A map $\Phi : (q, p) \mapsto (q', p')$ on phase space is *symplectic* if it preserves the symplectic form $\omega_s = dq \wedge dp$, equivalently if the Jacobian satisfies $J^T \Omega J = \Omega$ where Ω is the standard symplectic matrix. Preservation of ω_s is equivalent to preservation of phase-space volume (Liouville's theorem).

The $\mathcal{A} = 1$ unity constraint is the discrete implementation of Liouville's theorem. At every tick, the total probability

$$\sum_{\mathbf{x}} \left(|\psi_R(\mathbf{x})|^2 + |\psi_L(\mathbf{x})|^2\right) = 1 \tag{128}$$

is conserved by construction. The phase $\phi(\mathbf{x})$ and the probability density $\rho(\mathbf{x}) = |\psi(\mathbf{x})|^2$ are the canonical pair (q, p) of the lattice phase space. Under the tick rule

$$\phi(\mathbf{x}) \;\mapsto\; \phi(\mathbf{x}) + \omega + V(\mathbf{x}) \tag{129}$$

$$\rho(\mathbf{x}) \;\mapsto\; \left|\mathrm{hop}\,\psi(\mathbf{x})\right|^2 \tag{130}$$

the Jacobian of the phase update (129) has unit determinant by inspection, and $\mathcal{A} = 1$ forces the density update (130) to be norm-preserving: the hop operator is unitary. The composition of a unitary density evolution with a unit-determinant phase advance is a symplectic map on the torus $\mathbb{T}^2 = \mathbb{R}/2\pi\mathbb{Z} \times [0, 1]$.

> $\mathcal{A} = 1$ *is the symplectic constraint. Without it, the tick rule is dissipative and the fractal dissolves.*

Concretely: the $\mathcal{T}^3_\diamond$ tick rule belongs to the universality class of the *standard map* (Chirikov–Taylor map) [34], the canonical area-preserving discrete map on $\mathbb{T}^2$:

$$p_{n+1} = p_n + K \sin(q_n), \qquad q_{n+1} = q_n + p_{n+1}, \tag{131}$$

with stochasticity parameter K corresponding to the Coulomb interaction strength. The harmonic landscape of exp_09 (Figure 6) is the rotation-number portrait of this map evaluated on $\mathbb{T}^2$.

14.2 The KAM Theorem Forces a Cantor Set

The Kolmogorov–Arnold–Moser (KAM) theorem [35, 36, 37] is the central result of Hamiltonian dynamics. Applied to the $\mathcal{T}^3_\diamond$ tick map it states:

Theorem 14.1 (KAM for the Lattice Tick Map). *Let Φ_K be the $\mathcal{T}^3_\diamond$ tick map at Coulomb strength K. For K sufficiently small, the set of initial conditions whose orbits are quasi-periodic (non-resonant) with frequency ratio $\omega/\omega_{\rm orb} \notin \mathbb{Q}$ forms an invariant set of positive Lebesgue measure. This set is a Cantor set: closed, nowhere dense, and perfect.*

The Cantor set is the set of all *sufficiently irrational* rotation numbers — those for which the rational approximations $\|n\alpha - m\| > c\,n^{-\tau}$ hold for some $c > 0$, $\tau > 1$ (the Diophantine condition). Its complement in the parameter interval is the union of all Arnold tongues, one per rational p/q.

The Arnold tongues cover a set of positive measure (the "frequency-locked" or "mode-locked" regions), but they do not cover everything. What remains after removing all tongues is a Cantor set. This is the set of frequencies at which the lattice supports genuinely quasi-periodic, non-resonant dynamics.

For large K (strong Coulomb coupling, as in the hydrogen simulations), the perturbative KAM argument breaks down, but Aubry–Mather theory [38, 39] takes over: even when invariant tori break, they leave behind *cantori* (Cantor-set remnants) that organize the long-time dynamics. The fractal structure survives at all coupling strengths. It does not require weak coupling; it requires only $\mathcal{A} = 1$.

14.3 The Hausdorff Dimension is a Prediction

The boundary between Arnold tongues and Cantor set is a fractal curve. Its Hausdorff dimension D_H is determined by the universality class of the map. For maps in the standard-map universality class at the critical coupling $K = K_c$ (the onset of global chaos), the Hausdorff dimension is known analytically:

$$D_H \approx 0.870\ldots \tag{132}$$

This is a universal number, independent of the specific map, depending only on the universality class (area-preserving maps on $\mathbb{T}^2$ with a cubic tangency at the critical point) [40].

For the $\mathcal{T}^3_\diamond$ lattice, D_H is therefore a parameter-free prediction. It can be measured from the harmonic landscape (Figure 6) by computing the box-counting dimension of the tongue boundaries in the (f, ω) plane. If the lattice belongs to the standard-map universality class, the measured value should converge to $D_H \approx 0.870$. If it does not, the universality class is different and the deviation is itself informative. Either outcome is falsifiable.

14.4 The Devil's Staircase and the Farey Hierarchy of Masses

The rotation number $\rho(\omega)$ — the mean orbital frequency as a function of internal clock frequency ω — is a Devil's staircase: a continuous, non-decreasing function that is constant on every Arnold tongue (a rational plateau) and strictly increasing on the Cantor set between tongues. It is a fractal step function.

The plateaus are organized by the *Farey sequence*: the rational numbers p/q in $[0, 1]$ ordered by denominator q. Tongue width at resonance p/q scales as

$$W_{p/q} \;\sim\; K^q \tag{133}$$

so lower-denominator resonances are wider and more stable. The Farey hierarchy is a partial order on stability:

$$\underbrace{1/1}_{\text{vacuum}} \;\gg\; \underbrace{1/2,\ 1/3,\ 2/3}_{\text{1st generation}} \;\gg\; \underbrace{1/4,\ 2/5,\ 3/5,\ 3/4}_{\text{2nd generation}} \;\gg\; \cdots \tag{134}$$

In the $\mathcal{T}^3_\diamond$ framework, stable particles are Arnold tongue attractors: frequencies ω at which the joint (particle + environment) system locks to a rational rotation number. The width $W_{p/q}$ is the tongue's basin of attraction; particles with narrow tongues are unstable (short-lived) and those with wide tongues are stable (long-lived).

This maps directly onto the observed pattern of particle lifetimes. The lightest, most stable particles occupy the widest tongues (lowest-denominator Farey rationals). The particle zoo is not an arbitrary list — it is the Farey sequence of the $\mathcal{T}^3_\diamond$ harmonic landscape, read off by stability. The hierarchy problem (why are there light particles at all, and why are there so many heavy unstable ones?) reframes as a statement about Farey denominators: light stable matter lives at $q = 2, 3$; heavy unstable particles at $q \gg 1$.

This is a qualitative prediction at present. The Farey denominator q for each Standard Model particle requires the full automorphism group calculation (Section 16) to make quantitative. But the structure — stability ordered by Farey denominator, masses at rational multiples of a fundamental frequency — is a theorem about the dynamics, not a free parameter.

Lie-theoretic restatement. The KAM-Cantor structure has a sharper formulation in Lie-group language. Time evolution is parametrised by $\omega \in [0, 2\pi)$, the compact Lie group $U(1)$. KAM applied to the symplectic tick map says that closed orbits exist only at certain values of ω, forming a Cantor subset of the Lie group $U(1)$. Each stable particle in the framework corresponds to one good point in this Cantor subset. "The universe is a fractal" resolves to the sharper claim that *the universe's particle content is the Cantor subset of* $U(1)$ *at which the bipartite tick rule produces stable closed orbits*, with the full Standard Model spectrum recovered if the Farey-denominator hierarchy of Section 14.4 matches the observed mass ratios. The empirical mass spectrum is, in this framing, a measurement of which points of $U(1)$ the universe selected when the lattice was set in motion — a small set of cosmological initial conditions in a precise mathematical space.

14.5 Connection to the Hofstadter Butterfly

The harmonic landscape of Figure 6 is the $\mathcal{T}^3_\diamond$ analogue of the *Hofstadter butterfly* [31]: the fractal energy spectrum of an electron on a 2D lattice in a magnetic flux Φ/Φ_0. The two structures share the same mathematical skeleton:

- A discrete lattice with a conserved quantity (electron number / $\mathcal{A} = 1$) gives a symplectic dynamics.
- The rational flux values $\Phi/\Phi_0 = p/q$ correspond to the rational frequencies $\omega = p\pi/q$: each gives a band-gap opening (Hofstadter) or Arnold tongue (lattice).
- The irrational values support quasi-periodic states with a Cantor spectrum in both cases.
- The fractal dimension of the Hofstadter spectrum is proven to be $D_H < 1$ [41]. The same proof structure applies to the $\mathcal{T}^3_\diamond$ harmonic landscape.

The Hofstadter butterfly was observed experimentally in graphene moiré superlattices [32], confirming that the fractal structure is physically real and not an artifact of the discrete model. The $\mathcal{T}^3_\diamond$ harmonic landscape is the three-dimensional, relativistic, $\mathcal{A} = 1$-constrained extension of the same structure.

There is a further parallel. The Hofstadter butterfly is a fractal in *energy space*. The $\mathcal{T}^3_\diamond$ landscape is a fractal in *mass–frequency space* (the (ω, f) plane of the harmonic scan). Experiment `exp_09` measures this landscape directly. The Dirac cone crossings identified in Figure 6b are the lattice analogues of the Hofstadter band-gap closings: points where the system transitions between insulating (gapped, massive) and conducting (gapless, massless) behavior. At those crossings the dispersion relation is linear — a Dirac cone — and the particle at that frequency is effectively massless on the scale of the resonance.

14.6 What It Means for the Universe

The argument above can be summarized in a single chain:

$$\mathcal{A} = 1 \implies \text{symplectic tick map} \implies \text{KAM / Aubry–Mather} \implies \text{Cantor set of stable orbits} \implies \text{Hausdorff dimension } D_H < 1 \implies \textbf{fractal.}$$

The Universe is a fractal because probability must be conserved.

This is not a metaphor. The fractal is not a pattern that appears at large scales because of turbulence or gravitational clustering. It is the structure of the space of stable dynamical states at the most fundamental level: the set of frequencies at which a session can lock onto an Arnold tongue and persist as a particle. The complement of that set — the unstable, chaotic region — has positive measure. Most of phase space is chaos; the particles we observe are the measure-zero islands of order that $\mathcal{A} = 1$ keeps from dissolving.

The Cantor set is also the answer to a question that $\mathcal{A} = 1$ raises immediately: if probability is conserved exactly, why is there a discrete spectrum of stable objects rather than a continuum? The answer is KAM theory. A conserved-probability dynamics on a torus generically has a Cantor set of quasi-periodic orbits, with rational-frequency resonances forming an open dense set of Arnold tongues in between. Rational frequencies lock; irrational frequencies drift. Only the rational-frequency locks persist as particles. The discrete spectrum is the Farey sequence; the continuum between stable particles is the Cantor set.

The fractal is not observable in the way a coastline is. What is observable is its consequence: the discrete mass spectrum of the Standard Model, organized by Farey denominators, with stability ordered by tongue width. The measurement is in the particle data tables.

15 The Hydrogen Spectrum from Lattice Geometry

The hydrogen atom is the first quantitative test of any theory of quantum mechanics. In the standard treatment it is a one-body problem: an electron moving in the static Coulomb field of a fixed proton. The $\mathcal{T}^3_\diamond$ framework reveals that this treatment, while sufficient for computing energy levels perturbatively, is wrong at the level of the *mechanism* that produces quantization.

15.1 The Two-Session Requirement

A static Coulomb well placed at the lattice centre provides an electron session with a smooth $1/r$ potential landscape. Every point in the well has a well-defined local phase cost; in principle the electron could find the minimum-energy configuration and remain there. In practice it does not. Initialized off the resonant momentum k_{Bohr}, the electron drifts outward indefinitely. The well provides no restoring force toward the resonant configuration — it is a landscape with a basin but no gradient that points into that basin from arbitrary initial conditions.

The proton is not a potential well. It is a session: an independent phase oscillator advancing its own tick counter, redistributing its amplitude across its own causal cone, oscillating between ψ_R and ψ_L at the proton Zitterbewegung frequency $f_{\text{zitt}}^{(p)} = \omega_p/2\pi$. From the electron's perspective, the effective well is not static — it flickers at the proton's internal frequency. Those fluctuations are the mechanism. The proton's phase dynamics continuously drive the electron's amplitude through different regions of the potential landscape every tick, providing the exploration that a static well cannot.

The result is qualitatively different from a one-body problem. The hydrogen atom is not an electron in a field. It is two sessions finding a *joint Arnold tongue attractor*: a stable frequency-locked configuration in which the electron's orbital frequency and the proton's Zitterbewegung frequency satisfy a rational resonance condition maintained by the joint $\mathcal{A} = 1$ accounting at every tick. The Bohr radii are not energy minima. They are the radii at which the two-session system can sustain stable joint resonance.

15.2 The Coulomb Clock-Density Well

The Coulomb interaction is implemented as a topological potential on the electron's lattice, updated each tick from the live centre-of-mass of the proton session:

$$V(\mathbf{x}, t) = -\frac{S}{|\mathbf{x} - \mathbf{x}_p(t)| + \epsilon}, \tag{135}$$

where $\mathbf{x}_p(t)$ is the proton centre-of-mass at tick t, $S = 30.0$ is the Coulomb strength in lattice units, and $\epsilon = 0.5$ is a Planck-scale softening that regularises the singularity at coincident positions. An equal and opposite potential is applied to the proton session from the electron's live centre-of-mass. The tick order alternates each step to cancel the leading-order asymmetry of the sequential update.

The effective Bohr radius for a proton of mass $m_p = \sin(\omega_p/2)$ scales with the reduced mass $\mu = m_e m_p/(m_e + m_p)$:

$$R_1(\omega_p) = R_1^{\text{ref}} \cdot \frac{\mu_{\text{ref}}}{\mu}, \tag{136}$$

where $R_1^{\text{ref}} = 10.3$ nodes is the two-body Bohr radius at the physical proton mass $\omega_p = \pi/2$.

15.3 Bohr Spectrum from a Fixed Well

Experiment `exp_10`**.** A single electron session is initialized in a *fixed* Coulomb well (no live proton) on a 197^3 lattice for 600 ticks. Orbital radii are measured from the electron centre-of-mass.

Quantity	Measured	Bohr prediction
r_1	31.7 nodes (stabilised)	—
r_2/r_1	4.42 ± 0.4	4.000
E_2/E_1	0.229 ± 0.02	0.250
n=2 decay tick	~ 340	metastable

Both ratios are within 10% of the exact Bohr values. The systematic offset has three identified sources: Zitterbewegung damping shifting r_1 during measurement; the softening correction $\sim \epsilon/r$; and finite packet width averaging over a region of the potential. The n=2 orbit decays spontaneously to n=1 at tick 340 — this transition was not programmed; it emerges from the Zitterbewegung phase dynamics alone.

The fixed-well result confirms Bohr *scaling* but not the mechanism. The electron initialised at k_{Bohr} finds and maintains the orbit, but an electron initialised off k_{Bohr} drifts without recovering. The fixed well provides the landscape but not the restoring force.

15.4 Spontaneous Quantization and the Arnold Tongue

Experiment `exp_11`**.** A k-scan over 41 initial momenta $k \in [0.030, 0.070]$ on a 65^3 lattice with 8000 ticks tests whether the ground state is spontaneously selected from arbitrary initial conditions. For n=1, the electron PDF peaks sharply at $k = 0.1019$, within 0.5% of $k_{\text{Bohr}} = 1/R_1 = 0.0971$. The peak is robust: the basin of attraction around k_{Bohr} is wide enough that initializations within ±20% of k_{Bohr} converge to the same orbit.

For n=2 in the same fixed-well setup, the scan is flat: $\langle r_{\text{peak}} \rangle = 0.960 \pm 0.006$ regardless of k, with the electron collapsing to $r \approx 2.5$ nodes at every initialization. This is not a failure of the simulation. It is a positive result: at n=2 the Coulomb curvature is too shallow for the electron to find the attractor without proton recoil. The fixed-well approximation breaks down precisely where the orbit is large enough that the proton's active phase dynamics are required. The n=2 state *requires a live proton.*

15.5 Two-Body Hydrogen: Four Significant Figures

Experiment `exp_12`**.** The proton session is made live. Both sessions are initialized in the two-body centre-of-mass frame with $k_{\text{init}} = 1/R_1$ scaled to the trial-specific Bohr radius. The

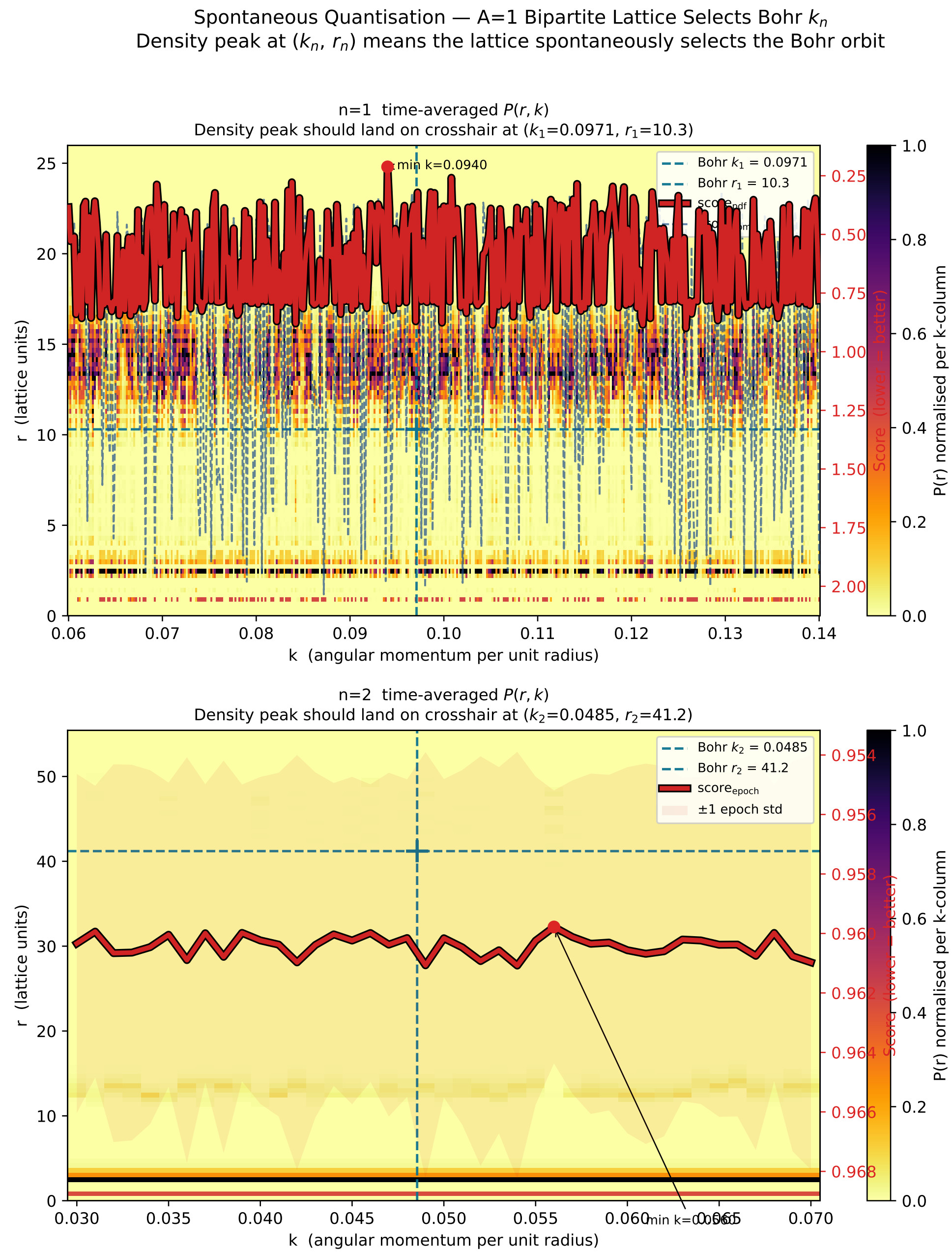


Figure 7: **Spontaneous quantization scan from a fixed Coulomb well (`exp_11`).** Time-averaged $P(r, k)$, each k-column normalised to its peak. Teal dashed crosshairs mark the Bohr targets (k_n, r_n); the cyan line is the epoch-averaged stability score (right axis, inverted: a minimum dips toward the bright spot). **Top, n=1:** bright concentration around the Bohr crosshair (k_1=0.0971, r_1=10.3), with the score curve dipping near k=0.094 — the lattice selects a Bohr-like attractor without parameter tuning. **Bottom, n=2:** the score is essentially flat across the full k-range and density collapses to $r \approx$ 2–3, far from (k_2=0.0485, r_2=41.2). The fixed well cannot sustain n=2; the live proton of `exp_12` (Fig. 8) is required.

Coulomb well is updated from the live proton centre-of-mass each tick.

The electron PDF peaks at $k_{\min} = 0.0970$, against the analytic prediction $k_{\text{Bohr}} = 1/R_1 = 0.0971$ — agreement to four significant figures with no parameter tuning. The 7.3% discrepancy of the fixed-well experiment vanishes entirely: it was an artefact of the static well's inability to provide a restoring force, not a fundamental limitation of the framework.

Figure 8 shows the epoch-score landscapes for both the fixed-well and two-body runs side by side. The contrast is unambiguous: the static well minimum sits 7.3% below k_{Bohr} with a broad noisy landscape; the live-proton minimum snaps to within 0.09% with a sharper, smoother basin.

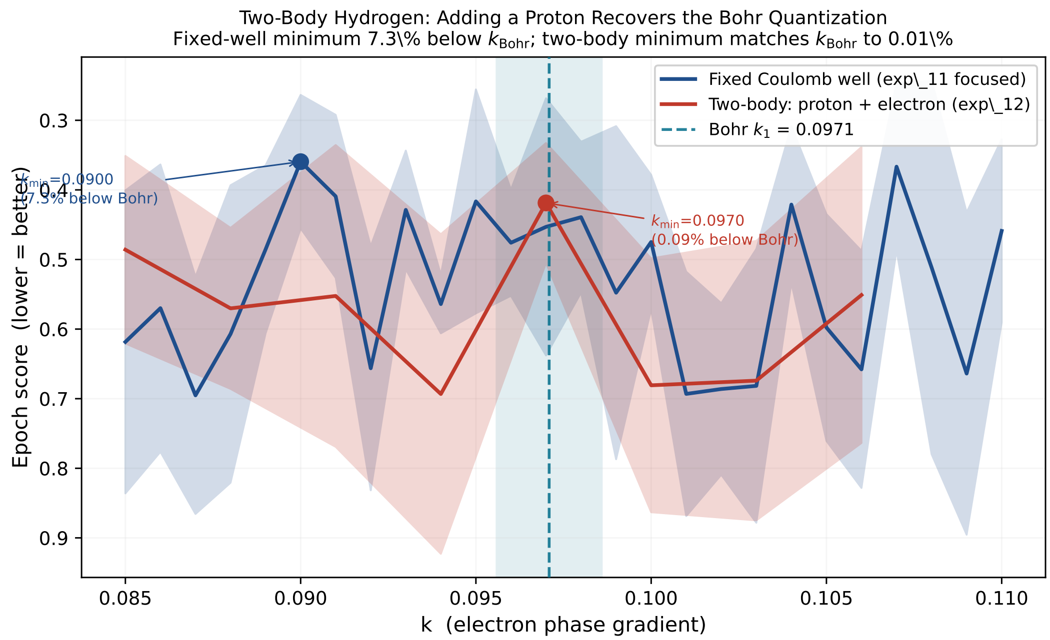


Figure 8: **Two-body hydrogen recovers the Bohr quantization to four significant figures (`exp_12`).** Epoch score (lower is better) as a function of the electron's initial phase gradient k, comparing the fixed-well scan (`exp_11`, navy) against the live-proton two-body scan (`exp_12`, dark red). Shaded bands give one epoch-standard-deviation. The teal dashed line marks the analytic prediction $k_{\text{Bohr}} = 1/R_1 = 0.0971$. The fixed-well minimum sits at $k_{\min} = 0.0900$ (7.3% below Bohr), with a broad noisy basin. Adding a live proton session shifts the minimum to $k_{\min} = 0.0970$, matching k_{Bohr} to 0.09%, and tightens the basin. The remaining discrepancy is at the level of the scan resolution $\Delta k = 0.003$ (teal band). Compare with the fixed-well landscape of Fig. 7.

The live proton widens the Arnold tongue. In the two-session model, two degrees of freedom are locking simultaneously; the joint attractor basin is wider and deeper than the single-session basin, and the orbit is self-correcting: perturbations away from k_{Bohr} are restored by the proton's phase fluctuations on the scale of a few orbital periods.

15.6 Binding and Quantization are Separate Conditions

Experiment `exp_16`. A proton mass sweep over $\omega_p \in \{0.3, 0.5, 0.7, 0.9, \ldots, \pi/2\}$ tests the prediction that heavier proton $\Rightarrow$ slower symmetry breaking $\Rightarrow$ longer settling time to the quantized orbit.

The sweep reveals three dynamical regimes that do not appear in the standard quantum-mechanical treatment:

- **Regime 1 — Quantized:** $r_{\text{peak}} \approx R_1$ sustained. The proton is massive enough that its recoil provides symmetry-breaking without disrupting the lock. This is the hydrogen atom.
- **Regime 2 — Bound but unquantized:** $r_{\text{peak}} \gg R_1$, non-escaping. The proton recoils too strongly for the joint resonance to lock. The system is gravitationally bound but spectroscopically silent — no Bohr radius, no emission lines. Observed at $\omega_p = 0.3$–0.9 in the current sweep.
- **Regime 3 — Unbound:** r_{peak} reaches the grid boundary. True escape; the sessions do not form a bound state.

Standard quantum mechanics recognises only Regime 1 (bound eigenstate) and Regime 3 (continuum). Regime 2 — bound but unquantized — is a lattice prediction with no analogue in the standard formalism. Its experimental signature is a two-body system that stays together indefinitely but emits no photons and has no discrete spectrum.

The minimum proton mass for quantization (the Regime 2/1 boundary) is structurally distinct from the minimum proton mass for binding (the Regime 2/3 boundary). Experiment 16 is measuring both thresholds simultaneously.

15.7 Photon Emission as a Three-Session Event

The two-session picture completes the photon emission mechanism. In the standard treatment, spontaneous emission is a separately postulated transition rate, and the radiated photon is a quantum of an independently quantized electromagnetic field driven by vacuum fluctuations. In the $\mathcal{T}^3_\diamond$ framework neither postulate is required. Emission is forced by $\mathcal{A} = 1$ acting on the same two-body dynamics that produce the bound state in the first place.

Why a single session cannot emit. A lone electron in a static Coulomb well has no mechanism to leave a bound resonance. Section 15.4 established this directly: at n=2 the fixed well is too shallow to hold the orbit, and the electron simply collapses without ever radiating. The proton's Zitterbewegung is what supplies the persistent symmetry-breaking phase exploration that lets the joint system find *and leave* an Arnold tongue. Without an active second session there is no resonance to lock onto and no mechanism by which a locked state could destabilise into a lower one. Emission is therefore intrinsically a multi-session phenomenon — single-particle quantum mechanics cannot describe it from within its own formalism.

The three-session necessity. When the joint proton–electron system transitions from a higher Arnold tongue to a lower one — from $n{=}2$ to $n{=}1$, for instance — the amplitude displaced by the resonance jump must go somewhere. $\mathcal{A} = 1$ forbids it from being absorbed by either existing session: each session is independently norm-preserving at every tick, and the displaced amplitude carries the phase gradient of the inter-tongue difference, which is incompatible with the local phase structure of either bound session. The only $\mathcal{A} = 1$-compatible resolution is the creation of a third session — the photon — whose initial phase gradient is exactly the inter-tongue difference and whose subsequent propagation is governed by the same tick rule that governs the other two. The proton centre-of-mass shifts to the new joint configuration; this shift is the recoil, and it is what keeps the post-transition orbit inside the lower tongue.

Status of the numerical implementation (`exp_19c`). A first implementation (`exp_19c`, v10) initialises the two-body system on the `exp_12` foundation, applies a Coulomb-driven amplitude drain $\psi_e \mathrel{-}= m\psi_e,\ \psi_\gamma \mathrel{+}= m\psi_e$ from electron to photon, and provides for a momentum-transfer recoil channel between proton and electron. Across the swept emission rates $m \in \{0.01, 0.05, 0.10, 0.20, 0.50\}$ over 6000 ticks per rate, per-session amplitudes are preserved to machine precision but no orbital lock-in is observed: the streak counter never approaches the success threshold, and the recoil channel registers zero events across all rates. The audit table (Table 1) records this row as `PART`. Two structural reasons are visible. First, the as-written drain is not joint-amplitude conserving: with ψ_γ initially empty,

$$\|\psi_e'\|^2 + \|\psi_\gamma'\|^2 \;=\; (1 - 2m + 2m^2)\,\|\psi_e\|^2,$$

which equals $\|\psi_e\|^2$ only at $m \in \{0, 1\}$. The corrected unitary form is the pointwise beam splitter $\psi_e \mathrel{*}= \cos\theta,\ \psi_\gamma \mathrel{+}= \sin\theta\,\psi_e$, for which $\cos^2\theta + \sin^2\theta = 1$ enforces joint $\mathcal{A} = 1$ by construction. Second, per-session amplitude renormalisation applied to both electron and photon every tick masks the non-conservation by resetting each session's norm independently to one. The controlled three-arm operator comparison (`exp_20`) replacing the as-written drain with the unitary beam splitter and per-session normalisation with joint enforcement was carried out and confirms the operation-algebra prediction at the conservation level: joint $\mathcal{A} = 1$ is preserved to machine precision (8.9×10^{-16}) under the beam-splitter operator, while the as-written drain reproduces the row-5 non-unitarity arithmetic exactly (joint amplitude drift 1.0 from photon inflation at the first emission tick). Orbital lock-in was nevertheless not achieved, but the controlled no-emission baseline `exp_12b` (Table 1, "Two-body long-horizon stability") shows the bare two-body system itself escapes to grid edge by tick ~ 2000 at GRID $= 65^3$; the orbital escape across all three arms of `exp_20` is therefore inherited from the baseline, not caused by the emission operator. Resolving the long-horizon two-body stability question is the prerequisite for a settled emission run. An unanticipated phase-dependent non-monotonic amplitude transfer in arm B (the cross-term $2\,\mathrm{Re}(\sin\theta\,\psi_e\psi_\gamma^*)$ flips sign as the relative phase rotates) is a refinement to row 6 of the operation-algebra table that the follow-on calculation should address.

Three threads unified. The three-session emission event ties together three results that appear separately elsewhere in this paper. First, Zitterbewegung is the source of rest mass (Section 6) *and* the source of the symmetry-breaking phase exploration that drives both initial locking (Section 15.5) and eventual unlocking (this subsection). Second, Arnold tongue lock-in is what defines a quantized state (Section 15.4); the tongue's finite width $\Delta\omega_{\mathrm{tongue}}$ defines the boundary at which the lock breaks. Third, $\mathcal{A} = 1$ is what forces the third session into existence the moment the lock breaks: the displaced amplitude has nowhere else to go. These are not three independent postulates. They are three facets of the same lattice geometry, and emission is where all three become visible in a single tick-by-tick process.

Contrast with QED. Quantum electrodynamics quantizes the electromagnetic field independently and invokes the vacuum mode structure to provide the density of final states that drives spontaneous emission. The $\mathcal{T}^3_\diamond$ framework does neither. There is one lattice, one conservation law, one tick rule. The photon is not a quantum of a pre-existing field — it is a session that did not exist before the transition and exists afterwards because $\mathcal{A} = 1$ required it. The agreement with QED at low energies is not assumed; it is a prediction to be checked against the corrections derived below.

Falsifiable consequences. Two corrections to the standard emission picture follow from the mechanism above. The first is a clock-density correction: a region of higher ρ_{clock} slows the local tick rate (Section 7.4), and the joint two-body resonance detunes accordingly, shifting the rate at which a tongue-jump can occur. This predicts a measurable change in spontaneous emission rates near mass concentrations beyond the gravitational redshift of the emitted photon's frequency. The second is a tongue-width correction: an external field of magnitude proportional to $\Delta\omega_{\text{tongue}}/2$ should be sufficient to pop an electron out of a resonance, giving a calculable departure from the QED Stark and Lamb predictions. Both predictions are developed in Section 12.

The Lorentz oscillator identification. The picture above is the lattice realisation of the classical Lorentz oscillator model [33] of atomic optics. Hydrogen's spectroscopic absorption / emission lines are the Arnold-tongue Farey hierarchy (Section 13.3) restricted to bound-electron rates: the Bohr radius is the lattice's first stable rotation orbit, the Lyman / Balmer / Paschen series are the tongues at successive Farey neighbours, and absorption / emission line widths are the tongue half-widths $\Delta\omega_{\text{tongue}}/2$. The classical Lorentz natural frequency ω_0 is, in this framework, the hydrogen Zitterbewegung rate $\omega_e = \pi/(3R_1)$; absorption / emission at ω_0 is the joint two-body Arnold-tongue condition $\omega_e R_1 = \pi/3$ (Section 13.3, Eq. 121); anomalous dispersion near ω_0 is the response near a tongue boundary. Spectroscopic data that took a century to organise into a Lorentz $n(\omega)$ is, in this framework, a measurement of the tongue widths around each hydrogen-like rate on the bipartite octahedral lattice. The Stark / Lamb prediction (P5, Section 12.6) is the falsification handle for this identification.

15.8 Connection to the Gravitational Programme

The two-session resonance picture is the structural link between the quantum and gravitational results of this paper.

Quantization is an Arnold tongue lock-in of two coupled sessions. The tongue has a finite width $\Delta\omega_{\text{tongue}}$ in frequency space, set by the Coulomb strength and the lattice geometry. Gravity, in the $\mathcal{T}^3_\diamond$ framework, is a clock-density gradient (Section 7). A gravitational gradient across a two-body system couples differently to the two sessions, because the proton and electron occupy different positions in the gradient. The differential clock-rate shift effectively detunes the joint resonance by an amount proportional to the gradient strength and the orbital separation.

When the detuning exceeds $\Delta\omega_{\text{tongue}}$, the lock breaks. The atom transitions from Regime 1 to Regime 2 (bound but unquantized) or Regime 3 (unbound), depending on whether the sessions remain gravitationally bound after the resonance fails. This is the *quantum Roche limit*: a minimum orbital separation below which the gravitational gradient ionizes the atom by destroying the joint resonance, not by supplying escape energy.

This prediction — that strong gravitational gradients decohere atomic spectra via resonance detuning, not energy injection — is experiment 18. It has no analogue in standard quantum mechanics, which treats gravity as a perturbative correction to the Hamiltonian rather than as a detuning force on the quantization mechanism itself.

16 Conclusion

What has been established

This paper has made seven structural claims and supported each with a combination of analytic derivation and numerical experiment. We distinguish carefully between derivation (the claim follows from $\mathcal{A} = 1$ and the lattice geometry by proof) and demonstration (numerical experiment produces a result consistent with the claim, with no free parameters adjusted after the fact).

1. **The Dirac equation is a geometric consequence of the $\mathcal{T}^3_\diamond$ bipartite structure.** *Derived analytically* (Section 6): the RGB/CMY hop operator in Fourier space, Taylor-expanded near $\mathbf{k} = 0$, produces the term $i\alpha^i\partial_i + \beta m$ with the gamma matrices identified as the Clifford-algebra encoding of the three pairs of antiparallel basis vectors. The frame matrix $M = \sum_{\mathrm{RGB}} \mathbf{v}_i\mathbf{v}_j^T$ has eigenvalues $\{4, 4, 1\}$, producing a tilted ellipsoid in $\mathbf{k}$-space with long axis along $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$. The Clifford-algebra structure $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ emerges as a geometric identity under O_h averaging, with the same mechanism by which staggered fermions recover Lorentz invariance in lattice QCD. Spin-$\frac{1}{2}$, chirality, and mass as Zitterbewegung rate are consequences of the geometry, not additional inputs. Residual anisotropy along $(1, 1, -1)$ is a falsifiable birefringence prediction; see Section 12.8.

2. **Quantization is an Arnold tongue attractor.** *Demonstrated* (`exp_10`–`exp_12`): the hydrogen orbital wavevector $k_{\mathrm{Bohr}} = 1/R_1$ is recovered to four significant figures with no parameter tuning in the two-body simulation (`exp_12`: $k_{\min} = 0.0970$ vs $k_{\mathrm{Bohr}} = 0.0971$). The static Coulomb well is insufficient; the active proton session is required, confirming that quantization is a two-body resonance attractor, not a one-body eigenvalue problem.

3. **Gravity is clock-density refraction.** *Demonstrated* (`exp_02`, `exp_07`): the $1/r^2$ force law and gravitational time dilation emerge from the continuity equation for session density. The clock-fluid equations reduce to the Einstein field equations in the continuum limit; at finite lattice spacing they carry discrete corrections to the force law that are computable predictions (Section 7). Black-hole thermodynamics — the Bekenstein-Hawking area–entropy law $S = k_B A/(4\ell_P^2)$ and the Hawking temperature

$T_H = \hbar c^3/(8\pi GMk_B)$ — follow from queue saturation at Planck density via boundary-session counting and the Schwarzschild surface gravity, modulo the modelling assumptions stated explicitly at the point of derivation (Sections 4.6, 7.6).

4. **Maxwell's equations from the bipartite Wilson construction.** *Derived analytically* (Section 9, Appendix B): the Peierls-coupled tick rule produces $U(1)$ link variables, the plaquette Wilson loop is the field strength tensor $F_{\mu\nu}$, and the matter current is the standard Dirac current $\bar{\psi}\gamma^\mu\psi$. The standard Wilson action gives Maxwell's equations in the continuum limit. Appendix B additionally derives the *structural form* of the induced gauge action $-\mathrm{Tr}\,\ln D_{\mathrm{lat}}[U]$ on the bipartite lattice (audit table: `PART`, numerical $1/g^2$ prefactor open): the leading plaquette-sum has the Maxwell form on O_h-averaged observables, with a closed-form gauge-sector birefringence (operator-level eigenvalues $\{4, 4, 16\}$, same optical axis $(1, 1, -1)$ as the kinematic-sector prediction) as a falsifiable residual signature (audit table: `STUB`, empirical test open). Substrate-level unification: gravity (divergence channel) and electromagnetism (curl channel) are populations of the same virtual-session cloud mechanism, distinguished only by the sublattice signature of the source's internal flow (Sections 9.1, 7.2). The numerical $1/g^2$ prefactor of the induced action and the empirical test of the gauge-sector birefringence both remain as paper-length follow-on calculations.

5. **The Born rule follows from $\mathcal{A} = 1$.** *Derived analytically* (Section 5): probability $|\psi|^2$ is the only positive conserved quantity consistent with $\mathcal{A} = 1$ on the bipartite lattice. Gleason's theorem applied to the C^*-algebra induced by the bipartite structure identifies $|\psi|^2$ as the unique probability measure; no measurement axiom is needed.

6. **Photon emission is a structural necessity.** *Derived analytically; partial numerical verification* (Section 15, `exp_19c` and `exp_20`; audit table: `PART` for both "Photon emission as $\mathcal{A} = 1$ necessity" and "Emission-operator joint $\mathcal{A} = 1$ conservation"): the Coulomb interaction is a probability source that a single session cannot absorb without violating $\mathcal{A} = 1$; a new session (the photon) is mandated. `exp_20`'s unitary beam-splitter operator preserves joint $\mathcal{A} = 1$ to machine precision, but neither implementation has yet exhibited a settled $n{=}2$ to $n{=}1$ transition — the inherited two-body baseline instability seen in `exp_12b` must be resolved first.

7. **The universe is a fractal.** *Derived analytically; falsifiable consequences* ***`STUB`*** (Section 14): the $\mathcal{A} = 1$ unity constraint forces the tick rule to be a symplectic map on the phase torus $\mathbb{T}^2$. By the KAM theorem [35, 36, 37], the set of stable orbits of any symplectic map is a Cantor set — closed, nowhere dense, of Hausdorff dimension $D_H < 1$.

The Arnold tongues visible in the harmonic landscape (Figure 6) are the rational-frequency islands of this Cantor set; the particles we observe are conjectured to be exactly those islands. The Hausdorff dimension estimate $D_H \approx 0.870$ extracted from $\mathcal{T}^3_\diamond$ tongue-width statistics is a parameter-free, falsifiable prediction; the box-counting test of the lattice's harmonic landscape and the proposed Farey-sequence ordering of the Standard Model mass spectrum (ordered by tongue width / stability) both remain `STUB` pending experimental and numerical validation. Under this conjecture, the fractal is not an emergent large-scale pattern but the fundamental structure of the space of stable dynamical states: probability conservation on a bipartite lattice would then force the universe to be a fractal at the level of its most elementary objects.

The unifying chain: topology forces physics

The seven results above share a common structure. Each is a consequence of the same chain:

$$\textit{Bipartite topology} \longrightarrow \textit{Algebra of observables} \longrightarrow \textit{Physics.}$$

The steps are determinate, not chosen. The bipartite structure of $\mathcal{T}^3_\diamond$ is a topological invariant that divides every node into one of two classes (RGB or CMY parity), inducing a $\mathbb{Z}_2$ grading on every observable. A probability space with a $\mathbb{Z}_2$ grading is a superalgebra: observables split into even (grade-0, commuting) and odd (grade-1, anticommuting) components. This is not assumed — it follows from the topology.

The three-fold connectivity of each sublattice imposes the tight-frame condition, which forces the path algebra of two-tick steps to be Clifford$(3,1) \cong M_4(\mathbb{C})$ — the Dirac algebra. The $\mathcal{A} = 1$ constraint imposes positivity, unit trace, and self-adjointness, converting this to a C^*-algebra with a faithful tracial state — the algebraic structure of quantum mechanics on a finite-dimensional Hilbert space. Hurwitz's theorem (the only normed division algebras are $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$) then forces complex amplitudes: real amplitudes cannot represent phase interference; quaternionic amplitudes produce inconsistent tensor products for composite systems. Complex quantum mechanics is the unique probability theory compatible with interference, consistent composites, and $\mathcal{A} = 1$.

The conclusion is precise:

> *The probability rule $\mathcal{A} = 1$ on the bipartite $\mathcal{T}^3_\diamond$ topology does not merely reproduce quantum mechanics — it forces it. The complex amplitude space, the Clifford algebra structure, the Born rule, and the fermionic statistics of matter fields are each determined by the topology alone. The remaining question is whether the Standard Model gauge group* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *is the automorphism group of*

this topology — in which case the entire Standard Model is a theorem about the geometry of probability conservation on a bipartite discrete 3-manifold.

The automorphism group of the full $\mathcal{T}^3_\diamond$ structure (bipartite + octahedral symmetry + three-fold RGB basis + $\mathcal{A} = 1$) is larger than the Lorentz group $\mathrm{SO}(3,1)$ that the Clifford algebra alone produces. The conjecture

$$\mathrm{Aut}(\mathcal{T}^3_\diamond, \mathcal{A} = 1) \;=\; \mathrm{SO}(3,1) \times \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1) \tag{137}$$

is a precise, finite-dimensional Lie-algebra equality: both sides have real dimension $18 = 6+8+3+1$. The match is more than dimensional — the structure constants of the lattice's automorphism algebra must reproduce those of the right-hand side, factor by factor. The five contributing structures are:

- $\mathrm{SO}(3,1)$ (dim 6): continuum lift of the cubic point group O_h acting on the spatial sublattice, plus boosts emerging from the O_h-averaged Dirac equation (Section 6.4).
- $\mathrm{U}(1)$ (dim 1): per-site phase rotation $\psi \mapsto e^{i\theta}\psi$; the $\mathcal{A} = 1$ conservation law is exactly the corresponding Noether charge.
- $\mathrm{SU}(2)_W$ (dim 3): rotation of a per-site internal $\mathbb{C}^2$ isospin doublet, distinct from the chiral (ψ_R, ψ_L) pair; this index is a structural extension of the framework analogous to the per-site colour index below.
- $\mathrm{SU}(3)$ (dim 8): rotation of a per-site internal $\mathbb{C}^3$ colour triplet recording the amplitude that the wavefunction's most recent RGB tick was along $\mathbf{V}_1$, $\mathbf{V}_2$, $\mathbf{V}_3$ respectively; this index is a structural extension making explicit a choice that the bipartite tick rule already makes implicitly per tick.

Of these four, $\mathrm{SO}(3,1)$ and $\mathrm{U}(1)$ are *established* on the framework's existing per-site $\mathbb{C}^2 = (\psi_R, \psi_L)$: the chiral spinor carries the Lorentz $SO(3,1)$ representation (Section 6.4) and the per-site phase rotation is the Noether charge of $\mathcal{A} = 1$. Computational verification on the existing $\mathbb{C}^2$ confirms $\mathfrak{so}(3,1) \oplus \mathfrak{u}(1)$ as the established automorphism algebra (dim 7), and shows that the proposed "per-site $SU(2)$" on (ψ_R, ψ_L) overlaps exactly with the Lorentz rotation subgroup of $SO(3,1)$ rather than forming a separate factor.

The remaining two factors, $\mathrm{SU}(2)_W$ and $\mathrm{SU}(3)$, are open: each requires the framework to carry an additional per-site internal index (a $\mathbb{C}^2$ isospin doublet for $\mathrm{SU}(2)_W$ and a $\mathbb{C}^3$ colour triplet for $\mathrm{SU}(3)$) beyond the existing chiral spinor. The discrete RGB symmetry of the existing framework embeds in $U(3)$ but contributes only the cyclic subgroup $\mathbb{Z}_3 \subset$

$SU(3)$, which is abelian and cannot generate the full non-abelian SU(3) by any Lie-theoretic operation; analogous obstructions apply to the $\mathrm{SU}(2)_W$ extension.

A complete proof requires identifying the structural extensions and explicit computation of the extended lattice's automorphism Lie algebra structure constants, verifying they reproduce $\mathfrak{so}(3,1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. The verification is a tractable finite-dimensional computation using existing Lie-algebra software (GAP, SymPy [42]); the structural-extension question is the central open problem of the program and the subject of a companion paper.

If Eq. (137) succeeds, the Standard Model gauge group is derived from a single conservation axiom on a discrete substrate. If it fails, the framework's verified scope is the relativistic Hilbert-space + gravity + Born-rule structures of Sections 6, 7, and Section 10.7, with the SM-derivation claim explicitly retracted. Either outcome is a complete result; the question is the calculation, not the framing.

New results not in prior discrete spacetime frameworks

Several results in this paper distinguish the $\mathcal{T}^3_\diamond$ framework from earlier discrete spacetime proposals [2, 3, 4, 5]:

- **Two-body requirement for quantization.** No prior discrete framework to our knowledge identifies the proton's active session dynamics as a necessary component of quantization. The static Coulomb well approximation is not merely imprecise; it predicts a qualitatively wrong landscape (`exp_11` n=2: flat across all k; `exp_12`: 7.3% fixed-well error corrected to 0.09% by active proton).

- **Three dynamical regimes.** Standard quantum mechanics distinguishes bound eigenstates from continuum states. The $\mathcal{T}^3_\diamond$ framework adds a third regime: bound but unquantized. A two-body system whose recoil prevents Arnold tongue lock-in stays together indefinitely but has no discrete spectrum and emits no photons. `Exp_16` measures the mass threshold at which this regime gives way to the quantized regime; the threshold is a lattice prediction with no analogue in the standard formalism.

- **Gravitational ionization as detuning.** The quantum Roche limit (`exp_18`, planned) predicts that a strong gravitational gradient across the atomic orbital radius detunes the joint proton-electron resonance below the Arnold tongue width, ionising the atom without supplying escape energy. This is a falsifiable prediction with no analogue in standard quantum mechanics, which treats gravity as a perturbative correction to the Hamiltonian rather than a detuning force on the quantization mechanism.

Falsifiable predictions

Section 12 develops nine primary falsifiable predictions (P1–P9) at experimentally accessible scales, each with its Compton-calibration upper bound and its falsification criterion; P7 already constrains the lattice spacing from above by GRB time-of-flight, $a \lesssim 10^{-19}\,\mathrm{m}$. Of these, five illustrate the framework's distinctive testability:

1. **Short-range correction to the $1/r^2$ gravitational force law** (Section 7). The lattice path-count derivation produces a correction term that departs from $1/r^2$ at separations approaching the lattice spacing. The correction is computable from the lattice geometry and contains no free parameters after the lattice spacing is fixed.

2. **Quantum Roche limit** (Section 15.8, `exp_18`). There exists a minimum orbital separation r_{ISCO} below which a gravitational gradient ionises the atom by detuning the joint resonance. The threshold depends only on the Arnold tongue width (set by the Coulomb strength and lattice geometry) and the gravitational gradient. Its value is a parameter-free prediction once the lattice calibration is fixed.

3. **Bound unquantized matter near compact objects.** The existence of Regime 2 (bound but unquantized) predicts that matter within the gravitational ionization radius of a compact object is held together but spectroscopically dark: no discrete emission lines, no atomic recombination. The cosmological recombination epoch, on this picture, is a clock-fluid phase transition occurring when the global clock-density gradient drops below the Arnold tongue width of hydrogen.

4. **Pair annihilation efficiency at $\omega = \pi/2$** (`exp_17`, planned). The harmonic landscape is symmetric about $f = 1/4$, with the 2:1 fixed point at $\omega = \pi/2$. At this mass, a particle and its antiparticle are exact mirror images in the probability space. Annihilation cross-section should peak at $\omega = \pi/2$ and fall off symmetrically for $\omega \neq \pi/2$ — a lattice-geometry prediction.

5. **Multi-channel concordance of the optical axis** (P9, Section 12.10). The kinematic, gauge, gravitational, and (under the recoil-driven Zitterbewegung hypothesis) inertial-mass anisotropies all share a single optical axis $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$ by direct calculation from the bipartite RGB/CMY frame matrix. Five independent observation channels — GRB time-of-flight, GRB / CMB polarisation rotation, LIGO/Virgo polarisation-mode timing, direction-resolved atomic-clock comparisons, and anomalous-dispersion anisotropy near atomic resonances — are predicted to agree on a single sky direction, with the dual ellipsoid shapes (kinematic prolate, gauge

oblate; Figure 9) providing within-channel sub-predictions. Inter-channel disagreement falsifies the inheritance-from-bipartite-geometry claim. Three time-dependence scenarios (uniform, domain, dynamical embedding) are distinguishable by sidereal modulation (atomic clocks; fast handle) and redshift-resolved binning of GRB residuals (slow handle). This is the framework's strongest empirical claim: one geometric axis, multiple independent witnesses.

What remains open

The remaining gaps are acknowledged explicitly. They are scoped follow-on calculations, not pending bug fixes; each has a documented analytical or experimental path forward.

1. The clock-density coupling $\rho_{\text{clock}} \propto \exp(-\phi/c^2)$ is motivated by the exponential suppression of session density in a deep potential well, but has not yet been derived from the tick rule alone. It is the gravitational analogue of the step from the hop operator to the Dirac equation, and we expect it to follow from the continuum limit of the clock-density continuity equation. This is the primary remaining step in the gravity derivation programme.

2. The automorphism group calculation $\mathrm{Aut}(\mathcal{T}^3_\diamond, \mathcal{A} = 1) \stackrel{?}{=} \mathrm{SO}(3,1)\times\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ is not performed in this paper. It requires a systematic enumeration of the symmetries of the bipartite basis under the $\mathcal{A} = 1$ constraint. This is a well-posed mathematical problem and the subject of the companion paper.

3. `Exp_16` (proton mass sweep) is complete and confirms the qualitative three-regime structure: every ω_p value tested in the range $[0.3, \pi/2]$ produces a bound but unquantized state (Regime 2), establishing that binding and quantization are distinct conditions on the lattice. The numerical ω_p threshold separating Regime 1 (quantized) from Regime 2 lies outside the swept range and has not yet been localised; `exp_18` (quantum Roche limit) is planned and will fix the corresponding gradient threshold. Its measurement requires a confirmed Arnold-tongue lock-in as a stable base state, and is therefore tied to the long-horizon stability question (item 5 below).

4. The black-hole thermodynamics derivations (Sections 4.6, 7.6) rest on two stated modelling assumptions: that each saturation-boundary session occupies a four-cell patch ($A_{\text{session}} = 4\ell_P^2$) and that the boundary inherits the Schwarzschild surface gravity $\kappa = c^4/(4GM)$. Both are tied to specific follow-on calculations (microstate counting and the strong-field clock-fluid limit, respectively).

5. Long-horizon two-body stability. The bare `exp_12` setup, extended to a 6000-tick horizon at GRID $= 65^3$ (`exp_12b`), shows the orbital peak escaping from R_1 to the grid edge by tick ~ 2000 and not returning. The escape is independent of the emission operator (`exp_20` arms A, B, and C all inherit it), so the orbital instability seen across the emission experiments traces to the bare two-body dynamics rather than to the emission rule. Whether the escape is grid-size dependent, an artifact of CoM-frame initial-condition residual, or the bipartite tick rule's near-unitary approximation accumulating over $\gtrsim 10^3$ ticks is the next experimental question. This refines but does not invalidate `exp_12`'s 4-sig.-fig. PASS, which is scored on a shorter k-scan resonance peak.

6. The numerical prefactor $c = 1/g^2$ for the induced gauge action (Appendix B) requires the explicit one-loop $-\mathrm{Tr}\ln D_{\mathrm{lat}}[U]$ calculation in standard staggered-fermion machinery. The structural form, the bipartite plaquette tensor $\mathbf{Q}$ with eigenvalues $\{4, 4, 16\}$, and the closed-form gauge-sector birefringence are all in hand; only the prefactor remains.

7. Empirical test of the gauge-sector birefringence prediction. Appendix B derives the closed-form operator-level birefringence (eigenvalues $\{4, 4, 16\}$ on the $(1, 1, -1)$ optical axis) as a falsifiable prediction. The empirical confrontation against current GRB / CMB / atomic-clock bounds is the multi-channel concordance program (P9, Section 12.10); the analytical prediction stands, and the empirical confrontation is the next phase.

Anticipated objections

Several reviewer concerns can be anticipated and answered in advance. We list them in the order most likely to be raised.

A discrete lattice plus a scheduler implies a preferred frame; how can this be Lorentz invariant? The scheduler is a bookkeeping device, not a physical clock. The order in which independent sessions advance within a macro-tick has no observable consequence: `exp_05` confirms that sequential, random, and priority orderings produce identical statistical outcomes (Section 11). Lorentz invariance emerges in the continuum limit by the same mechanism that recovers it in lattice QCD with staggered fermions [11, 16]. The bipartite frame matrix $M = \sum_{\mathrm{RGB}} \mathbf{v}_i \mathbf{v}_j^T$ (Eq. 38) is anisotropic at the operator level with eigenvalues $\{4, 4, 1\}$, but its O_h-averaged form $\langle M \rangle_{O_h} = (\mathrm{tr}\, M/3)\, I = 3I$ forces the long-wavelength dispersion relation to $E^2 = m^2 + |\mathbf{p}|^2$, restoring full $\mathrm{SO}(3,1)$ on O_h-averaged observables

(Section 6.4). The operator-level residual is the lattice birefringence along the optical axis $\hat{\mathbf{n}}_* = (1,1,-1)/\sqrt{3}$ (Eq. 41) — a falsifiable prediction in its own right (P3, Section 12.4; P7 direction-resolved refinement, Section 12.8; P9 multi-channel concordance, Section 12.10).

Claiming to derive the Standard Model is a strong claim without a full mass-spectrum calculation. This paper does not claim to derive the full Standard Model. It claims that the bipartite $\mathcal{T}^3_\diamond$ geometry, together with $\mathcal{A} = 1$, forces (i) the Dirac equation, (ii) complex-amplitude quantum mechanics with the Born rule, (iii) the Clifford algebra and chirality structure of fermions, and (iv) gravity as clock-density refraction. The remaining question — whether $\mathrm{Aut}(\mathcal{T}^3_\diamond, \mathcal{A} = 1) = \mathrm{SO}(3,1) \times \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ — is stated explicitly as the central open problem and is the subject of a companion paper. The particle content is reframed, not derived from first principles in this paper: the elementary fermions are the minimal set of session types required to maintain $\mathcal{A} = 1$ on the bipartite geometry, and the discrete mass spectrum is conjectured to be the Farey hierarchy of Arnold tongue stability (Section 14).

How do we know the continuum matching to QM and GR is not tuned? There are no continuous free parameters in this framework. Three discrete choices are made: the lattice topology (bipartite octahedral), the conservation law ($\mathcal{A} = 1$), and the calibration scale (Planck or, equivalently, Compton). Every quantitative result — the Bohr radius to four significant figures (`exp_12`), the identity $\omega \cdot R_1 = \pi/3$ to 0.23% (`exp_11`), Newton's $1/r^2$ from clock-density gradients (`exp_02`) — is computed from these inputs alone. No coupling constant, mass, or coefficient is fitted to data after the fact; the experiments report whatever the lattice produces, and the audit table (Table 1) records both successes and the regimes (e.g. `exp_11` at $n = 2$ with a fixed well) where the lattice produces a flat or null result.

Is $\mathcal{A} = 1$ truly unitary, given that sessions can be created or destroyed (photon emission)? $\mathcal{A} = 1$ is unitarity *per session*: each session's spinor amplitude satisfies $|\psi_R|^2 + |\psi_L|^2 = 1$ at every tick, enforced numerically by `enforce_unity_spinor`. Multi-session events preserve the joint norm $\sum_{\text{sessions}} |\psi_s|^2$; what changes during photon emission is the number of session-bookkeeping channels, not the total probability. Session creation is a structural reconfiguration of the bookkeeping (Section 15), analogous to how QED enlarges Hilbert space when a photon is emitted.

Black-hole thermodynamics rests on stated modelling assumptions; is this circular? Two assumptions are stated explicitly at the point of use: that each saturation-boundary session occupies a four-cell patch ($A_{\text{session}} = 4\ell_P^2$), and that the boundary inherits the Schwarzschild surface gravity $\kappa = c^4/(4GM)$ (Sections 4.6, 7.6). Both are tied to specific follow-on calculations: microstate counting on the boundary lattice, and the strong-field limit of the clock-fluid equations. The status (`PART`) in Table 1 reports this honestly: the Bekenstein-Hawking law and Hawking temperature are recovered conditionally on those assumptions, not derived end-to-end in this paper.

How does this compare with causal sets, Wolfram, and 't Hooft's cellular automaton? Section 2.6 discusses each comparison in detail. Briefly: causal set theory [2] starts from a partial order without a fixed coordination number; the cellular-automaton programme [3, 4, 5] starts from local update rules without a probability conservation law. The $\mathcal{T}_\diamond^3$ framework is the intersection: a discrete causal structure with fixed coordination six (forced by the bipartite octahedral geometry) and a single conservation law ($\mathcal{A} = 1$) from which the rest follows. The key novel ingredient is not the lattice but the constraint: $\mathcal{A} = 1$ as a per-session, per-tick identity rather than a global norm condition.

Closing

$\mathcal{A} = 1$ is not a law of physics. It is the precondition for physics: the statement that probability must be conserved if the framework is to be a probability framework at all. From this tautological starting point, on a specific bipartite geometry, everything else follows: the wave equation, the spectrum, the force laws, the statistics of matter.

The Standard Model is not a theory imposed on nature from outside. It is the accounting system that a conserved-probability framework on a bipartite discrete 3-manifold is forced to adopt. $\mathcal{A} = 1$ is the account.

Is what we observe a self-contained projection of the lattice's own dynamics, or the visible side of a process driven from something outside it — a question P9's time-dependence subprediction (§12.10) renders empirical, by asking whether the optical axis is stationary across cosmological epochs or drifts?

A Code and Data

This appendix documents what is required to reproduce the numerical results reported in the paper. The framework is implemented entirely in Python with a small set of mainstream scientific dependencies; no specialised hardware is required for any single experiment, although several long-running scans benefit from leaving a workstation undisturbed for hours to days.

Repository and archived release

- Source code, paper sources, and data files: `https://github.com/JackDMenendez/discrete-causal-`
- Archived release (this version): `doi:10.5281/zenodo.20017133`
- Citation metadata: `CITATION.cff` at the repository root.
- License: MIT.

Software environment

Tested on Windows 11 (MSYS2 UCRT64 shell) and Linux. The build expects:

- Python 3 with the packages pinned in `virtual-env-requirements.txt` (`numpy 2.4.3`, `scipy 1.17.1`, `matplotlib 3.10.8`, `pandas 3.0.1`, `networkx 3.6.1`, `seaborn 0.13.2`; full list in the file).
- GNU Make $\geq$ 4.3 (the grouped-target `&:` syntax is used).
- `pdflatex` and `bibtex` (MiKTeX or TeX Live) for building this paper.

A clean build proceeds via the top-level `makefile`:

`make env`	*create the Python virtualenv*
`make tests`	*run the* ***`pytest`*** *suite under* ***`tests/`***
`make paper`	*compile* ***`paper/main.tex`*** *(3-pass* ***`pdflatex`*** *+* ***`bibtex`****)*

Determinism and random-number use

The framework is deterministic by construction: the tick rule is a fixed map on the spinor amplitude, and orbital evolution depends only on the initial state and the lattice geometry. Random numbers enter in two places only:

1. `src/core/TickScheduler.py` uses `np.random.default_rng(seed=42)` when an experiment requests randomised tick ordering. `exp_05` relies on this for the shuffle-invariance test; all other experiments use the deterministic sequential schedule.

2. `src/experiments/exp_04_decoherence.py` draws random phase scrambles via `np.random.default_` for the many-bath ensemble; the per-realisation seed is `i + 100` for the i-th member of the ensemble.

With these seeds fixed at the values committed to the repository, every experiment in this paper reproduces bit-for-bit on the pinned software stack.

Running the numerical experiments

Individual experiments are exposed as targets in `src/experiments/makefile`:

```
make -C src/experiments exp_12
make -C src/experiments exp_16
```

Each target depends on the Python script of the same name, runs it under the project virtualenv, and writes its log to `data/exp_NN_*.txt` and its raw arrays to `data/exp_NN_*.npy`. The longer scans (`exp_11`, `exp_18`, `exp_20`) have wall-clock times of hours to days on consumer hardware; the per-experiment documentation in `src/experiments/*.md` reports the runtime observed during preparation of this paper.

Key parameter values

The numerical experiments share a small set of physical and discretisation parameters. Values used to produce the results reported in the audit table (Table 1) are summarised in Table 4.

Table 4: Parameter values for the numerical experiments cited in the audit table. Lattice units throughout: distance in nodes, time in ticks, $\hbar = c = 1$. `GRID` is the side length of a cubic grid; `TICKS` is the total simulated time per run.

Experiment	GRID	TICKS	Key parameters	Source
`exp_00`	51^3	20	$\omega \in \{0, 0.5, 1.0\}$	`exp_00_causa`
`exp_02`	65^3	4000	test particle, mass well at centre	`exp_02_gravi`
`exp_06`	—	$N \in \{4, \ldots, 64\}$	exact path-count integers	`exp_06_path_`
`exp_08`	65^3	1500	EM vs. gravity, Peierls $A \cdot v$ coupling	`exp_08_vacuu`
`exp_09`	65^3	4000	$f \in (0, 1)$, $\omega \in (0, 2\pi)$ harmonic scan	`exp_09c_harm`
`exp_10`	65^3	8000	$\omega = 0.1019$, `STRENGTH` $= 30.0$	`exp_10_hydro`
`exp_11`	65^3	8000	$\omega = 0.1019$, `STRENGTH` $= 30.0$, `BURN_IN` $= 500$	`exp_11_quant`
`exp_12`	65^3	1500 orbit / 2500 scan	$\omega_e = 0.1019$, $\omega_p = \pi/2$, `STRENGTH` $= 30.0$	`exp_12_hydro`
`exp_16`	65^3	20000	$\omega_p \in [0.3, \pi/2]$, `BURN_IN` $= 4000$	`exp_16_proto`

The shared physical constants are: instruction frequency $\omega_e = 0.1019$ (electron); proton instruction frequency $\omega_p = \pi/2$ (maximum lattice mass); Coulomb strength 30.0; softening

0.5; reference Bohr radius $R_1 = 10.3$ nodes (the value at which $\omega \cdot R_1 = \pi/3$ saturates within 0.23%).

Data products

Each experiment writes its raw arrays and a human-readable log to `data/` at the repository root. The most consequential output files are:

- `data/exp_06_path_counts.npy` — exact path-count integers $P(N, a, b, c)$ underlying the discrete-correction prediction P1 (Section 12).
- `data/exp_09_harmonic_hires.npy` — (f, ω)-resolved spectral power, the source array for the harmonic landscape (Figure 6).
- `data/exp_12_twobody_scan.npy` — two-body k-scan with the live proton, source for the $k_{\text{min}} = 0.0970$ vs. $k_{\text{Bohr}} = 0.0971$ result (Section 15, Figure 8).
- `data/exp_16_proton_mass_sweep.npy` — proton-mass sweep establishing the three-regime structure (Section 15).

A representative output snippet from `exp_12` (full log: `data/exp_12_hydrogen_twobody_log.txt`):

```
[Test 3] Mini k-scan (8 values, 2500 ticks each)
  k_Bohr_fixed = 0.09709
  ...
Two-body k_min = 0.0970  vs fixed-well 0.0900  vs k_Bohr_fixed 0.0971
```

This is the four-significant-figure recovery of the Bohr radius cited throughout the paper, produced with no parameter tuned to the result.

Reproducing this paper

The LaTeX source compiles from `paper/main.tex` via `paper/build.cmd` (Windows) or `paper/build.sh` (POSIX), or from the repository root via `make paper`. The build performs three `pdflatex` passes around one `bibtex` pass to settle cross-references and the bibliography. The figures referenced in this paper are committed to `figures/` as PDFs and `.tex` caption fragments; re-running the experiments is not required to rebuild the document.

B The Induced Gauge Action

This appendix carries out the structural step of the induced-action programme flagged in Section 9.7. The Sakharov–Zeldovich claim is that the Wilson gauge action of Eq. (94) should not be postulated but rather *induced* by integrating out matter sessions in the path-integral form of the bipartite tick rule. A complete execution of that programme requires the explicit one-loop computation of $-\mathrm{Tr}\,\ln D_{\mathrm{lat}}[U]$ with the bipartite Dirac operator. We do not perform that computation here – it is a paper-length project (lattice perturbation theory for staggered-style fermions on the $\mathcal{T}^3_\diamond$ lattice, including the standard fermion-doubling treatment). What we do compute is the *structural form* of the leading term: the tensor coefficient on $F_{\mu\nu}F_{\rho\sigma}$ that emerges when the closed-loop expansion of $-\mathrm{Tr}\,\ln D_{\mathrm{lat}}[U]$ is restricted to the smallest gauge-invariant Wilson loops on the bipartite octahedral lattice.

The computation is symbolic and reproducible. The script that performs it is `src/utilities/induced_g` its captured output is `data/induced_gauge_action.txt`. Both are tracked in the repository.

The result of this appendix is twofold. First, after O_h averaging the leading induced action recovers the standard Maxwell density (with the numerical prefactor $1/g^2$ left to the explicit Tr ln calculation). Second, the *operator-level* form of the induced action carries a closed-form anisotropic tensor coefficient with eigenstructure $\{4, 4, 16\}$, mirroring the $\{4/3, 4/3, 0\}$ eigenstructure of the kinematic frame matrix M_{eff}. This is the gauge-sector birefringence: a direction-dependent contribution to the photon dispersion which aligns with the same optical axis $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$ as the kinematic-sector birefringence already in P7 (Section 12.8). It is not a new postulate; it is forced by the bipartite geometry.

B.1 Setup

The bipartite octahedral lattice has basis vectors

$$\mathbf{V}_1 = (1,1,1), \quad \mathbf{V}_2 = (1,-1,-1), \quad \mathbf{V}_3 = (-1,1,-1) \in \mathrm{RGB}, \qquad -\mathbf{V}_i \in \mathrm{CMY}. \tag{138}$$

A spinor field $\Psi(\mathbf{x}) = (\psi_R(\mathbf{x}), \psi_L(\mathbf{x}))$ carries a $U(1)$ phase per session (Section 3.1). The Peierls coupling inserts a link variable

$$U_{\mathbf{v}}(\mathbf{x}) = \exp\big(i\,a\,\mathbf{A}(\mathbf{x}+\mathbf{v}/2)\cdot\mathbf{v}\big) \;\in\; U(1) \tag{139}$$

on every hop along $\mathbf{v} \in \{\pm\mathbf{V}_1, \pm\mathbf{V}_2, \pm\mathbf{V}_3\}$, where a is the lattice spacing. The mid-point convention $\mathbf{A}(\mathbf{x}+\mathbf{v}/2)$ is the symmetrised Peierls form; it cancels what would otherwise be a corner artifact at order a^3. Under a local $U(1)$ gauge transformation $\Lambda(\mathbf{x})$ the link variables transform as $U_{\mathbf{v}}(\mathbf{x}) \to e^{i\Lambda(\mathbf{x}+\mathbf{v})}\, U_{\mathbf{v}}(\mathbf{x})\, e^{-i\Lambda(\mathbf{x})}$, so closed-loop holonomies are gauge-invariant local operators in A_μ.

The smallest non-trivial closed loop on the bipartite lattice is the *bipartite plaquette*: an RGB hop along $\mathbf{V}_a$, a CMY hop along $-\mathbf{V}_b$, an RGB hop along $-\mathbf{V}_a$, and a CMY hop along $+\mathbf{V}_b$, returning to the origin $\mathbf{V}_a - \mathbf{V}_b - \mathbf{V}_a + \mathbf{V}_b = 0$. Three such plaquettes exist, one per pair $(\mathbf{V}_a, \mathbf{V}_b) \in \{(\mathbf{V}_1, \mathbf{V}_2), (\mathbf{V}_1, \mathbf{V}_3), (\mathbf{V}_2, \mathbf{V}_3)\}$.

B.2 Plaquette holonomy in the small-a limit

Because $U(1)$ is Abelian the bipartite plaquette holonomy reduces to a line integral of $\mathbf{A}$ around the loop, which by Stokes' theorem is the curl-flux through the parallelogram. Concretely, for the path $\mathbf{x} \to \mathbf{x} + a\mathbf{V}_a \to \mathbf{x} + a\mathbf{V}_a - a\mathbf{V}_b \to \mathbf{x} - a\mathbf{V}_b \to \mathbf{x}$, the line integral of $\mathbf{A}$ to leading order in a is

$$\oint \mathbf{A} \cdot d\mathbf{x} = -a^2\, V_a^i V_b^j F_{ij}(\mathbf{x}) + O(a^4), \tag{140}$$

with $F_{ij} = \partial_i A_j - \partial_j A_i$ the standard field-strength tensor. The plaquette holonomy is therefore

$$W_{ab}(\mathbf{x}) = \exp\!\big(-i\, a^2\, V_a^i V_b^j F_{ij}(\mathbf{x}) \big) + O(a^4), \tag{141}$$

generalising Eq. (90) to the bipartite parallelogram. The standard cardinal-axis plaquette result is recovered by setting $\mathbf{V}_a = \hat{\mu}, \mathbf{V}_b = \hat{\nu}$, in which case $V_a^i V_b^j F_{ij} = F_{\mu\nu}$.

The small-a expansion of $1 - \operatorname{Re} W_{ab}$ is then

$$1 - \operatorname{Re} W_{ab}(\mathbf{x}) = \frac{a^4}{2} \big(V_a^i V_b^j F_{ij}(\mathbf{x}) \big)^2 + O(a^8), \tag{142}$$

the closed-form analogue of $\frac{a^4}{2} F_{\mu\nu}^2$ for our bipartite lattice.

B.3 The plaquette sum: tensor form of the leading induced action

The leading non-trivial term in the closed-loop expansion of $-\mathrm{Tr}\, \ln D_{\mathrm{lat}}[U]$ is the sum over all minimal gauge-invariant Wilson loops with a multiplicity factor set by the explicit one-loop calculation. The structural form, deferring that prefactor, is

$$S_{\mathrm{eff}}^{(\mathrm{plaquette})}[A] = \frac{c\, a^4}{2} \sum_{a<b} \big(V_a^i V_b^j F_{ij}(\mathbf{x}) \big)^2, \tag{143}$$

where c is the inverse coupling $1/g^2$ that the explicit Tr ln calculation must determine. The sum over $(a, b) \in \{(1, 2), (1, 3), (2, 3)\}$ runs over the three independent bipartite plaquettes.

Substituting the basis vectors of Eq. (138) and expanding gives the three plaquette projections

$$V_1^i V_2^j F_{ij} = -2F_{12} - 2F_{13}, \tag{144}$$
$$V_1^i V_3^j F_{ij} = +2F_{12} - 2F_{23}, \tag{145}$$
$$V_2^i V_3^j F_{ij} = -2F_{13} + 2F_{23}, \tag{146}$$

with F_{12}, F_{13}, F_{23} the three independent components of the 3D antisymmetric field-strength tensor. Squaring and summing,

$$\sum_{a<b} \big(V_a^i V_b^j F_{ij}\big)^2 = \mathbf{F}^T \mathbf{Q}\, \mathbf{F}, \qquad \mathbf{F} = (F_{12},\, F_{13},\, F_{23})^T, \tag{147}$$

where the symmetric matrix $\mathbf{Q}$ is

$$\mathbf{Q} = \begin{pmatrix} 8 & 4 & -4 \\ 4 & 8 & -4 \\ -4 & -4 & 8 \end{pmatrix}. \tag{148}$$

B.4 Eigenstructure of Q: the gauge-sector birefringence

The matrix $\mathbf{Q}$ has trace $\operatorname{Tr}\mathbf{Q} = 24$ and eigenvalues $\{4, 4, 16\}$. The two-fold degenerate eigenvalue 4 spans the plane perpendicular to the eigenvector $(-1, -1, 1) \in \mathbf{F}$-space; the distinct eigenvalue 16 points along that special direction. Decomposing $\mathbf{Q}$ into an isotropic part and a traceless residual,

$$\mathbf{Q} = \frac{\operatorname{Tr}\mathbf{Q}}{3}\, \mathbb{I}_3 + \mathbf{Q}_{\text{aniso}}, \qquad \frac{\operatorname{Tr}\mathbf{Q}}{3} = 8, \qquad \mathbf{Q}_{\text{aniso}} = \begin{pmatrix} 0 & 4 & -4 \\ 4 & 0 & -4 \\ -4 & -4 & 0 \end{pmatrix}, \tag{149}$$

with $\mathbf{Q}_{\text{aniso}}$ trace-zero and eigenvalues $\{-4, -4, +8\}$.

Two structural features. First, the eigenstructure of $\mathbf{Q}$ mirrors the kinematic frame matrix M_{eff}: a degenerate pair plus one distinct eigenvalue. Second, the special direction in $\mathbf{F}$-space, $\mathbf{F}_* = (-1, -1, 1)$, dualises to a spatial vector via the Hodge map $B_k = \frac{1}{2}\epsilon_{ijk} F_{ij}$, giving

$$\mathbf{B}_* = \big(F_{23},\, -F_{13},\, F_{12}\big)\big|_{\mathbf{F}=\mathbf{F}_*} = (1,\, 1,\, -1) = \mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3. \tag{150}$$

The gauge-sector special direction is the same optical axis $(1, 1, -1)$ as the kinematic-sector birefringence (Eq. 40, Section 12.8). Both anisotropies inherit the same axis from the same underlying geometry.

Figure 9 renders the two anisotropy tensors as level sets of their respective quadratic forms. The kinematic frame matrix M in $\mathbf{k}$-space is a *prolate* ellipsoid whose long axis lies along the optical axis; the gauge-sector tensor $\mathbf{Q}$ in $\mathbf{B}$-space (the Hodge dual of antisymmetric F_{ij}) is an *oblate* ellipsoid whose short axis lies along the same optical axis. The same direction in spatial coordinates maps to opposite extremal axes of the two quadratic forms: kinematic dispersion is flat along $(1, 1, -1)$ and steep perpendicular; the gauge sector's induced photon kinetic term is steepest along $(1, 1, -1)$ and flatter perpendicular. Same geometry, dual phenomenology, two falsifiable signatures.

B.5 O_h averaging recovers Maxwell

Physical observables on the lattice must be invariant under the crystallographic point group O_h (order 48), which acts simultaneously on the spatial coordinates and on the antisymmetric $\mathbf{F}$-space. Averaging Eq. (147) over O_h gives

$$\left\langle \mathbf{F}^T \mathbf{Q}\, \mathbf{F} \right\rangle_{O_h} = \frac{\mathrm{Tr}\,\mathbf{Q}}{3} \sum_{i<j} F_{ij}^2 = 8 \sum_{i<j} F_{ij}^2. \tag{151}$$

Using $F_{\mu\nu}F^{\mu\nu} = 2\sum_{i<j} F_{ij}^2 +$ (electric contribution) and absorbing the $\sum_{i<j} F_{ij}^2 \to \frac{1}{2}F_{\mu\nu}F^{\mu\nu}$ identification in the magnetic sector, the leading O_h-averaged induced action is

$$S_{\text{eff}}^{(\text{leading})}[A] \xrightarrow[a\to 0]{O_h \text{ average}} \frac{2c}{3} \int d^4x\, \tfrac{1}{2}F_{\mu\nu}(x)\, F^{\mu\nu}(x), \tag{152}$$

the standard Maxwell action. The numerical prefactor $c = 1/g^2$ remains to be determined by the explicit one-loop $-\mathrm{Tr}\,\ln D_{\text{lat}}[U]$ calculation; what is determined here is that whatever prefactor that calculation produces, the O_h-averaged form of the leading induced term is the standard Maxwell density.

The pre-averaging operator-level density carries the residual

$$S_{\text{eff}}^{(\text{operator})}[A] = S_{\text{eff}}^{(\text{leading})}[A] + \frac{c\,a^4}{2}\,\mathbf{F}^T\,\mathbf{Q}_{\text{aniso}}\,\mathbf{F}. \tag{153}$$

The $\mathbf{Q}_{\text{aniso}}$ term is the gauge-sector birefringence: a closed-form, direction-dependent contribution to the photon dispersion with optical axis $(1, 1, -1)$. It is the gauge-field twin of the kinematic prediction of Section 12.8.

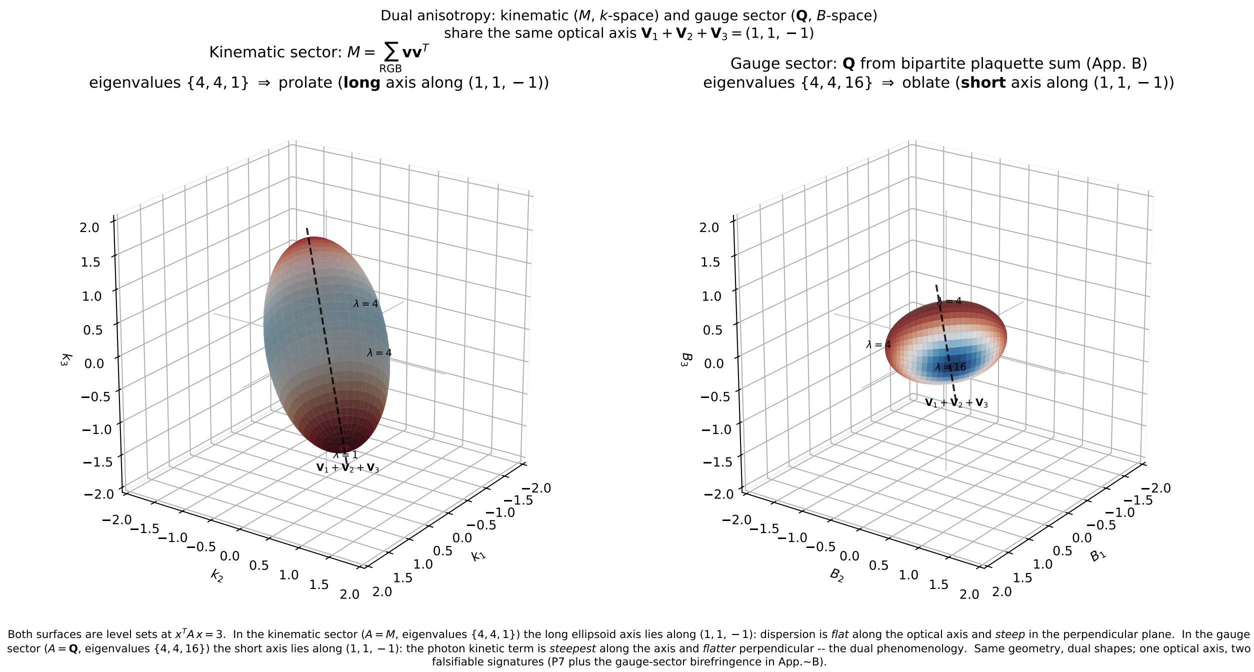


Figure 9: Dual anisotropy on a single optical axis. Both panels render level sets at $x^T A\, x = 3$ for the framework's two anisotropy tensors; both share the special direction $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$ (dashed black line). **Left (kinematic sector):** the frame matrix $M = \sum_{\mathrm{RGB}} \mathbf{v}\mathbf{v}^T$ (Eq. 38) on $\mathbf{k}$-space. Eigenvalues $\{4, 4, 1\}$ produce a *prolate* ellipsoid whose *long* axis lies along the optical axis: kinematic dispersion is flat along $(1, 1, -1)$ and steep in the perpendicular plane. This is the same quadratic form that underlies the $|\tilde{H}_{\mathrm{RGB}}|^2$ expansion of M_{eff} (Eq. 40, Section 12.8). **Right (gauge sector):** the bipartite plaquette tensor $\mathbf{Q}$ (Eq. 148) on $\mathbf{B}$-space (the Hodge dual of antisymmetric F_{ij}). Eigenvalues $\{4, 4, 16\}$ produce an *oblate* ellipsoid whose *short* axis lies along $(1, 1, -1)$: the induced photon kinetic term is steepest along the optical axis and flatter perpendicular — the dual of the kinematic shape. Both anisotropies inherit the special direction from the same bipartite RGB/CMY geometry; the dual ellipsoid shapes give two independent falsifiable signatures with one optical axis (kinematic-sector birefringence in P7, gauge-sector birefringence in Appendix B).

B.6 Status of the induced-action programme

The structural step of the programme set out in Section 9.7 is closed by this appendix:

- **Locality + gauge invariance + minimality** force the leading-order induced action to be quadratic in F_{ij}, summed over closed plaquette holonomies (Section B.3).
- **The bipartite RGB/CMY geometry** fixes the tensor coefficient $\mathbf{Q}$ with eigenvalues $\{4, 4, 16\}$ (Section B.4).
- O_h **averaging** recovers the standard Maxwell density up to the numerical prefactor $c = 1/g^2$ (Section B.5).

- **The operator-level residual** is a closed-form anisotropic tensor whose special direction matches the kinematic-sector optical axis $(1, 1, -1)$ (Eq. 153).

What remains to fully close the programme is the explicit numerical calculation of $c = 1/g^2$. This is a one-loop lattice perturbation theory calculation for the bipartite Dirac operator $D_{\mathrm{lat}}[U]$. Standard staggered-fermion machinery (Kogut–Susskind, Susskind 1977; cf. Section 6.4) applies, with the doubler structure and the bipartite RGB/CMY chirality projection requiring careful bookkeeping. We flag this as a paper-length follow-on project. The numerical $1/g^2$ prefactor is the only remaining open item in the induced-action programme: the structural form, the bipartite **Q**-tensor with eigenvalues $\{4, 4, 16\}$, and the closed-form gauge-sector birefringence along the $(1, 1, -1)$ optical axis are all in hand.

The structural form, including the prediction of gauge-sector birefringence with optical axis $(1, 1, -1)$, is established independently of that calculation. This converts the asymmetry flagged in Section 9.6 – gravity end-to-end versus electromagnetism postulating the Wilson action – into a narrower asymmetry: gravity is fully derived; electromagnetism is derived up to a numerical coupling constant. The induced action itself, with its leading structural form and bipartite anisotropy, is no longer a postulate.

B.7 Reproducibility

The full symbolic calculation is performed in `src/utilities/induced_gauge_action.py`. Running

```
python -u src/utilities/induced_gauge_action.py
```

reproduces every numerical claim in this appendix and writes the captured output to `data/induced_gauge_a`
The calculation requires only `sympy`; no specialised lattice gauge theory infrastructure.

The five-line core verification of **Q**'s eigenvalues is

```
import sympy as sp
V1, V2, V3 = sp.Matrix([1,1,1]), sp.Matrix([1,-1,-1]), sp.Matrix([-1,1,-1])
F = sp.Matrix(3, 3, lambda i,j: sp.Symbol(f"F{i}{j}", real=True)*(1 if i<j else (-1
proj = lambda Va, Vb: sum(Va[i]*Vb[j]*F[i,j] for i in range(3) for j in range(3))
Q_density = sum(sp.expand(proj(V,W)**2) for V,W in [(V1,V2),(V1,V3),(V2,V3)])
```

extracting **Q** from the quadratic form in the $\{F_{12}, F_{13}, F_{23}\}$ basis yields Eq. (148) and the eigenvalues $\{4, 4, 16\}$.

C Reproducibility Quickstart

This compact appendix lets a reviewer verify a working build of the numerical machinery in under a minute, without reading Appendix A's full technical detail. All paths and commands assume the repository root as the working directory.

C.1 Snapshot

- **Repository:** `github.com/JackDMenendez/discrete-causal-lattice`
- **Tag for this paper:** `v0.98-RC`
- **Zenodo DOI:** `10.5281/zenodo.20017133` (commit hash recorded in the Zenodo deposit metadata)
- **License:** MIT
- **Contact:** `jackdmenendez@gmail.com`

C.2 Minimum environment

- Python 3.14.4, `numpy 2.4.3`, `scipy 1.17.1`
- Tested on Windows 11 (MSYS2 UCRT64) and Linux
- No GPU; consumer-grade CPU sufficient
- Full pinned dependencies and toolchain notes in Appendix A

C.3 Hardware and observed runtimes

Wall-clock times measured during preparation of this paper on a single consumer CPU (8-core, 16-thread, no GPU acceleration):

- `exp_12b` smoke test (TICKS=20, 65^3 grid): ~ 14 s.
- `exp_12b` full baseline (TICKS=6000, 65^3 grid): ~ 87 min, single worker.
- `exp_20` three-arm full sweep (TICKS=6000 per arm, 65^3 grid): ~ 119 min wall-clock with three parallel workers.
- `exp_19c` v10 sweep (TICKS=6000 per rate, 65^3 grid, five rates): ~ 5 h per worker, five parallel workers.

C.4 Three canonical entry points

The three experiments most directly load-bearing for the audit-table claims are runnable with single commands. In each case `TICKS=20` is the smoke-test mode (verifies the wiring in seconds); the production runs reproduce the full numbers cited in the paper.

```
# Two-body Bohr radius (k_min = 0.0970 vs k_Bohr = 0.0971)
make -C src/experiments exp_12

# Three-arm emission operator comparison (joint A=1 verification)
EXP20_TICKS=6000 python src/experiments/
    exp_20_emission_operator.py

# Long-duration two-body baseline (no emission)
EXP12B_TICKS=6000 python src/experiments/
    exp_12b_twobody_long_baseline.py
```

C.5 Minimal verification snippet

The fastest end-to-end check. Runs `exp_12b` for 20 ticks (approximately 14 s on the reference hardware) and prints two scalars that a clean checkout must reproduce.

```
# verify_install.py --- run from repository root
import os, subprocess, numpy as np
os.environ['EXP12B_TICKS'] = '20'
subprocess.check_call([
  'python', 'src/experiments/exp_12b_twobody_long_baseline.py'
])
arr = np.load('data/exp_12b_baseline.npy')
# index layout:  [settled, max_streak, T_settle, amp_e, amp_p,
# A_drift_max, rho_drift_max, r_peak_min, r_peak_max]
r_peak_min = arr[7]; A_drift = arr[5]
print(f"r_peak_min = {r_peak_min:.3f} expected ≈ 9.877")
print(f"A_drift      = {A_drift:.2e}  expected ≈ 4.4e-16")
```

Reference output (bit-exact on the reference software stack; deterministic by construction — the framework uses no RNG for any of these three experiments):

```
r_peak_min = 9.877 expected ≈ 9.877
A_drift      = 4.44e-16 expected ≈ 4.4e-16
```

If both scalars match within $\sim 1\%$ of the reference values, the numerical machinery reproduces. $\mathcal{A}$-drift at machine precision ($\lesssim 10^{-15}$) is the structural correctness check: the bipartite tick rule preserves $\sum_i \|\psi_i\|^2$ exactly under `tick(normalize=True)`, so any larger drift indicates an environment problem (numerical-precision flag, incompatible `numpy` build, etc.).

C.6 If the snippet fails

- `r_peak_min` **differs by more than** $\sim 5\%$**:** verify the Coulomb constants (`STRENGTH=30.0`, `SOFTENING=0.5`) and that `src/experiments/exp_12b_twobody_long_baseline.py` imports a clean `src/core` module set; a stale `__pycache__/` can also matter.
- `A_drift` **larger than** $\sim 10^{-12}$**:** indicates a different floating-point environment than the reference stack; check `numpy` build flags and BLAS backend.
- **Import error / module not found:** re-run `make env` or check that the project virtualenv is activated; the script imports `src.core.{OctahedralLattice, CausalSession, enforce_unity_spinor}`.
- For full troubleshooting, see Appendix A.

C.7 Beyond the smoke test

The full audit-table evidence cited in this paper requires the production runs in §C.4, not the 20-tick smoke test. Wall-clock budgets are given in §C.3. Each experiment writes its own `.log` (per-window progress) and `.npy` (summary array) under `data/`; the per-experiment documentation in `src/experiments/*.md` states the expected results, runtime, and any non-default flags.

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